Computations in Enumerative Geometry

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Contents

1 Moduli spaces **5**

1.1 Enumerative problems . . . 5

1.1.1 Transversality, and counting lines through two points . . . 7

1.2 Warming up: lines on hypersurfaces . . . 10

1.3 Hilbert and Quot schemes of points . . . 12

1.3.1 Existence, and Quot-to-Chow . . . 12

1.3.2 What's (un)known? . . . 13

1.3.3 The local case . . . 14

1.3.4 Quot-to-Chow in the local case . . . 16

1.3.5 Smoothness . . . 17

1.3.6 Torus action on the Hilbert scheme of points . . . 17

1.3.7 The character of the tangent space . . . 21

2 Motivic techniques **27**

2.1 Grothendieck ring of varieties and attached gadgets . . . 27

2.1.1 Absolute setup: motivic measures and zeta functions . . . 27

2.1.2 Relative setup: main structures and operations . . . 29

2.1.3 Power structures . . . 31

2.1.4 Motivic exponential . . . 32

2.2 The Denef–Loeser motivic zeta function . . . 33

2.3 Białynicki-Birula decomposition . . . 35

2.4 The motive of the Quot scheme of points on a curve . . . 37

2.5 The motive of the Hilbert scheme of points on $\mathbb{A}^2$ . . . 39

2.6 The motive of the Quot scheme of points . . . 43

2.6.1 Power structure formula for Quot schemes . . . 43

2.6.2 Computation of Hilb_S . . . 47

2.7 Motivic DT invariants of $\mathbb{A}^3$ . . . 47

2.7.1 The virtual motive of a critical locus . . . 48

2.7.2 Quivers: framings, and motivic quantum torus . . . 48

2.7.3 Quiver representations and their stability . . . 49

2.7.4 Quivers with potential . . . 50

2.7.5 Framed motivic DT invariants . . . 51

2.7.6 Dimensional reduction . . . 52

2.7.7 Stability on the framed 3-loop quiver . . . 52

2.7.8 The virtual motive of the Quot scheme of points . . . 53

2.7.9 Calculation via wall-crossing 54

3 Localisation techniques **57**

3.1 Equivariant cohomology 57

3.1.1 Universal principal bundles and classifying spaces 57

3.1.2 Definition of equivariant cohomology 61

3.1.3 Moving to algebraic geometry: approximation spaces 65

3.1.4 Equivariant vector bundles 67

3.1.5 Two computations on $\mathbb{P}^{n-1}$ 69

3.2 The Atiyah–Bott localisation formula 70

3.2.1 A glimpse of the self-intersection formula 70

3.2.2 Equivariant pushforward 72

3.2.3 Trivial torus actions 73

3.2.4 Torus fixed loci 73

3.2.5 The localisation formula 75

3.3 Applications of the localisation formula 78

3.3.1 How not to compute the simplest intersection number 78

3.3.2 The 27 lines on a smooth cubic surface 80

3.3.3 The 2875 ines on the quintic 3-fold 82

3.3.4 The degree of $\mathbb{G}(1, n+1)$ and its enumerative meaning 83

1 Moduli spaces

1.1 Enumerative problems

The typical question in classical enumerative geometry asks how many objects satisfy a given list of geometric conditions. The presence of this 'list' makes the subject tightly linked to the subject of intersection theory.

Examples of classical problems in the subject are the following:

(1) Given an integer $n > 0$, how many lines $\ell \subset \mathbb{P}^{n+1}$ are incident to $2n$ general $(n-1)$-planes $\Lambda_1, \ldots, \Lambda_{2n} \subset \mathbb{P}^{n+1}$? (Answer in Section 3.3.4)

(2) How many lines $\ell \subset \mathbb{P}^3$ lie on a general cubic surface $S \subset \mathbb{P}^3$? (Answer in Section 3.3.2)

(3) How many lines $\ell \subset \mathbb{P}^4$ lie on a general quintic 3-fold $Y \subset \mathbb{P}^4$? (Answer in Section 3.3.3)

In the questions above the objects we want to count are lines in some projective space, on which we impose some geometric constraints, such as intersecting other linear spaces or lying on a smooth hypersurface. In (1), we impose incidence with $2n$ linear spaces. Why exactly $2n$? We immediately see that in order to even get started we have to ask ourselves the following:

> **Question 1**. How do we know how many constraints we should put on our objects in order to *expect* a finite answer? In other words, how do we ask the right question?

See Section 1.2 for a full treatment of the topic 'expectations' in the case of lines on hypersurfaces: we will confirm that the expected number of lines on a general hypersurface $Y \subset \mathbb{P}^n$ of degree d is finite precisely when $d = 2n-3$.

The main idea to guide our geometric intuition in formulating and solving an enumerative problem should be the following recipe:

(A) construct a moduli[1] space $\mathcal{M}$ for the objects we are interested in,

(B) compactify $\mathcal{M}$ if necessary,

(C) impose $\dim \mathcal{M}$ conditions to expect a finite number of solutions, and

(D) count these solutions via intersection theory methods.

[1] The latin word *modulus* means *parameter*, and its plural is *moduli*. Thus a *moduli space* is to be thought of as a parameter space for objects of some kind.

None of these steps is a trivial one, in general. The last two, in a little more detail, would ideally go as follows: each 'condition' we impose in step (C) is described by a cycle $Z_i \subset \mathcal{M}$ which is Poincaré dual to a Chow class $\alpha_i \in A^*\mathcal{M}$, and the intersection of these cycles is represented by the product $\alpha = \alpha_1 \cup \cdots \cup \alpha_r \in A^{\dim \mathcal{M}}\mathcal{M}$, where '$\cup$' is the ring multiplication in the Chow ring $A^*\mathcal{M}$.[2] The fact that α lies is the top degree (i.e. in maximal codimension) means that

$$\sum_{i=1}^{r} \operatorname{codim}(Z_i, \mathcal{M}) = \dim \mathcal{M},$$

which we should have achieved in the third step if we have 'asked the right question'. The final step asks us to compute the number

$$\int_{\mathcal{M}} \alpha = \deg_{\mathcal{M}}(\alpha \cap [\mathcal{M}]) \in \mathbb{Z},$$

where $\deg_{\mathcal{M}} : A_*\mathcal{M} \to A_*\mathrm{pt} = \mathbb{Z}$ is the pushforward on Chow groups along the structure morphism $\mathcal{M} \to \mathrm{pt}$, which exists by compactness of $\mathcal{M}$.

We are therefore facing the following question:

> **Question 2**. How do we know this intersection-theoretic degree is the answer to our original question? In other words, how to ensure that our algebraic solution is actually *enumerative*?

Concretely, take Problem (2): how do we make sure that each line $\ell \subset S$ appears as a point in the moduli space with multiplicity one? The truth is that we cannot *always* be sure that this is the case. It will be, both for Problem (2) and Problem (3) (by Lemma 1.2.4 and Theorem 3.3.5 respectively), but not in general. However, we should get used to the idea that this is not something to be worried about: if a solution comes with multiplicity bigger than one, there must be some good geometric reason for this, and we should not disregard it. More precisely (but not too precisely), if a point on the moduli space is 'fat', i.e. nonreduced, it means that the geometric object it corresponds to has nontrivial deformations, and thus it is natural to count it with some multiplicity — in this sense, one may say that our original enumerative question was too naive.

Remark 1.1.1. Compactness of $\mathcal{M}$ (in the above example, the Grassmannian) is used in order to make sense of taking the *degree* of cycles. Intuitively, we need compactness in order to prevent the solutions of our enumerative problem to escape to infinity, like for instance it would occur if we were to intersect two *parallel* lines in $\mathbb{A}^2$.

We emphasise that compatcness really is a non-negotiable condition we have to ask of our moduli space — with an important exception, that will be treated in later sections: the case when the moduli space has a torus action. In this case, if the torus fixed locus $\mathcal{M}^{\mathbb{T}} \subset \mathcal{M}$ is compact, a sensible enumerative solution to a counting problem can be *defined* by means of the *localisation formula*, one of the most important tools in enumerative geometry (and in these notes). The original formula due to Atiyah and Bott will be proved in Theorem 3.2.8.

One more fundamental notion in counting problems, *transversality*, is discussed in the next subsection, by means of an elementary example.

[2] The reader can more or less safely replace A_* with homology H_* throughout these notes.

1.1.1 Transversality, and counting lines through two points

Let X be an algebraic variety, $x \in X$ a point. Denote by $T_x X = (\mathfrak{m}_x/\mathfrak{m}_x^2)^*$ the Zariski tangent space of X at x, where $\mathfrak{m}_x \subset \mathscr{O}_{X,x}$ is the maximal ideal in the local ring attached to x.

Definition 1.1.2 (Transverse intersection)**.** Let Y be a quasiprojective variety. We say that subvarieties $Z_1, \dots, Z_m$ of Y *intersect transversely* at a smooth point $y \in Y$ if y is a smooth point on each Z_i and $\operatorname{codim}(\bigcap_i T_y Z_i, T_y Y) = \sum_i \operatorname{codim}(T_y Z_i, T_y Y)$.

Consider the enumerative problem of counting the number of lines in $\mathbb{P}^2$ through two given points $p, q \in \mathbb{P}^2$. Let N_{pq} be this number. Then, of course,

$$\mathsf{N}_{pq} = \begin{cases} 1 & \text{if } p \neq q, \\ \infty & \text{if } p = q. \end{cases}$$

The answer $\mathsf{N}_{pp} = \infty$ corresponds to the cardinality of the pencil $Z_p \cong \mathbb{P}^1$ of lines through p (see Figure 1.1). For the sake of 'completeness', the formula $\mathsf{N}_{pq} = 1$ will be proved in Section 3.3.1.

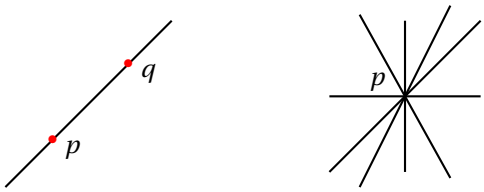


Figure 1.1: The unique line through two distinct points, and the infinitely many lines through one point in the plane.

The point we want to make in this section is that

$$\boxed{\text{the answer '1' can be recovered in the degenerate setting } p = q.}$$

Remark 1.1.3 (Let us not cheat)**.** To obtain the answer relative to the picture on the left, we have to notice that the two cycles

$$Z_p = \left\{ \ell \subset \mathbb{P}^2 \,\middle|\, p \in \ell \right\} \subset \mathbb{P}^{2*}, \quad Z_q = \left\{ \ell \subset \mathbb{P}^2 \,\middle|\, q \in \ell \right\} \subset \mathbb{P}^{2*}$$

intersect transversely in the dual projective space $\mathbb{P}^{2*}$, and we can use the intersection product on $\mathbb{P}^{2*}$ to compute $Z_p \cdot Z_q = \#(Z_p \cap Z_q) = 1$. Now, using basic intersection theory, it is clear how to obtain the answer '1' also in the case $p = q$. Since we are working in $\mathbb{P}^{2*} \cong \mathbb{P}^2$, we know that any $q \neq p$ yields a *homologous* cycle $Z_q \sim Z_p$, and again the intersection product yields $Z_p^2 = Z_p \cdot Z_q = 1$. But in general we will not be working in such a pleasant ambient space and thus we will not know whether algebraic deformations such as $Z_p \rightsquigarrow Z_q$, leading to a transverse setup, are available. For instance, a (-1)-curve[3] C on a surface S cannot be 'moved' algebraically on S to another curve C' such that C and C' intersect transversely!

Now, the case $p = q$ is a 'degeneration' of the case $p \neq q$, and we certainly want our enumerative answer not to depend on small perturbations of the geometry of the problem. Why do we want that?

[3] A (-1)-curve on a surface S is a curve $C \subset S$ such that $C.C = -1$, where the intersection number $C.C$ can be seen as the degree of the normal bundle $\mathscr{N}_{C/S} = \mathscr{O}_S(C)|_C$ to C in S.

Just because we are reasonable people: we were already taught how to be reasonable when our first Calculus teacher told us that a decent function is *at least* continuous.[4]

Next we explain how to get the 'correct' answer

$$\mathsf{N}_{pp}^{\text{corrected}} = 1$$

by means of the *excess intersection formula*, one of the most important tools in classical enumerative geometry. We mention it not only because it is a beautiful piece of intersection theory, but also because it lies at the very roots of *modern* enumerative geometry, lying right at the foundation of the idea of virtual classes.

Before we start, we need to recall the following important notion.

Definition 1.1.4 ((Co)normal sheaf)**.** The *conormal sheaf* of a closed immersion of schemes $X \hookrightarrow M$ defined by an ideal $\mathscr{I} \subset \mathscr{O}_M$ is the quasicoherent[5] $\mathscr{O}_X$-module

$$\mathscr{C}_{X/M} = \mathscr{I}/\mathscr{I}^2,$$

and the *normal sheaf* is its $\mathscr{O}_X$-linear dual,

$$\mathscr{N}_{X/M} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{I}/\mathscr{I}^2, \mathscr{O}_X).$$

The sheaves $\mathscr{C}_{X/M}$ and $\mathscr{N}_{X/M}$ are locally free (of rank d) when $X \hookrightarrow M$ is a regular immersion (of codimension d).

Example 1.1.5. If $X \hookrightarrow M = \mathbb{P}^r$ is a hypersurface of degree d, then the ideal sheaf of X in $\mathbb{P}^r$ is the invertible sheaf $\mathscr{O}_{\mathbb{P}^r}(-d)$, so $\mathscr{N}_{X/\mathbb{P}^r} = \mathscr{O}_{\mathbb{P}^r}(d)|_X$.

Notation 1.1.6. Let $X \hookrightarrow M$ be a closed immersion. We set

$$N_{X/M} = \operatorname{Spec}_{\mathscr{O}_X} \operatorname{Sym} \mathscr{C}_{X/M}.$$

It is naturally a scheme over X. With a slight terminology abuse, we will also refer to it as the *normal sheaf* to X in M.

Exercise 1.1.7. Let $X \hookrightarrow M$ be a closed immersion, $\widetilde{M} \to M$ a morphism, set $\widetilde{X} = X \times_M \widetilde{M}$ and let $g\colon \widetilde{X} \to X$ be the induced map. Show that there is a natural injective map of sheaves $\mathscr{N}_{\widetilde{X}/\widetilde{M}} \hookrightarrow g^*\mathscr{N}_{X/M}$, which is an isomorphism whenever $\widetilde{M} \to M$ is flat. Deduce that there is a closed immersion

$$N_{\widetilde{X}/\widetilde{M}} \hookrightarrow g^* N_{X/M} = N_{X/M} \times_X \widetilde{X}$$

of schemes over $\widetilde{X}$. (**Hint**: Try to construct a surjection $g^*\mathscr{C}_{X/M} \twoheadrightarrow \mathscr{C}_{\widetilde{X}/\widetilde{M}}$ involving the conormal sheaves. If in need of further hints, see [85, Tag 01R1]).

Also recall (see [50, II.8.13] for a reference) that on any projective space $\mathbb{P}^r$ we have the *Euler sequence*

$$0 \to \mathscr{O}_{\mathbb{P}^r} \to \mathscr{O}_{\mathbb{P}^r}(1)^{\oplus r+1} \to \mathscr{T}_{\mathbb{P}^r} \to 0, \tag{1.1.1}$$

where $\mathscr{O}_{\mathbb{P}^r}(1)$ is the hyperplane bundle and $\mathscr{T}_{\mathbb{P}^r}$ is the algebraic tangent bundle.

Now back to our problem. The $\mathbb{P}^1$ of lines through p can be neatly seen as the exceptional divisor E in the blowup $B = \operatorname{Bl}_p \mathbb{P}^2$, cf. Figure 1.2.

[4] Ultimately, we are going to study *invariants*, e.g. Donaldson–Thomas invariants. They deserve to be called that precisely because they don't change if we slightly (but holomorphically) deform the variety they are attached to.

[5] It is coherent as long as M is locally noetherian.

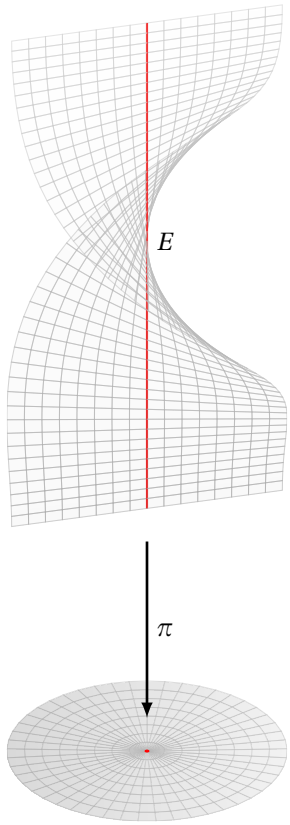


Figure 1.2: The blowup $\pi\colon B \to \mathbb{P}^2$ of the plane in one point.

Looking at the fibre diagram

$$\begin{tikzcd} E \arrow[r, hook] \arrow[d, "g"'] \arrow[dr, phantom, "\square"] & B \arrow[d, "\pi"] \\ p \arrow[r, hook] & \mathbb{P}^2 \end{tikzcd} \tag{1.1.2}$$

we know by Exercise 1.1.7 that there is an injection of locally free sheaves $\mathcal{N}_{E/B} = \mathcal{O}_E(-1) \subset g^*\mathcal{N}_{p/\mathbb{P}^2}$. The *excess bundle* (or *obstruction bundle*)

$$\mathrm{Ob}_{pp} \to \mathbb{P}^1$$

of the fibre diagram (1.1.2) is defined as the quotient of these two bundles [42, Section 6.3]. But the short exact sequence

$$0 \to \mathcal{O}_E(-1) \to \mathcal{O}_E \otimes T_p\mathbb{P}^2 \to \mathrm{Ob}_{pp} \to 0$$

is nothing but the Euler sequence (1.1.1) on $\mathbb{P}^1$ twisted by $\mathcal{O}_E(-1)$. Therefore

$$\mathrm{Ob}_{pp} = \mathcal{T}_{\mathbb{P}^1}(-1) = \mathcal{O}_{\mathbb{P}^1}(2-1) = \mathcal{O}_{\mathbb{P}^1}(1).$$

Note that we can repeat the process with $q \neq p$, which would yield $\pi^{-1}(q) = \mathrm{pt}$. In this case we get $\mathrm{Ob}_{pq} = 0$. We can now write a universal formula for our counting problem: if $\mathcal{M}_{pq} = \pi^{-1}(q)$ is the 'moduli space of lines' through p and q, the *virtual number* of lines through p and q is

$$\int_{\mathcal{M}_{pq}} e(\mathrm{Ob}_{pq}) = 1,$$

for all $(p,q) \in \mathbb{P}^2 \times \mathbb{P}^2$. The *Euler class* $e(V)$ of a vector bundle V is its top Chern class. Note that the rank of Ob_{pq} is the difference between the actual dimension of the moduli space, and the expected one.

Remark 1.1.8. Note that, unlike in Remark 1.1.3, we have now obtained

$$\mathsf{N}_{pp}^{\text{corrected}} = 1$$

as an intersection number on the *actual* moduli space $Z_p \cong \mathbb{P}^1 \cong E$.

This discussion allows us to formulate another intrinsic difficulty in enumerative geometry. Suppose, just to dream for a second, that we are able to solve *all* enumerative problems in generic (transverse) situations, and we know that the answer does not change after a small perturbation of the initial data.

> **Question 3**. How do we 'pretend' we can work in a transverse situation when there is none available?

The modern way to do this is to use *virtual fundamental classes*.

1.2 Warming up: lines on hypersurfaces

Let $Y \subset \mathbb{P}^n$ be a general hypersurface of degree d. We want to show the following:

We should expect a finite number of lines on Y if and only if $d = 2n-3$.
We should expect *no lines* on Y if $d > 2n-3$.
We should expect infinitely many lines on Y if $d < 2n-3$.

To understand the condition

$$\ell \subset Y$$

for a given line $\ell \subset \mathbb{P}^n$ and a hypersurface $Y \subset \mathbb{P}^n$, we give the following concrete example.

Example 1.2.1. Let $\ell \subset \mathbb{P}^3$ be the line cut out by $L_1 = L_2 = 0$, where $L_i = L_i(z_0, z_1, z_2, z_3)$ are linear forms on $\mathbb{P}^3$. To fix ideas, set $L_1 = z_0$ and $L_2 = z_0 + z_2 + z_3$. Let $Y \subset \mathbb{P}^3$ be defined by a homogeneous equation $f = 0$, for instance the cubic polynomial

$$f = z_0^3 + 3z_0 z_1^2 - z_2^2 z_3.$$

Then we see that plugging $L_1 = L_2 = 0$ into f does not give zero, for

$$f\big|_\ell = 0 + 0 - z_2^2(-z_0 - z_2) = z_2^3.$$

This means that ℓ is not contained in Y. On the other hand, the line cut out by L_1 and $L_2' = z_3$ lies entirely on Y.

Let $Y \subset \mathbb{P}^n$ be the zero locus of a general homogeneous polynomial

$$f \in \mathrm{H}^0(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(d)).$$

As we anticipated in Example 1.2.1, a line $\ell \subset \mathbb{P}^n$ is contained in Y if and only if $f|_\ell = 0$. This condition can be rephrased by saying that the image of f under the restriction map

$$\mathrm{H}^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) \xrightarrow{\mathsf{res}_\ell} \mathrm{H}^0(\ell, \mathcal{O}_\ell(d)) \tag{1.2.1}$$

vanishes. We want to determine when we should expect Y to contain a finite number of lines. We set, informally,

$$\mathsf{N}_1(Y) = \text{expected number of lines in } Y.$$

Let us consider the Grassmannian

$$\mathbb{G} = \mathsf{Gr}(2, n+1) = \mathbb{G}(1, n) = \left\{ \text{Lines } \ell \subset \mathbb{P}^n \right\},$$

a smooth complex projective variety of dimension $2n-2$. Recall the universal structures living on $\mathbb{G}$. First of all, the tautological exact sequence

$$\begin{tikzcd}
 & \text{rank } 2 \arrow[d, dash] & \text{rank } n+1 \arrow[d, dash] & \text{rank } n-1 \arrow[d, dash] & \\
0 \arrow[r] & \mathscr{S} \arrow[r] & \mathcal{O}_{\mathbb{G}} \otimes_{\mathbb{C}} \mathrm{H}^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))^* \arrow[r] & \mathscr{Q} \arrow[r] & 0
\end{tikzcd}$$

where the fibre of $\mathscr{S}$ over a point $[\ell] \in \mathbb{G}$ is the 2-dimensional vector space $\mathrm{H}^0(\ell, \mathcal{O}_\ell(1))^*$. Let, also,

$$\mathcal{L} = \left\{ (p, [\ell]) \in \mathbb{P}^n \times \mathbb{G} \,\middle|\, p \in \ell \right\} \subset \mathbb{P}^n \times \mathbb{G}$$

be the universal line. Consider the two projections

$$\begin{tikzcd}
\mathcal{L} \arrow[r, "q"] \arrow[d, "\pi"'] & \mathbb{P}^n \\
\mathbb{G} &
\end{tikzcd}$$

and the coherent sheaf

$$\mathscr{E}_d = \pi_* q^* \mathcal{O}_{\mathbb{P}^n}(d).$$

Exercise 1.2.2. Show that $\mathscr{E}_d$ is locally free of rank $d+1$, and that one has an isomorphism of locally free sheaves

$$\mathscr{E}_d \cong \mathsf{Sym}^d \mathscr{S}^*.$$

(**Hint**: use cohomology and base change, e.g. [31, Theorem B.5]).

Dualising the universal inclusion $\mathscr{S} \hookrightarrow \mathcal{O}_{\mathbb{G}} \otimes_{\mathbb{C}} \mathrm{H}^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))^*$ and applying Sym^d, we obtain a surjection

$$\mathcal{O}_{\mathbb{G}} \otimes_{\mathbb{C}} \mathrm{H}^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) \twoheadrightarrow \mathsf{Sym}^d \mathscr{S}^*,$$

which is just a global version of (1.2.1). The association

$$\mathbb{G} \ni [\ell] \mapsto f|_\ell \in \mathrm{H}^0(\ell, \mathcal{O}_\ell(d)) \cong \mathsf{Sym}^d \mathrm{H}^0(\ell, \mathcal{O}_\ell(1))$$

defines a section τ_f of $\mathscr{E}_d \to \mathbb{G}$. The zero locus of $\tau_f = \pi_* q^* f \in \mathrm{H}^0(\mathbb{G}, \mathscr{E}_d)$ is the locus of lines contained in Y.

Definition 1.2.3 (Fano scheme of lines)**.** Let $Y \subset \mathbb{P}^n$ be a hypersurface defined by $f = 0$. The zero scheme

$$F_1(Y) = Z(\tau_f) \hookrightarrow \mathbb{G} = \mathbb{G}(1, n)$$

is called the *Fano scheme of lines* in Y.

Since f is generic, $\tau_f \in \Gamma(\mathbb{G}, \mathscr{E}_d)$ is also generic. Therefore, the fundamental class of the Fano scheme of lines in Y is Poincaré dual to the Euler class

$$e(\mathscr{E}_d) \in A^{d+1}\mathbb{G}.$$

Thus $[F_1(Y)] = e(\mathscr{E}_d) \cap [\mathbb{G}] \in A_*\mathbb{G}$ is a 0-cycle if and only if $d+1 = 2n-2$, i.e.

$$d = 2n-3.$$

A rigorous definition of $\mathsf{N}_1(Y)$ is then

$$\mathsf{N}_1(Y) = \int_{\mathbb{G}} e(\mathscr{E}_d) = \int_{\mathbb{G}} c_{d+1}(\mathsf{Sym}^d \mathscr{S}^*).$$

Such degree is the *actual* number of lines on Y whenever $\mathrm{H}^0(\ell, \mathscr{N}_{\ell/Y}) = 0$ for all $\ell \subset Y$. This condition means that the Fano scheme is reduced at all its points $[\ell]$, since $\mathrm{H}^0(\ell, \mathscr{N}_{\ell/Y})$ is its tangent space at the point $[\ell]$.

LEMMA 1.2.4. *If $S \subset \mathbb{P}^3$ is a smooth cubic surface and $\ell \subset S$ is a line, then* $\mathrm{H}^0(\ell, \mathscr{N}_{\ell/S}) = 0$.

Proof. It is enough to show that $\mathscr{N} = \mathscr{N}_{\ell/S}$, viewed as a line bundle on $\ell \cong \mathbb{P}^1$, has negative degree. By the adjunction formula,

$$K_\ell = K_S|_\ell \otimes_{\mathscr{O}_\ell} \mathscr{N}.$$

Using that $K_\ell = \mathscr{O}_\ell(-2)$ and $K_S = K_{\mathbb{P}^3}|_S \otimes_{\mathscr{O}_S} \mathscr{N}_{S/\mathbb{P}^3} = \mathscr{O}_S(d-4)$ for a surface of degree d in $\mathbb{P}^3$, by taking degrees we obtain

$$-2 = (3-4) + \deg \mathscr{N},$$

so that $\deg \mathscr{N} = -1 < 0$. □

Exercise 1.2.5. Let $Y \subset \mathbb{P}^n$ be a general hypersurface of degree $d \leqslant 2n-3$. Show that $F_1(Y) \subset \mathbb{G}(1,n)$ is smooth of dimension $2n-3-d$.

1.3 Hilbert and Quot schemes of points

1.3.1 Existence, and Quot-to-Chow

Let X be a smooth quasiprojective variety of dimension $d \geqslant 1$, and let F be a locally free sheaf of rank $r \geqslant 1$ over X. Fix an integer $n \geqslant 0$. The *Quot functor* attached to this data is the functor $\underline{\mathsf{Quot}}_X(F,n)\colon \mathsf{Sch}_{\mathbb{C}}^{\mathrm{op}} \to \mathsf{Sets}$ sending a $\mathbb{C}$-scheme B to the set of isomorphism classes of surjections $F_B \twoheadrightarrow T$ in $\mathsf{Coh}(X \times B)$, where T is a B-flat sheaf such that T_b is 0-dimensional and satisfies $\chi(T_b) = n$ for every geometric point $b \in B$. Two surjections are considered isomorphic if they share the same kernel. One can also say that the functor parametrises *short exact sequences*

$$0 \longrightarrow K \longrightarrow F_B \longrightarrow T \longrightarrow 0$$

in $\mathsf{Coh}(X \times B)$, with no equivalence relation imposed. A classical result of Grothendieck states that this functor is represented by a quasiprojective scheme (projective as soon as X is projective)

$$\mathsf{Quot}_X(F,n).$$

If $F = \mathscr{O}_X$, we obtain the *Hilbert scheme of points*

$$\mathsf{Hilb}^n(X) = \mathsf{Quot}_X(\mathscr{O}_X, n).$$

It parametrises closed subschemes $Z \subset X$ such that $\chi(\mathscr{O}_Z) = n$. Its B-valued points, namely the short exact sequences

$$0 \longrightarrow \mathscr{I}_{\mathcal{Z}} \longrightarrow \mathscr{O}_{X\times B} \longrightarrow \mathscr{O}_{\mathcal{Z}} \longrightarrow 0,$$

with $\mathscr{O}_{\mathcal{Z}}$ flat over B of relative degree n, correspond precisely to closed subschemes $\mathcal{Z} \hookrightarrow X \times B$ such that $\mathcal{Z} \to B$ is flat of degree n.

Notation 1.3.1. The closed point in $\mathsf{Hilb}^n(X)$ corresponding to a closed subscheme $Z \subset X$ with ideal sheaf $\mathscr{I}_Z \subset \mathscr{O}_Z$ will be denoted by $[Z]$ or by $[\mathscr{I}_Z]$.

Morphisms *into* the Quot scheme are classified by the Quot functor, thanks to its representability. The most important morphism *out* of the Quot scheme is the *Quot-to-Chow morphism*, namely the morphism

$$\mathsf{Quot}_X(F, n) \xrightarrow{\sigma_{F,n}} \mathsf{Sym}^n(X) = X^n/\mathfrak{S}_n, \tag{1.3.1}$$

taking the class of a quotient $[F \twoheadrightarrow T]$ to the 0-cycle

$$[\mathrm{Supp}\, T] = \sum_{x\in X} \mathsf{length}_{\mathscr{O}_{X,x}} T_x \cdot x.$$

See [82, Cor. 7.15] or [38, 37] for the construction. The *punctual Quot scheme*

$$\mathsf{Quot}_X(F, n)_x \subset \mathsf{Quot}_X(F, n)$$

is defined to be the fibre of $\sigma_{F,n}$ over the 'fat cycle' $[n\,x] \in \mathsf{Sym}^n(X)$. If F has rank 1, we denote it by $\mathsf{Hilb}^n(X)_x$.

Definition 1.3.2. The *smoothable component* $\Gamma_{\mathsf{sm}} \subset \mathsf{Hilb}^n(X)$ is the closure of the locus of reduced subschemes $Z \subset X$. If $X = \mathbb{A}^d$, we denote it by $\Gamma^{n,d}_{\mathsf{sm}}$.

PROPOSITION 1.3.3 ([21, Prop. 4.15]). *All monomial ideals $I \subset \mathbb{C}[x_1, \dots, x_d]$ of colength n lie in the smoothable component $\Gamma^{n,d}_{\mathsf{sm}} \subset \mathsf{Hilb}^n(\mathbb{A}^d)$.*

1.3.2 What's (un)known?

Not much is known about the geometry of $\mathsf{Hilb}^n(X)$ for $d \geqslant 3$. Here is a recap on some known properties of Hilbert and Quot scheme of points:

1. $\mathsf{Hilb}^n(\mathbb{A}^d)$ is connected for all n and d. It is smooth (and irreducible) if and only if $d \leqslant 2$ or $n \leqslant 3$.

2. $\mathsf{Hilb}^n(\mathbb{A}^d)$ is irreducible for all d and $n \leqslant 7$, see [67]. If $d \geqslant 4$, then $\mathsf{Hilb}^n(\mathbb{A}^d)$ is irreducible if and only if $n \leqslant 7$.

3. $\mathsf{Hilb}^n(\mathbb{A}^3)$ is irreducible for $n \leqslant 11$, see [53, 29] and the references therein. See also [90] for $n = 9, 10$. Moreover, $\mathsf{Hilb}^n(\mathbb{A}^3)$ is reducible for $n \geqslant 78$, see [56].

4. $\mathsf{Hilb}^{13}(\mathbb{A}^6)$ is nonreduced by Szachniewicz's work [87].

5. The punctual Hilbert scheme $\mathsf{Hilb}^n(\mathbb{A}^2)_0$ is irreducible of dimension $n-1$, see [17, Corollaire V.3.3].

6. The punctual Hilbert scheme $\mathsf{Hilb}^n(\mathbb{A}^3)_0$ is irreducible for all $n \leqslant 11$, see [58].

7. If F is locally free, then $\mathsf{Quot}_X(F,n)$ is connected [73, Theorem 1.4].

8. If $r>1$, then $\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r},n)$ is smooth if and only if $n=1$.

9. $\mathsf{Quot}_{\mathbb{A}^2}(\mathscr{O}^{\oplus r},n)$ is irreducible of dimension $(r+1)n$, see [32].

10. If $d,r>3$, then $\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r},8)$ has a generically nonreduced component [59, §6.5].

On the other hand, the following questions are open since a long time:

1. What is the smallest n such that $\mathsf{Hilb}^n(\mathbb{A}^3)$ is reducible?

2. If $d \geqslant 3$, what is an upper bound for the number of irreducible components of $\mathsf{Hilb}^n(\mathbb{A}^d)$?

3. If $d \geqslant 3$ and $p \in \Gamma^{n,d}_{\mathsf{sm}} \subset \mathsf{Hilb}^n(\mathbb{A}^d)$ belongs to the smoothable component, what is, at most, the dimension of the tangent space of $\mathsf{Hilb}^n(\mathbb{A}^d)$ at p?

4. Is $\mathsf{Hilb}^n(\mathbb{A}^3)$ generically reduced? That is, if $W \subset \mathrm{H} = \mathsf{Hilb}^n(\mathbb{A}^3)$ is an irreducible component with generic point ξ, is $\mathscr{O}_{\mathrm{H},\xi}$ reduced?

1.3.3 The local case

Let us set $X = \mathbb{A}^d$, so that necessarily F is a trivial bundle (of rank r). We now give a hands-on (scheme-theoretic) construction of

$$\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r},n) = \left\{ K \subset \mathbb{C}[x_1,\dots,x_d]^{\oplus r} \;\middle|\; \dim_{\mathbb{C}} \mathbb{C}[x_1,\dots,x_d]^{\oplus r}/K = n \right\},$$

that will turn out useful later. More precisely, we will provide equations cutting out the Quot scheme inside a smooth quasiprojective variety (the so-called *noncommutative Quot scheme*), cf. Theorem 1.3.7.

Let us get started with our description of $\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r},n)$. To ease notation, let us put $R_d = \mathbb{C}[x_1,\dots,x_d]$. The condition defining a point $[K] \in \mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r},n)$ is that the R_d-linear quotient

$$R_d^{\oplus r} \twoheadrightarrow R_d^{\oplus r}/K$$

is a vector space of dimension n. To construct a point in the Quot scheme, we need:

(1) a vector space $V_n \cong \mathbb{C}^n$,

(2) an R_d-module structure

$$\vartheta\colon R_d \longrightarrow \mathrm{End}_{\mathbb{C}}(V_n)$$

with the property that

(3) such structure is induced by an R_d-linear surjection from $R_d^{\oplus r}$.

So let us fix an n-dimensional vector space V_n. Later we will have to remember that we made such a choice, and since all we wanted was "$\dim V_n = n$" we will have to quotient out all equivalent choices. Let us forget about this for the moment. In (2), we need ϑ to be a ring homomorphism, so we need to specify one endomorphism of V_n for each coordinate $x_i \in R_d$. All in all, ϑ gives us d matrices

$$A_1, A_2, \dots, A_d \in \mathrm{End}_{\mathbb{C}}(V_n).$$

The matrix A_i will be responsible for the R_d-linear operator "multiplication by x_i" for the resulting module structure on V_n. Also in this step we should note a reminder for later: strictly speaking, what we have defined so far is a $\mathbb{C}\langle x_1, x_2, \dots, x_d\rangle$-module structure on V_n. But in R_d the variables commute with one another. So we will eventually have to impose the relations $[A_i, A_j] = 0$ for all $1 \leqslant i < j \leqslant d$.

Condition (3) is a little more tricky. Let us reason backwards, assuming we already have an R_d-linear quotient $\phi : R_d^{\oplus r} \twoheadrightarrow V_n$. Then it is clear that the images of the coordinate vectors $\boldsymbol{e}_1, \dots, \boldsymbol{e}_r \in R_d^{\oplus r}$ must generate V_n as a $\mathbb{C}\langle x_1, x_2, \dots, x_d\rangle$-module. In other words, every element $w \in V_n$ belongs to

$$\mathsf{Span}_{\mathbb{C}}\left\{ A_1^{m_1} A_2^{m_2} \cdots A_d^{m_d} \cdot \phi(\boldsymbol{e}_i) \,\middle|\, m_i \geqslant 0,\ 1 \leqslant i \leqslant r \right\}.$$

This tells us exactly what we should add to the picture to obtain Condition (3): for a fixed $\mathbb{C}\langle x_1, x_2, \dots, x_d\rangle$-module structure, i.e. d-tuple of matrices $(A_1, A_2, \dots, A_d)$, we need to specify r vectors $v_1, \dots, v_r \in V_n$ with the property that

$$\mathsf{Span}_{\mathbb{C}}\left\{ A_1^{m_1} A_2^{m_2} \cdots A_d^{m_d} \cdot v_i \,\middle|\, m_i \in \mathbb{Z}_{\geqslant 0},\ 1 \leqslant i \leqslant r \right\} = V_n. \tag{1.3.2}$$

Let us consider the $(dn^2 + rn)$-dimensional affine space

$$W_{r,d}^n = \mathrm{End}_{\mathbb{C}}(V_n)^{\oplus d} \oplus V_n^{\oplus r}. \tag{1.3.3}$$

Exercise 1.3.4. Show that the locus

$$U_{r,d}^n = \{ (A_1, A_2, \dots, A_d, v_1, \dots, v_r) \mid (1.3.2) \text{ holds} \} \subset W_{r,d}^n$$

is a Zariski open subset.

Consider the GL_n-action on $W_{r,d}^n$ given by

$$g \cdot (A_1, A_2, \dots, A_d, v_1, \dots, v_r) = (A_1^g, A_2^g, \dots, A_d^g, g v_1, \dots, g v_r) \tag{1.3.4}$$

where $M^g = g M g^{-1}$ is the standard conjugation action.

LEMMA 1.3.5. *The* GL_n*-action* (1.3.4) *is free on* $U_{r,d}^n$.

Proof. If $g \in \mathrm{GL}_n$ fixes a point $(A_1, A_2, \dots, A_d, v_1, \dots, v_r) \in U_{r,d}^n$, then $v_i = g v_i$ lies in the invariant subspace $\ker(g - \mathrm{id}) \subset V_n$ for $i = 1, \dots, r$. But by definition of $U_{r,d}^n$, the smallest invariant subspace containing $v_1, \dots, v_r$ is V_n itself, thus $g = \mathrm{id}$. □

Definition 1.3.6 (Noncommutative Quot scheme)**.** The (geometric) GIT quotient

$$\mathsf{ncQuot}_{r,d}^n = U_{r,d}^n / \mathrm{GL}_n$$

is called the *noncommutative Quot scheme*.

Note that $\mathsf{ncQuot}_{r,d}^n$ is smooth by [85, Tag 02K5] and has dimension

$$\dim U_{r,d}^n - \dim \mathrm{GL}_n = (d-1)n^2 + rn.$$

The following result is of crucial importance for us (and can be taken as the *definition* of the local Quot scheme).

THEOREM 1.3.7 ([88, 7]). *There is a closed immersion*

$$\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r}, n) \hookrightarrow \mathsf{ncQuot}^n_{r,d}$$

cut out by the relations $[A_i, A_j] = 0$ *for* $1 \leqslant i < j \leqslant d$.

Furthermore, if $d = 3$*, such relations cut out the zero scheme of the exact 1-form*

$$\mathrm{d}f_{r,n} \in \mathrm{H}^0(\mathsf{ncQuot}^n_{r,3}, \Omega^1)$$

defined by the regular function $f_{r,n} \colon \mathsf{ncQuot}^n_{r,3} \to \mathbb{A}^1$ *sending*

$$[A_1, A_2, A_3, v_1, \dots, v_r] \longmapsto \operatorname{Tr} A_1[A_2, A_3].$$

In particular, $\mathsf{Quot}_{\mathbb{A}^3}(\mathscr{O}^{\oplus r}, n)$ *is a critical locus.*

Exercise 1.3.8. Let $d = r = 1$. Show that $\mathsf{Hilb}^n(\mathbb{A}^1) = \mathsf{ncQuot}^n_{1,1} = \mathbb{A}^n$.

Remark 1.3.9. If $d = 2$ the description of $\mathsf{Hilb}^n(\mathbb{A}^2)$ is equivalent to Nakajima's description [77, Theorem 1.14]. See also [53] for an equivalent description of $\mathsf{Hilb}^n(\mathbb{A}^d)$, in terms of *perfect extended monads*. See [22] for a description of $\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r}, n)$ in terms of framed sheaves on $\mathbb{P}^d$, available when $d \geqslant 3$.

1.3.4 Quot-to-Chow in the local case

Fix $n \geqslant 0$ and $d, r \geqslant 1$. The Quot-to-Chow morphism

$$\sigma_{\mathscr{O}^{\oplus r}, n} \colon \mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r}, n) \longrightarrow \mathsf{Sym}^n \mathbb{A}^d$$

introduced in (1.3.1) can be reinterpreted as follows. Pick a point

$$[A_1, \dots, A_d, v_1, \dots, v_r] \in \mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r}, n)$$

and notice that since the matrices pairwise commute, they can be simultaneously made upper triangular. So, since the tuple is defined up to GL_n, we may assume they are in the form

$$A_\ell = \begin{pmatrix} a_{11}^{(\ell)} & * & * & \cdots & * \\ 0 & a_{22}^{(\ell)} & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn}^{(\ell)} \end{pmatrix}, \quad 1 \leqslant \ell \leqslant d.$$

Then $\sigma_{\mathscr{O}^{\oplus r}, n}$ is given by

$$[A_1, \dots, A_d, v_1, \dots, v_r] \longmapsto \sum_{1 \leqslant i \leqslant n} \left(a_{ii}^{(1)}, \dots, a_{ii}^{(d)}\right).$$

When all the matrices are *nilpotent*, and only in this case, the corresponding quotient $\mathscr{O}^{\oplus r} \twoheadrightarrow T$ is entirely supported at the origin. In other words,

$$\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r}, n)_0 = \{[A_1, \dots, A_d, v_1, \dots, v_r] \mid A_1, \dots, A_d \text{ are nilpotent}\}$$

is a way to describe the punctual Quot scheme.

1.3.5 Smoothness

We shall make use of the following classical result from deformation theory.

PROPOSITION 1.3.10 ([83, 36]). *Let $p \in \mathsf{Quot}_X(F,n)$ be a closed point corresponding to a short exact sequence*

$$0 \longrightarrow K \longrightarrow F \longrightarrow T \longrightarrow 0.$$

Then

(i) *The tangent space to $\mathsf{Quot}_X(F,n)$ at p is the vector space*

$$T_p \mathsf{Quot}_X(F,n) = \mathrm{Hom}_X(K,T).$$

(ii) *If $\mathrm{Ext}^1_X(K,T)=0$, then $\mathsf{Quot}_X(F,n)$ is smooth at p.*

PROPOSITION 1.3.11. *If X is a smooth quasiprojective surface, then $\mathsf{Hilb}^n(X)$ is smooth and irreducible of dimension $2n$. In particular, it is equal to the smoothable component.*

Proof. We may assume X to be projective, since the general result will follow from this one via a smooth compactification $X \hookrightarrow \overline{X}$, inducing an open immersion $\mathsf{Hilb}^n(X) \hookrightarrow \mathsf{Hilb}^n(\overline{X})$.

Suppose $\dim T_p \mathsf{Hilb}^n(X) = 2n$ for any $p \in \mathsf{Hilb}^n(X)$, then $\mathsf{Hilb}^n(X)$ is smooth along the smoothable component, because its interior is a smooth open subset of dimension $2n$. Since $\mathsf{Hilb}^n(X)$ is connected by [49], the presence of any other irreducible component W would yield singularities along $W \cap \Gamma_{\mathsf{sm}}$. Thus we are reduced to showing that $\dim T_p \mathsf{Hilb}^n(X) = 2n$ for any $p \in \mathsf{Hilb}^n(X)$.

Let $Z \subset X$ be the closed subscheme corresponding to p. By Proposition 1.3.10, we have $T_p \mathsf{Hilb}^n(X) = \mathrm{Hom}_X(\mathscr{I}_Z, \mathscr{O}_Z)$, so we get an exact sequence

$$0 \longrightarrow \mathrm{Hom}_X(\mathscr{O}_Z,\mathscr{O}_Z) \xrightarrow{e} \mathrm{Hom}_X(\mathscr{O}_X,\mathscr{O}_Z) \xrightarrow{u} T_p \mathsf{Hilb}^n(X) \xrightarrow{v} \mathrm{Ext}^1_X(\mathscr{O}_Z,\mathscr{O}_Z)$$

where e is an isomorphism between n-dimensional vector spaces, so that $u=0$ and v is injective. We next show that $\mathrm{ext}^1_X(\mathscr{O}_Z,\mathscr{O}_Z) \leqslant 2n$. Note that $\chi(\mathscr{O}_Z,\mathscr{O}_Z) = \mathrm{ext}^0(\mathscr{O}_Z,\mathscr{O}_Z) - \mathrm{ext}^1(\mathscr{O}_Z,\mathscr{O}_Z) + \mathrm{ext}^2(\mathscr{O}_Z,\mathscr{O}_Z) = 2n - \mathrm{ext}^1(\mathscr{O}_Z,\mathscr{O}_Z)$, where we used Serre duality to compute ext^2. Note that, for any locally free sheaf E on X, one has $\chi(E,\mathscr{O}_Z) = \dim \mathrm{H}^0(Z,\mathscr{O}_Z)^{\mathrm{rk}\,E} = n \cdot \mathrm{rk}\,E$. So, let $E^\bullet \to \mathscr{O}_Z$ be a resolution consisting of finitely many locally free sheaves E^i on X, so that $0 = \mathrm{rk}\,\mathscr{O}_Z = \mathrm{rk}\,E^\bullet = \sum_\ell (-1)^\ell \,\mathrm{rk}\,E^\ell$. This implies

$$\chi(\mathscr{O}_Z,\mathscr{O}_Z) = \sum_\ell (-1)^\ell \chi(E^\ell,\mathscr{O}_Z) = \sum_\ell (-1)^\ell n \cdot \mathrm{rk}\,E^\ell = 0,$$

proving $\mathrm{ext}^1(\mathscr{O}_Z,\mathscr{O}_Z) = 2n$. □

COROLLARY 1.3.12. *Let X be a smooth surface. Then, the Hilbert–Chow morphism*

$$\sigma_{\mathscr{O}_X,n} \colon \mathsf{Hilb}^n(X) \to \mathsf{Sym}^n(X)$$

is a resolution of singularities.

1.3.6 Torus action on the Hilbert scheme of points

In this chapter we work with the Hilbert scheme

$$\mathsf{Hilb}^n(\mathbb{A}^d).$$

We view it as a fine moduli space of ideals $I \subset \mathbb{C}[x_1,\dots,x_d]$ of colength n, i.e. such that the quotient algebra $\dim_{\mathbb{C}} \mathbb{C}[x_1,\dots,x_d]/I$ has dimension n as a complex vector space.

Consider the d-dimensional torus

$$\mathbb{T} = \mathbb{G}_m^d$$

acting on $\mathbb{A}^d$ by

$$t \cdot (a_1,\dots,a_d) = (t_1 a_1,\dots,t_d a_d). \tag{1.3.5}$$

Exercise 1.3.13. Show that the action (1.3.5) lifts to a $\mathbb{T}$-action $\mathbb{T} \times \mathsf{Hilb}^n(\mathbb{A}^d) \to \mathsf{Hilb}^n(\mathbb{A}^d)$. Upgrade this replacing $\mathbb{A}^d$ with any smooth toric d-fold (If in need of a hint, open [81, Section 4.2] or [39, Section 9.1]).

Exercise 1.3.14. Show that a $\mathbb{T}$-fixed subscheme $Z \subset \mathbb{A}^d$ is entirely supported at the origin $0 \in \mathbb{A}^d$. (**Hint**: Show that $\mathrm{Supp}(t \cdot [Z]) = t \cdot \mathrm{Supp}(Z)$).

PROPOSITION 1.3.15. *An ideal $I \in \mathsf{Hilb}^n(\mathbb{A}^d)$ is $\mathbb{T}$-fixed if and only if it is a monomial ideal. In particular, the fixed locus $\mathsf{Hilb}^n(\mathbb{A}^d)^{\mathbb{T}}$ is finite.*

Proof. Recall that the character lattice of the torus $\mathbb{T}^* = \mathrm{Hom}(\mathbb{T},\mathbb{G}_m)$ is isomorphic to $\mathbb{Z}^d$, since each character $\mathbb{T} \to \mathbb{G}_m$ is necessarily of the form

$$\chi_{\boldsymbol{m}} : (t_1,\dots,t_d) \longmapsto t_1^{m_1} \cdots t_d^{m_d}$$

for some $\boldsymbol{m} = (m_1,\dots,m_d) \in \mathbb{Z}^d$. As an initial step, we notice that the geometric action (1.3.5) dualises to a $\mathbb{T}$- action on $\mathbb{C}[x_1,\dots,x_d]$ via the rule

$$t \cdot f(x_1,\dots,x_d) = f(t_1^{-1}x_1,\dots,t_d^{-1}x_d). \tag{1.3.6}$$

This already shows that a monomial ideal is necessarily $\mathbb{T}$-fixed, so it remains to prove the converse.

We next show that the monomials

$$\boldsymbol{x}^{\boldsymbol{m}} = x_1^{m_1} \cdots x_d^{m_d}$$

form an eigenbasis of $\mathbb{C}[x_1,\dots,x_d]$ as a $\mathbb{T}$-representation. An *eigenvector* of a $\mathbb{T}$-representation V, in this context, is an element $v \in V$ for which there exists a character $\chi \in \mathbb{T}^*$ such that $t \cdot v = \chi(t)v$ for all $t \in \mathbb{T}$. (This χ plays the role of "classical" eigenvalues in linear algebra.) For us, $V = \mathbb{C}[x_1,\dots,x_d]$. Pick $v = \boldsymbol{x}^{\boldsymbol{m}}$. Then according to (1.3.6) one has

$$t \cdot \boldsymbol{x}^{\boldsymbol{m}} = (t_1^{-1}x_1)^{m_1} \cdots (t_d^{-1}x_d)^{m_d} = t_1^{-m_1} \cdots t_d^{-m_d}(x_1^{m_1} \cdots x_d^{m_d}) = \chi_{-\boldsymbol{m}}(t)\boldsymbol{x}^{\boldsymbol{m}}.$$

So each monomial $\boldsymbol{x}^{\boldsymbol{m}}$ is an eigenvector with respect to the weight $\chi_{-\boldsymbol{m}}$. In particular, each corresponds to a different weight, therefore all weight spaces

$$V_{\boldsymbol{m}} = \left\{ f \in \mathbb{C}[x_1,\dots,x_d] \,\middle|\, t \cdot f = \chi_{\boldsymbol{m}}(t) f \text{ for all } t \in \mathbb{T} \right\}$$

are 1-dimensional $\mathbb{T}$-subrepresentations (each spanned by $\boldsymbol{x}^{\boldsymbol{m}}$) and the action (1.3.6) is diagonalisable by monomials.

Now pick a $\mathbb{T}$-fixed ideal $I \subset V$. In particular, I is a $\mathbb{T}$-subrepresentation. But a $\mathbb{T}$-subrepresentation of a diagonalisable $\mathbb{T}$-representation is again diagonalisable (prove this!), so I has a basis of eigenvectors. But each eigenvector is a multiple of a monomial, thus I is a monomial ideal. □

For $d \in \mathbb{Z}_{\geqslant 1}$, the lattice $\mathbb{N}^d$ is endowed with its standard component-wise poset structure throughout.

Definition 1.3.16. Let $d \geqslant 1$ and $n \geqslant 0$ be integers. A $(d-1)$-dimensional partition of size n is a collection of n points $\lambda = \{\boldsymbol{a}_1, \dots, \boldsymbol{a}_n\} \subset \mathbb{N}^d$ with the following property: whenever a point $\boldsymbol{y} \in \mathbb{N}^d$ satisfies $\boldsymbol{y} \leqslant \boldsymbol{a}_i$ for all $i = 1, \dots, n$, one has that $\boldsymbol{y} \in \lambda$. We set

$$\mathsf{p}_k(n) = \left| \{ k\text{-dimensional partitions of } n \} \right|.$$

A 1-dimensional partition ($d = 2$) is simply called a *partition*. A 2-dimensional partition ($d = 3$) is called a *plane partition*.

Example 1.3.17. One has that $\mathsf{p}_0(n) = 1$ for all $n \geqslant 0$.

Remark 1.3.18. Partitions correspond bijectively to Young diagrams (also known as Ferrers diagrams), so we will use both terminologies and notations interchangeably. We adopt the English convention for Young diagrams, with the x-axis growing downwards, the y-axis growing rightwards and the box in the corner corresponding to the origin $(0,0) \in \mathbb{A}^2$. See the figures below.

Definition 1.3.19. Let $\lambda \subset \mathbb{N}^2$ be a partition. The *arm-length*, resp. *leg-length* of a box $\square \in \lambda$ is the number of boxes sitting on the right, resp. below $\square$.

Notation 1.3.20. Set $d = 2$. There are two different (equivalent) ways to represent a partition $\lambda \subset \mathbb{N}^2$, namely

(1) $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_p)$ where $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_p > 0$ with $n = \sum_j \lambda_j$, and

(2) $\lambda = (1^{\alpha_1} 2^{\alpha_2} \cdots n^{\alpha_n})$ with $n = \sum_i i\alpha_i$ (and some α_i might be 0, in which case the term i^{α_i} is omitted).

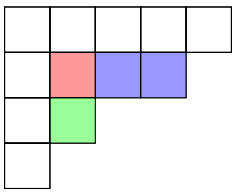
In both cases, we write '$\lambda \vdash n$' to say that λ is a (1-dimensional) partition of n. The notation in (2) means that λ consists of α_i parts of size i for all $i = 1, \dots, n$. Furthermore:

(i) The quantity
$$||\lambda|| = \sum_i \alpha_i$$
counts the number of rows in the Young diagram attached to λ. It agrees with the number of boxes $\square \in \lambda$ such that $a(\square) = 0$.

(ii) The quantity λ_1 counts the number of columns in the Young diagram attached to λ. It agrees with the number of boxes $\square \in \lambda$ such that $l(\square) = 0$.

(iii) The *automorphism group* of $\lambda \subset \mathbb{N}^2$ is the group $\mathrm{Aut}(\lambda) = \prod_i \mathfrak{S}_{\alpha_i}$, where $\mathfrak{S}_a$ is the symmetric group on a letters.

Example 1.3.21. In the partition $\lambda = (5,4,2,1) \vdash n = 12$, written also $\lambda = (1^1 2^1 4^1 5^1)$, the red box $\square = (1,1)$ has arm-length 2 and leg-length 1.



The partition has $\mathrm{Aut}(\lambda) = \mathrm{id}$, $||\lambda|| = 4$ and $\lambda_1 = 5$.

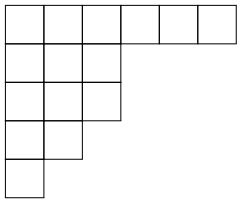


Figure 1.3: the partition $\lambda = (6,3,3,2,1) = (1^1 2^1 3^2 6^1)$ has $\|\lambda\| = 1+1+2+1 = 5$, $\lambda_1 = 6$, $\mathrm{Aut}(\lambda) = \mathfrak{S}_2$.

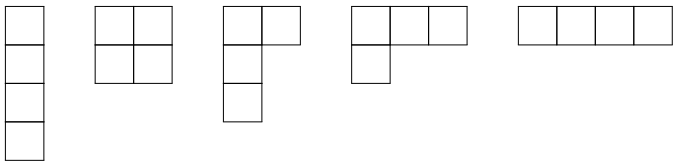


Figure 1.4: The five partitions $\lambda_1, \dots, \lambda_5$ of 4. According to Notation 1.3.20, $\lambda_1 = (1,1,1,1) = (1^4)$, $\lambda_2 = (2,2) = (2^2)$, $\lambda_3 = (2,1,1) = (1^2 2^1)$, $\lambda_4 = (3,1) = (1^1 3^1)$, $\lambda_5 = (4) = (4^1)$.

Example 1.3.22. If $d = 3$, a *plane partition* of n is the same thing as a way of stacking n boxes in the corner of a room (assuming gravity points in the $(-1,-1,-1)$ direction!).

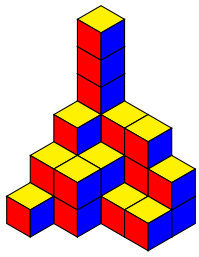


Figure 1.5: A plane partition.

The following result is a direct consequence of Proposition 1.3.15. The 'proof' is given informally via Figure 1.6.

COROLLARY 1.3.23. *There is a bijective correspondence between $\mathbb{T}$-fixed subschemes $Z \subset \mathbb{A}^d$ of length n and $(d-1)$-dimensional partitions of n.*

The generating function of the numbers of partitions (for $d = 2$) is given by the following formula.

THEOREM 1.3.24 (Euler [35, Chapter 16]). *There is an identity*

$$\sum_{n \geqslant 0} \mathsf{p}_1(n) q^n = \prod_{m \geqslant 1} (1 - q^m)^{-1}.$$

Proof. By contemplation. For instance, say we want to compute $\mathsf{p}_1(4)$, that we know equals 5 (cf. Figure 1.4). Then expanding

$$\prod_{m \geqslant 1} (1-q^m)^{-1} = (1 + q^1 + q^{1+1} + q^{1+1+1} + q^{1+1+1+1} + \cdots)$$
$$\cdot (1 + q^2 + q^{2+2} + q^{2+2+2} + \cdots) \cdot (1 + q^3 + q^{3+3} + \cdots) \cdot (1 + q^4 + q^{4+4} + \cdots) \cdots$$

1	x_2	x_2^2	x_2^3	x_2^4	x_2^5	x_2^6	x_2^7
x_1	x_1x_2	$x_1x_2^2$	$x_1x_2^3$				
x_1^2	$x_1^2x_2$	$x_1^2x_2^2$					
x_1^3	$x_1^3x_2$	$x_1^3x_2^2$					
x_1^4	$x_1^4x_2$						
x_1^5							
x_1^6							

Figure 1.6: A 1-dimensional partition (Young diagram) draws a staircase and determines (and is determined by) a monomial ideal, in this case $I_\lambda = (x_1^6, x_1^4x_2, x_1^3x_2^2, x_1x_2^3, x_2^7)$. Note that the colength of I_λ is $|\lambda| = 17$.

we see that to compute the coefficient of q^4 we have to sum the coefficients of

$$\begin{aligned} &q^4 \\ &q^1 \cdot q^3 \\ &q^{1+1} \cdot q^2 \\ &q^{1+1+1+1} \\ &q^{2+2}. \end{aligned}$$

These clearly correspond to partitions of 4. □

Definition 1.3.25 (MacMahon function)**.** The *MacMahon function* is the infinite product

$$\mathsf{M}(q) = \prod_{m \geqslant 1} (1 - q^m)^{-m}.$$

The generating function of the numbers of plane partitions (for $d = 3$) is given by the following formula.

THEOREM 1.3.26 (MacMahon [65]). *There is an identity*

$$\sum_{n \geqslant 0} \mathsf{p}_2(n) q^n = \mathsf{M}(q).$$

1.3.7 The character of the tangent space

The goal of this section is to compute the character of the tangent representation at a torus fixed point (monomial ideal) of $\mathsf{Hilb}^n(\mathbb{A}^d)$.

Consider the torus $\mathbb{T} = \mathbb{G}_m^d$. Let $K_0^{\mathbb{T}}(\mathsf{pt})$ be the K-group of the abelian category of $\mathbb{T}$-representations. Taking a representation to its weight decomposition is a ring isomorphism

$$\chi : K_0^{\mathbb{T}}(\mathsf{pt}) \xrightarrow{\sim} \mathbb{Z}[t_1^{\pm}, \ldots, t_d^{\pm}],$$

so we will identify the two rings without further mention. In order to allow infinite-dimensional representations, and also (just to be safe) rational coefficients, we define

$$\widehat{K}_0^{\mathbb{T}}(\mathsf{pt}) = \mathbb{Q}(\!(t_1, \dots, t_d)\!).$$

Define, for F, G two equivariant coherent sheaves on $\mathbb{A}^d$, the representations

$$\chi(F, G) = \sum_{i \geqslant 0} (-1)^i \operatorname{Ext}^i(F, G) \in \widehat{K}_0^{\mathbb{T}}(\mathsf{pt}),$$

and set $\chi(G) = \chi(\mathcal{O}_{\mathbb{A}^d}, G)$. These representations satisfy the usual relation

$$\chi(F, G) = \chi(F^* \otimes G). \tag{1.3.7}$$

Example 1.3.27. The canonical line bundle $K_{\mathbb{A}^d}$ is equivariantly nontrivial, as

$$K_{\mathbb{A}^d} = \mathcal{O}_{\mathbb{A}^d} \otimes (t_1 \cdots t_d)^{-1}.$$

Indeed, the sheaf of Kähler differentials splits as $\Omega_{\mathbb{A}^d} = \mathcal{O}_{\mathbb{A}^d} \mathrm{d}x_1 \oplus \cdots \oplus \mathcal{O}_{\mathbb{A}^d} \mathrm{d}x_d$ and x_i carries a -1 weight (recall that the geometric action on $\mathbb{A}^d$ has weights 1, thus the algebraic action on $x_i \in \mathbb{C}[x_1, \dots, x_d]$ has weights -1). This also gives the relation

$$\chi(T_0 \mathbb{A}^d) = t_1 + \cdots + t_d. \tag{1.3.8}$$

Example 1.3.28. Since the action on coordinates is given by $t_i \cdot x_i = t_i^{-1} x_i$, and monomials span $\mathbb{C}[x_1, \dots, x_d]$, we have

$$\chi(\mathcal{O}_{\mathbb{A}^d}) = \sum_{m_1, \dots, m_d \geqslant 0} t_1^{-m_1} \cdots t_d^{-m_d} = \prod_{1 \leqslant i \leqslant d} \frac{1}{(1 - t_i^{-1})}.$$

Denote by $\overline{(\cdot)}$ the involution on $K_0^{\mathbb{T}}(\mathsf{pt})$ sending t_i to t_i^{-1}. Then one has

$$\frac{1}{\overline{\chi(\mathcal{O}_{\mathbb{A}^d})}} = (1 - t_1) \cdots (1 - t_d) = (-1)^d (t_1 \cdots t_d)(1 - t_1^{-1}) \cdots (1 - t_d^{-1}) = \frac{(-1)^d (t_1 \cdots t_d)}{\chi(\mathcal{O}_{\mathbb{A}^d})},$$

which is equivalent to the relation

$$\chi(\mathcal{O}_{\mathbb{A}^d}) = (-1)^d (t_1 \cdots t_d) \overline{\chi(\mathcal{O}_{\mathbb{A}^d})}. \tag{1.3.9}$$

Definition 1.3.29. The tautological (rank n) bundle over $\mathsf{Hilb}^n(\mathbb{A}^d)$ is the pushdown

$$\mathcal{V} = q_* \mathcal{O}_{\mathcal{Z}}$$

of the structure sheaf of the universal subscheme $\mathcal{Z} \subset \mathbb{A}^d \times \mathsf{Hilb}^n(\mathbb{A}^d)$ along the projection map $q \colon \mathbb{A}^d \times \mathsf{Hilb}^n(\mathbb{A}^d) \to \mathsf{Hilb}^n(\mathbb{A}^d)$. The restriction $\mathcal{V}_\lambda = \mathcal{V}|_{\mathscr{I}_\lambda}$ to a point $[\mathscr{I}_\lambda] \in \mathsf{Hilb}^n(\mathbb{A}^d)$ corresponding to a partition $\lambda \subset \mathbb{N}^d$ is naturally a finite dimensional $\mathbb{T}$-representation (of rank n), i.e. an element of $K_{\mathbb{T}}^0(\mathsf{pt})$.

LEMMA 1.3.30. *Let S be a nonsingular quasiprojective toric surface. Lift the natural action of $\mathbb{T} = \mathbb{G}_m^2$ on S to $\mathsf{Hilb}^n(S)$. Then, at a $\mathbb{T}$-fixed point $\mathscr{I}_Z \in \mathsf{Hilb}^n(S)$, one can write the tangent space as*

$$T_{\mathscr{I}_Z} \mathsf{Hilb}^n(S) = \chi(\mathcal{O}_S) - \chi(\mathscr{I}_Z, \mathscr{I}_Z) \in K_0^{\mathbb{T}}(\mathsf{pt}).$$

Proof. Let us compute

$$\begin{aligned}\chi(\mathscr{O}_S)-\chi(\mathscr{I}_Z,\mathscr{I}_Z)&=\chi(\mathscr{O}_S)-\chi(\mathscr{O}_S-\mathscr{O}_Z,\mathscr{O}_S-\mathscr{O}_Z)\\&=\chi(\mathscr{O}_S,\mathscr{O}_Z)+\chi(\mathscr{O}_Z,\mathscr{O}_S)-\chi(\mathscr{O}_Z,\mathscr{O}_Z)\\&=\chi(\mathscr{I}_Z,\mathscr{O}_Z)+\chi(\mathscr{O}_Z,\mathscr{O}_Z)+\chi(\mathscr{O}_Z,\mathscr{O}_S)-\chi(\mathscr{O}_Z,\mathscr{O}_Z)\\&=\chi(\mathscr{I}_Z,\mathscr{O}_Z)+\chi(\mathscr{O}_Z,\mathscr{O}_S).\end{aligned}$$

Applying $\mathrm{Hom}(-,\mathscr{O}_Z)$ to the ideal sheaf exact sequence defining $Z\subset S$, we obtain

$$\begin{tikzcd}0 \arrow[r] & \mathrm{Hom}(\mathscr{O}_Z,\mathscr{O}_Z) \arrow[r, "\sim"] & \mathrm{Hom}(\mathscr{O}_S,\mathscr{O}_Z) \arrow[r, "0"] & \mathrm{Hom}(\mathscr{I}_Z,\mathscr{O}_Z) \arrow[dll, rounded corners] \\ & \mathrm{Ext}^1(\mathscr{O}_Z,\mathscr{O}_Z) \arrow[r] & 0 \arrow[r] & \mathrm{Ext}^1(\mathscr{I}_Z,\mathscr{O}_Z) \arrow[dll, rounded corners] \\ & \mathrm{Ext}^2(\mathscr{O}_Z,\mathscr{O}_Z) \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0.\end{tikzcd}$$

where we have used the vanishings $\mathrm{Ext}^{>0}(\mathscr{O},\mathscr{O}_Z)=0$ and $\mathrm{Ext}^2(\mathscr{I}_Z,\mathscr{O}_Z)=\mathrm{Hom}(\mathscr{O}_Z,\mathscr{I}_Z\otimes\omega_S)^\vee=0$. We also have

$$\mathrm{Ext}^1(\mathscr{I}_Z,\mathscr{O}_Z)\xrightarrow{\sim}\mathrm{Ext}^2(\mathscr{O}_Z,\mathscr{O}_Z)\cong\mathrm{Hom}(\mathscr{O}_Z,\mathscr{O}_Z)^\vee\cong\mathrm{Hom}(\mathscr{O}_S,\mathscr{O}_Z)^\vee$$

to be read off from the long exact sequence. Therefore

$$\begin{aligned}\chi(\mathscr{O}_S)-\chi(\mathscr{I}_Z,\mathscr{I}_Z)&=\chi(\mathscr{I}_Z,\mathscr{O}_Z)+\chi(\mathscr{O}_Z,\mathscr{O}_S)\\&=T_{\mathscr{I}_Z}\mathsf{Hilb}^n(S)-\mathrm{Ext}^1(\mathscr{I}_Z,\mathscr{O}_Z)+\mathrm{Ext}^2(\mathscr{O}_Z,\mathscr{O}_S)\\&=T_{\mathscr{I}_Z}\mathsf{Hilb}^n(S)-\mathrm{Hom}(\mathscr{O}_S,\mathscr{O}_Z)^\vee+\mathrm{Hom}(\mathscr{O}_S,\mathscr{O}_Z)^\vee\\&=T_{\mathscr{I}_Z}\mathsf{Hilb}^n(S).\end{aligned}$$

□

LEMMA 1.3.31. *Let F be an equivariant coherent sheaf on $\mathbb{A}^d$. Then one has the relation*

$$\chi(F)=\chi(F\otimes\mathscr{O}_0)\chi(\mathscr{O}_{\mathbb{A}^d}). \tag{1.3.10}$$

Proof. Set $R=\mathbb{C}[x_1,\dots,x_d]$ and $M=T_0^*\mathbb{A}^d$, which as a representation is just $t_1^{-1}+\cdots+t_d^{-1}$ by (1.3.8). Note that $0\in\mathbb{A}^d$ is the unique torus fixed point. The structure sheaf $\mathscr{O}_0=R/\mathfrak{m}$ has a torus equivariant Koszul resolution

$$0\longrightarrow R\otimes\bigwedge^d M\longrightarrow R\otimes\bigwedge^{d-1}M\longrightarrow\cdots\longrightarrow R\otimes\bigwedge^1 M\longrightarrow R\longrightarrow\mathscr{O}_0\longrightarrow 0.$$

Out of this, we get

$$\chi(\mathscr{O}_0)=\chi(\wedge^\bullet M)=\sum_{1\leqslant i\leqslant d}(-1)^i\chi(\wedge^i M)=\prod_{1\leqslant i\leqslant d}(1-t_i^{-1})=\frac{1}{\chi(\mathscr{O}_{\mathbb{A}^d})}.$$

Now, $\chi(F\otimes\mathscr{O}_0)=\chi(F)\chi(\mathscr{O}_0)$ gives the result. □

The character of the tangent space at a fixed point of the Hilbert scheme (i.e. its decomposition into weight spaces) can be computed combining Lemma 1.3.30 with the following formula.

LEMMA 1.3.32. *Let F,G two $\mathbb{T}$-equivariant coherent sheaves on $\mathbb{A}^d$. Denote by $\overline{(\cdot)}$ the involution on $K_0^{\mathbb{T}}(\mathsf{pt})$ sending t_i to t_i^{-1}. Then one has the relation*

$$\chi(F,G)=\frac{\overline{\chi(F)}\chi(G)}{\overline{\chi(\mathscr{O}_{\mathbb{A}^d})}}. \tag{1.3.11}$$

Proof. Since stalks and tensor products commute, we have

$$\begin{aligned}\chi((F\otimes F')\otimes \mathscr{O}_0) &= \chi((F\otimes \mathscr{O}_0)\otimes (F'\otimes \mathscr{O}_0))\\ &= \chi(F\otimes \mathscr{O}_0)\chi(F'\otimes \mathscr{O}_0).\end{aligned} \tag{1.3.12}$$

Also note that Serre duality $(-1)^d\overline{\chi(F',F)} = \chi(F, F'\otimes K_{\mathbb{A}^d})$ specialises (when F' is trivial) to

$$(-1)^d\overline{\chi(F)} = \chi(F, K_{\mathbb{A}^g}) = \chi(F^*\otimes K_{\mathbb{A}^g}) = \chi(F^*)(t_1\cdots t_d)^{-1}. \tag{1.3.13}$$

So we can compute

$$\begin{aligned}\chi(F,G) &= \chi(F^*\otimes G) && \text{by (1.3.7)}\\ &= \chi((F^*\otimes G)\otimes \mathscr{O}_0)\chi(\mathscr{O}_{\mathbb{A}^g}) && \text{by (1.3.10)}\\ &= \chi(F^*\otimes \mathscr{O}_0)\chi(G\otimes \mathscr{O}_0)\chi(\mathscr{O}_{\mathbb{A}^g}) && \text{by (1.3.12)}\\ &= \chi(F^*)\chi(G\otimes \mathscr{O}_0) && \text{by (1.3.10)}\\ &= (-1)^d(t_1\cdots t_d)\overline{\chi(F)}\chi(G\otimes \mathscr{O}_0) && \text{by (1.3.13)}\\ &= \frac{\chi(\mathscr{O}_{\mathbb{A}^d})}{\overline{\chi(\mathscr{O}_{\mathbb{A}^d})}}\overline{\chi(F)}\chi(G\otimes \mathscr{O}_0) && \text{by (1.3.9)}\\ &= \frac{\overline{\chi(F)}\chi(G)}{\overline{\chi(\mathscr{O}_{\mathbb{A}^d})}} && \text{by (1.3.10)}\end{aligned}$$

as required. □

Let us go back to the case $d=2$. The torus action is conceived so that one has

$$\begin{aligned}\chi(\mathscr{O}_{\mathbb{A}^2}) &= \sum_{\square\in\mathbb{Z}^2_{\geqslant 0}} t^{-\square} = \frac{1}{(1-t_1^{-1})(1-t_2^{-1})}\\ \overline{\chi(\mathscr{O}_{\mathbb{A}^2})} &= \frac{1}{(1-t_1)(1-t_2)}\\ \overline{\chi(\mathscr{O}_{\mathbb{A}^2})}^{-1} &= (1-t_1)(1-t_2) = (1-t_1^{-1})(1-t_2^{-1})t_1t_2.\end{aligned}$$

If $Z\subset\mathbb{A}^2$ defines a torus fixed point, it is determined by a monomial ideal $\mathscr{I}_\lambda\subset\mathbb{C}[x_1,x_2]$, corresponding to a partition (Young diagram) $\lambda\vdash n$. The character associated to the $\mathbb{T}$-module $\mathscr{O}_Z$ (the restriction of the tautological bundle to the point determined by Z) is

$$\mathcal{V}_\lambda = \chi(\mathscr{O}_Z) = \sum_{\square\in\lambda} t^{-\square} = \sum_{(i,j)\in\lambda} t_1^{-i}t_2^{-j}.$$

It follows from Formula (1.3.11) that one can compute

$$\begin{aligned}T_{\mathscr{I}_\lambda}\operatorname{Hilb}^n(\mathbb{A}^2) &= \chi(\mathscr{O}_{\mathbb{A}^2}) - \chi(\mathscr{I}_\lambda,\mathscr{I}_\lambda)\\ &= \chi(\mathscr{O}_{\mathbb{A}^2}) - \chi(\mathscr{O}_{\mathbb{A}^2}-\mathscr{O}_Z, \mathscr{O}_{\mathbb{A}^2}-\mathscr{O}_Z)\\ &= \chi(\mathscr{O}_{\mathbb{A}^2}) - \overline{\chi(\mathscr{O}_{\mathbb{A}^2}-\mathscr{O}_Z)}\chi(\mathscr{O}_{\mathbb{A}^2}-\mathscr{O}_Z)\overline{\chi(\mathscr{O}_{\mathbb{A}^2})}^{-1}\\ &= \chi(\mathscr{O}_{\mathbb{A}^2}) - \left[\overline{\chi(\mathscr{O}_{\mathbb{A}^2})}\chi(\mathscr{O}_{\mathbb{A}^2}) - \overline{\chi(\mathscr{O}_{\mathbb{A}^2})}\mathcal{V}_\lambda - \overline{\mathcal{V}}_\lambda\chi(\mathscr{O}_{\mathbb{A}^2}) + \overline{\mathcal{V}}_\lambda\mathcal{V}_\lambda\right]\overline{\chi(\mathscr{O}_{\mathbb{A}^2})}^{-1}\\ &= \chi(\mathscr{O}_{\mathbb{A}^2}) - \left[\chi(\mathscr{O}_{\mathbb{A}^2}) - \mathcal{V}_\lambda - \overline{\mathcal{V}}_\lambda t_1t_2 + \overline{\mathcal{V}}_\lambda\mathcal{V}_\lambda(1-t_1)(1-t_2)\right]\\ &= \mathcal{V}_\lambda + \overline{\mathcal{V}}_\lambda t_1t_2 - \overline{\mathcal{V}}_\lambda\mathcal{V}_\lambda(1-t_1)(1-t_2).\end{aligned} \tag{1.3.14}$$

Exercise 1.3.33. Let $\lambda\vdash n$ be a partition. Prove[6] that

$$T_{\mathscr{I}_\lambda}\operatorname{Hilb}^n(\mathbb{A}^2) = \sum_{\square\in\lambda} t_1^{-l(\square)}t_2^{a(\square)+1} + t_1^{l(\square)+1}t_2^{-a(\square)}. \tag{1.3.15}$$

[6]This has been known for a while but I do not know an "old" reference; a recent proof is given in [69, Lemma 2.1].

Example 1.3.34. Let us confirm the conclusion of Exercise 1.3.33 for $n \leqslant 3$.

<u>$n=1$.</u> Equation (1.3.14) gives

$$T_{\square}\mathsf{Hilb}^1(\mathbb{A}^2) = 1 + t_1 t_2 - (1-t_1)(1-t_2) = t_1 + t_2.$$

On the other hand, Equation (1.3.15) gives

$$T_{\square}\mathsf{Hilb}^1(\mathbb{A}^2) = t_1^0 t_2^1 + t_1^1 t_2^0 = t_1 + t_2,$$

as desired.

<u>$n=2$.</u> Here we have two partitions to consider. We have

$$\mathcal{V}_{\begin{smallmatrix}\square\\\square\end{smallmatrix}} = 1 + t_1^{-1}, \qquad \mathcal{V}_{\square\square} = 1 + t_2^{-1}.$$

So Equation (1.3.14) gives

$$\begin{aligned}
T_{\begin{smallmatrix}\square\\\square\end{smallmatrix}}\mathsf{Hilb}^2(\mathbb{A}^2) &= 1 + t_1^{-1} + (1+t_1)t_1 t_2 - (1+t_1)(1+t_1^{-1})(1-t_1)(1-t_2) \\
&= t_1 + t_2 + t_1^2 + t_1^{-1} t_2 \\
T_{\square\square}\mathsf{Hilb}^2(\mathbb{A}^2) &= 1 + t_2^{-1} + (1+t_2)t_1 t_2 - (1+t_2)(1+t_2^{-1})(1-t_1)(1-t_2) \\
&= t_1 + t_2 + t_2^2 + t_1 t_2^{-1}.
\end{aligned}$$

On the other hand, Equation (1.3.15) gives

$$\begin{aligned}
T_{\begin{smallmatrix}\square\\\square\end{smallmatrix}}\mathsf{Hilb}^2(\mathbb{A}^2) &= (t_1^{-1} t_2 + t_1^2 t_2^0) + (t_1^0 t_2 + t_1^1 t_2^0) \\
&= t_1 + t_2 + t_1^2 + t_1^{-1} t_2 \\
T_{\square\square}\mathsf{Hilb}^2(\mathbb{A}^2) &= (t_1^0 t_2^2 + t_1 t_2^{-1}) + (t_1^0 t_2 + t_1^1 t_2^0) \\
&= t_1 + t_2 + t_2^2 + t_1 t_2^{-1}.
\end{aligned}$$

<u>$n=3$.</u> Here we have three partitions to consider. We have

$$\mathcal{V}_{\square\square\square} = 1 + t_2^{-1} + t_2^{-2}, \qquad \mathcal{V}_{\begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}} = 1 + t_1^{-1} + t_1^{-2}, \qquad \mathcal{V}_{\begin{smallmatrix}\square&\square\\\square&\end{smallmatrix}} = 1 + t_1^{-1} + t_2^{-1}.$$

So Equation (1.3.14) gives

$$\begin{aligned}
T_{\square\square\square} &= 1 + t_2^{-1} + t_2^{-2} + (1+t_2+t_2^2)t_1 t_2 - (1+t_2+t_2^2)(1+t_2^{-1}+t_2^{-2})(1-t_1)(1-t_2) \\
&= t_1 + t_2 + t_2^2 + t_2^3 + t_1 t_2^{-1} + t_1 t_2^{-2} \\
T_{\begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}} &= 1 + t_1^{-1} + t_1^{-2} + (1+t_1+t_1^2)t_1 t_2 - (1+t_1+t_1^2)(1+t_1^{-1}+t_1^{-2})(1-t_1)(1-t_2) \\
&= t_1 + t_2 + t_1^2 + t_1^3 + t_1^{-1} t_2 + t_1^{-2} t_2 \\
T_{\begin{smallmatrix}\square&\square\\\square&\end{smallmatrix}} &= 1 + t_1^{-1} + t_2^{-1} + (1+t_1+t_2)t_1 t_2 - (1+t_1+t_2)(1+t_1^{-1}+t_2^{-1})(1-t_1)(1-t_2) \\
&= 2t_1 + 2t_2 + t_1^2 t_2^{-1} + t_1^{-1} t_2^2.
\end{aligned}$$

On the other hand, Equation (1.3.15) gives

$$
\begin{aligned}
T_{\ydiagram{3}} &= (t_1^0 t_2^3 + t_1 t_2^{-2}) + (t_1^0 t_2^2 + t_1 t_2^{-1}) + (t_1^0 t_2 + t_1 t_2^0) \\
&= t_1 + t_2 + t_2^2 + t_2^3 + t_1 t_2^{-1} + t_1 t_2^{-2} \\
T_{\ydiagram{1,1,1}} &= (t_1^{-2} t_2 + t_1^3 t_2^0) + (t_1^{-1} t_2 + t_1^2 t_2^0) + (t_1^0 t_2 + t_1 t_2^0) \\
&= t_1 + t_2 + t_1^2 + t_1^3 + t_1^{-1} t_2 + t_1^{-2} t_2 \\
T_{\ydiagram{2,1}} &= (t_1^{-1} t_2^2 + t_1^2 t_2^{-1}) + (t_1^0 t_2 + t_1 t_2^0) + (t_1^0 t_2 + t_1 t_2^0) \\
&= 2t_1 + 2t_2 + t_1^2 t_2^{-1} + t_1^{-1} t_2^2.
\end{aligned}
\tag{1.3.16}
$$

2 | Motivic techniques

2.1 Grothendieck ring of varieties and attached gadgets

2.1.1 Absolute setup: motivic measures and zeta functions

The Grothendieck ring of complex varieties $K_0(\mathsf{Var}_{\mathbb{C}})$ is the free abelian group generated by isomorphism classes $[X]$ of $\mathbb{C}$-varieties, modulo the *scissor relations*, namely the relations

$$[X] = [Y] + [X \setminus Y]$$

whenever $Y \subset X$ is a closed subvariety. The group structure agrees, on generators, with the disjoint union of varieties and has neutral element $0 = [\emptyset]$. The fibre product defines a ring structure on $K_0(\mathsf{Var}_{\mathbb{C}})$, with neutral element $1 = [\operatorname{Spec}\mathbb{C}]$. Every constructible subset $Z \subset X$ of a variety X has a well-defined motivic class $[Z]$, independent upon the decomposition of Z into locally closed subsets. One could also perform a similar construction with schemes (or algebraic spaces) over $\mathbb{C}$ instead of varieties, but the resulting Grothendieck rings would come out isomorphic to $K_0(\mathsf{Var}_{\mathbb{C}})$. By the scissor relations, the motive of a scheme agrees with the motive of its reduction, so $[-]$ is insensitive to scheme structures.

The classes $[X]$ of honest $\mathbb{C}$-varieties in $K_0(\mathsf{Var}_{\mathbb{C}})$ are called *effective*. A key example is the *Lefschetz motive*

$$\mathbb{L} = [\mathbb{A}^1] \in K_0(\mathsf{Var}_{\mathbb{C}}).$$

The ring $K_0(\mathsf{Var}_{\mathbb{C}})$ has the following universal property. Suppose R is a ring and $w(-)$ is an R-valued invariant of algebraic varieties, such that

- $w(\mathsf{pt}) = 1$,
- $w(\emptyset) = 0$,
- $w(X \times Y) = w(X)w(Y)$ for every two varieties X and Y, and
- $w(X) = w(Y) + w(X \setminus Y)$ for every variety X and closed subvariety $Y \subset X$.

Then there is precisely one ring homomorphism $w\colon K_0(\mathsf{Var}_{\mathbb{C}}) \to R$ sending an effective class $[X]$ to $w(X) \in R$. Ring homomorphisms out of $K_0(\mathsf{Var}_{\mathbb{C}})$ are called *motivic measures*, or generalised Euler characteristics, or realisations [27, 64]. The key examples are the following. If Y is a complex variety, one can consider

(i) The *Hodge characteristic*

$$\chi_{\mathrm{Hodge}}(Y) = \sum_{i \geqslant 0} (-1)^i [\mathrm{H}^i_c(Y, \mathbb{Q})] \in K_0(\mathsf{HS}),$$

where HS is the abelian category of Hodge structures, and the vector space $\mathrm{H}^i_c(Y,\mathbb{Q})$ is equipped with Deligne's mixed Hodge structure [25, 26].

(ii) The *Hodge–Deligne polynomial*

$$\mathsf{E}(Y;u,v)=\sum_{p,q,i}(-1)^i h^{p,q}(\mathrm{H}^i_c(Y,\mathbb{Q}))u^p v^q \in \mathbb{Z}[u,v],$$

where $h^{p,q}(\mathrm{H}^i_c(Y,\mathbb{Q}))$ is the dimension of the (p,q)-component of the mixed Hodge structure $\mathrm{H}^i_c(Y,\mathbb{Q})$.

(iii) The *weight polynomial* $(u,v\mapsto z^{1/2})$

$$\mathsf{w}(Y,z^{1/2})=\sum_{p,q,i}(-1)^i h^{p,q}(\mathrm{H}^i_c(Y,\mathbb{Q}))z^{(p+q)/2} \in \mathbb{Z}[z^{1/2}],$$

which coincides with the signed *Poincaré polynomial*

$$\mathsf{p}(Y,-z^{1/2})=\sum_{i\geqslant 0}\dim_{\mathbb{Q}}\mathrm{H}^i(Y,\mathbb{Q})(-z^{1/2})^i$$

if Y is smooth and projective.

(iv) The *compactly supported Euler characteristic* $(z^{1/2}\mapsto 1)$

$$\chi_c(Y)=\mathsf{w}(Y,1)=\sum_{i\geqslant 0}(-1)^i\dim_{\mathbb{Q}}\mathrm{H}^i_c(Y,\mathbb{Q})\in\mathbb{Z},$$

which can be shown to equal the topological Euler characteristic $\chi(Y)$ [43, p. 95 and pp. 141–142].

Remark 2.1.1. It is possible to obtain homomorphisms

$$K_0(\mathsf{Var}_{\mathbb{C}})\to K_0(\mathsf{MHS})\to\mathbb{Z}[u,v]\to\mathbb{Z}[z^{1/2}]\to\mathbb{Z}$$

with slightly different specialisations, but with our choice we ensured that all specialisations are homomorphisms of pre-λ-rings, just as in [24, Sec. 1.4]. Note that these homomorphisms, the way they have been defined, send

$$\mathbb{L}\mapsto[\mathrm{H}^2_c(\mathbb{A}^1,\mathbb{Q})]\mapsto uv\mapsto z\mapsto 1.$$

The key features of $K_0(\mathsf{Var}_{\mathbb{C}})$ to keep in mind, which will be used constantly throughout, are the following:

(a) If $X\to B$ is a bijective morphism, then $[X]=[B]$ in $K_0(\mathsf{Var}_{\mathbb{C}})$.

(b) If $X\to B$ is a Zariski locally trivial fibration with fibre F, then $[X]=[B][F]$ in $K_0(\mathsf{Var}_{\mathbb{C}})$.

Exercise 2.1.2. Prove (a) and (b).

As a special case of (a), consider a scheme X along with locally closed subschemes $Z_1,\dots,Z_r$ such that the disjoint union of the locally closed immersions $Z_1\amalg\cdots\amalg Z_r\to X$ is a bijective morphism (in this case we say that $Z_1,\dots,Z_r$ form a *stratification* of X, and we write $X=\coprod_i Z_i$ with a slight abuse of notation). One then has an identity $[X]=\sum_{1\leqslant i\leqslant r}[Z_i]$ in $K_0(\mathsf{Var}_{\mathbb{C}})$. For instance, $\mathbb{P}^n$ has the same class as the scheme $\mathrm{pt}\amalg\mathbb{A}^1\amalg\cdots\amalg\mathbb{A}^n$, namely $1+\mathbb{L}+\cdots+\mathbb{L}^n$.

Example 2.1.3. The class of the Grassmannian $\mathsf{Gr}(k,n)$ of k-planes in $\mathbb{C}^n$ is the '$\mathbb{L}$-binomial coefficient', namely

$$[\mathsf{Gr}(k,n)] = \frac{[n]_{\mathbb{L}}!}{[k]_{\mathbb{L}}![n-k]_{\mathbb{L}}!}, \qquad [a]_{\mathbb{L}}! = \prod_{k=1}^{a}\left(\mathbb{L}^k - 1\right). \tag{2.1.1}$$

Throughout we adopt the conventions $[\mathsf{Gr}(k,n)] = 0 = [\mathbb{P}^e]$ if $k > n$ or $e < 0$.

Every complex variety X has an associated *zeta function* $\zeta_X(t)$, introduced by Kapranov [61]. It is defined as

$$\zeta_X(t) = \sum_{d\geqslant 0}[\mathsf{Sym}^d(X)]t^d \in 1 + t\,K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!]. \tag{2.1.2}$$

It satisfies, for every integer $n \geqslant 0$, the identities

$$\zeta_{\mathbb{A}^n\times X}(t) = \zeta_X(\mathbb{L}^n t), \qquad \zeta_{X\amalg Y}(t) = \zeta_X(t)\zeta_Y(t),$$

from which one deduces

$$\zeta_{\mathbb{A}^n}(t) = \frac{1}{1-\mathbb{L}^n t}, \qquad \zeta_{\mathbb{P}^n}(t) = \prod_{i=0}^{n}\frac{1}{1-\mathbb{L}^i t}.$$

2.1.2 Relative setup: main structures and operations

Relative motives

Fix a scheme S locally of finite type over $\mathbb{C}$. The Grothendieck ring of S-varieties

$$K_0(\mathsf{Var}_S)$$

is the free abelian group generated by isomorphism classes $[X \to S]$ of S-varieties modulo the *scissor relations*, namely the identities

$$\left[X \xrightarrow{f} S\right] = \left[Y \xrightarrow{f|_Y} S\right] + \left[X \setminus Y \xrightarrow{f|_{X\setminus Y}} S\right]$$

imposed whenever $Y \subset X$ is a closed S-subvariety of X. The ring structure is given on generators by fibre product over S,

$$[X \to S]\cdot[Y \to S] = [X \times_S Y \to S]. \tag{2.1.3}$$

The neutral element for the sum is $[\emptyset \to S]$, whereas for the product it is the class $[\mathrm{id}\colon S \to S]$ of the identity. Clearly, the case $S = \operatorname{Spec}\mathbb{C}$ recovers the ring $K_0(\mathsf{Var}_{\mathbb{C}})$ of Section 2.1.1. The element

$$\mathbb{L} = [\mathbb{A}^1 \times_{\mathbb{C}} S \to S] \in K_0(\mathsf{Var}_S)$$

is again called the *Lefschetz motive* (over S). If S' is another complex scheme, there is an external product

$$K_0(\mathsf{Var}_S) \times K_0(\mathsf{Var}_{S'}) \xrightarrow{\boxtimes} K_0(\mathsf{Var}_{S\times S'}) \tag{2.1.4}$$

defined on generators by sending $([f\colon X \to S],[g\colon X' \to S']) \mapsto [f\times g\colon X\times X' \to S\times S']$.

A morphism $f\colon S \to T$ induces a ring homomorphism $f^*\colon K_0(\mathsf{Var}_T) \to K_0(\mathsf{Var}_S)$ by base change and a $K_0(\mathsf{Var}_T)$-linear map $f_!\colon K_0(\mathsf{Var}_S) \to K_0(\mathsf{Var}_T)$ defined on generators by composition with f.

Definition 2.1.4. We denote by $S_0(\mathsf{Var}_S)$ the semigroup of *effective* motives, i.e. the semigroup generated by isomorphism classes $[X \to S]$ of complex quasiprojective S-varieties modulo the scissor relations. The product (2.1.3) turns $S_0(\mathsf{Var}_S)$ into a semiring. There is a natural semiring map $S_0(\mathsf{Var}_S) \to K_0(\mathsf{Var}_S)$, and we say that a motivic class $\alpha \in K_0(\mathsf{Var}_S)$ is effective if it lies in the image of this map.

Equivariant motives and the quotient map

Recall that if S is a scheme with a *good* action by a finite group G (i.e. an action such that every point of S has an affine G-invariant open neighborhood), the quotient S/G exists as a scheme. For instance, finite group actions on quasiprojective varieties are good.

Definition 2.1.5. Let G be a finite group, S a scheme with good G-action. We denote by $\widetilde{K}_0^G(\mathsf{Var}_S)$ the free abelian group generated by isomorphism classes $[X \to S]$ of G-equivariant S-varieties with good action, modulo the G-equivariant scissor relations. We denote by $K_0^G(\mathsf{Var}_S)$ the quotient of $\widetilde{K}_0^G(\mathsf{Var}_S)$ by the relations

$$[V \to X \to S] = [\mathbb{A}^r_X \to X \to S],$$

where $V \to X$ is a G-equivariant vector bundle of rank r over a G-equivariant S-variety $X \to S$.

Notation 2.1.6. Sometimes we write $[X \to S, G]$ to remember the group action.

There is a natural ring structure on $\widetilde{K}_0^G(\mathsf{Var}_S)$, where the product of two classes $[X \to S]$ and $[Y \to S]$ is given by taking the diagonal action on $X \times_S Y$. The structures f^*, $f_!$ and $\boxtimes$ naturally extend to the equivariant setting, along with their basic compatibilities. For instance, if $f : S \to T$ (resp. $g : S' \to T'$) is a G-equivariant (resp. G'-equivariant) map, and α, β are equivariant motives over S, S', then

$$(f \times g)_!(\alpha \boxtimes \beta) = f_!\alpha \boxtimes g_!\beta \tag{2.1.5}$$

in the $(G \times G')$-equivariant K-group over $T \times T'$.

Recall that, since the G-action on S is good, S/G exists as a scheme. One can define a $K_0(\mathsf{Var}_{S/G})$-linear map (cf. [24, Lemma 1.5])

$$\widetilde{K}_0^G(\mathsf{Var}_S) \xrightarrow{\pi_G} K_0(\mathsf{Var}_{S/G}) \tag{2.1.6}$$

given on generators by taking the orbit space,

$$\pi_G[X \to S] = [X/G \to S/G].$$

This map does not always extend to $K_0^G(\mathsf{Var}_S)$. It does when G acts freely on S, by [14, Lemma 3.2].

Lambda ring structures

Let $n > 0$ be an integer, and let $\mathfrak{S}_n$ be the symmetric group of n elements. By [24, Lemma 1.6], namely the relative version of [9, Lemma 2.4], there exist "n-th power" maps

$$K_0(\mathsf{Var}_S) \xrightarrow{(\cdot)^{\otimes n}} \widetilde{K}_0^{\mathfrak{S}_n}(\mathsf{Var}_{S^n}) \tag{2.1.7}$$

where $S^n = S \times \cdots \times S$ is endowed with the natural $\mathfrak{S}_n$-action. The power map takes $[f : X \to S]$ to the class of the equivariant function $f^n : X^n \to S^n$. For $A \in K_0(\mathsf{Var}_S)$, consider the classes

$$\pi_{\mathfrak{S}_n}(A^{\otimes n}) \in K_0(\mathsf{Var}_{S^n/\mathfrak{S}_n}) = K_0(\mathsf{Var}_{\mathsf{Sym}^n(S)}).$$

The ring $K_0(\mathsf{Var}_{\mathbb{C}})$ is a *lambda ring*, in which the lambda ring operations are defined by

$$A \mapsto \sigma^n(A) = \pi_{\mathfrak{S}_n}(A^{\otimes n}) \in K_0(\mathsf{Var}_{\mathbb{C}})$$

for effective classes $A \in K_0(\mathsf{Var}_{\mathbb{C}})$, and then taking the unique extension to a lambda ring structure on $K_0(\mathsf{Var}_{\mathbb{C}})$, determined by the relation

$$\sum_{i=0}^{n} \sigma^i([X]-[Y])\sigma^{n-i}[Y] = \sigma^n[X]. \tag{2.1.8}$$

If S comes with a commutative associative map $\nu\colon S \times S \to S$, we likewise define

$$\sigma^n_\nu(A) = \overline{\nu}_! \pi_{\mathfrak{S}_n}(A^{\otimes n}) \in K_0(\mathsf{Var}_S)$$

on effective classes $A = [X \to S]$, where $\overline{\nu}$ is the map $\mathsf{Sym}^n(S) = S^n/\mathfrak{S}_n \to S$. One then uses the analogue of the relation (2.1.8) to find a unique set of lambda ring operators σ^n_ν restricting to the previous identity on effective motives.

As a special case, one can consider $(S, \nu) = (\mathbb{N}, +)$, viewed as a symmetric monoid in the category of schemes. We obtain lambda operations $\sigma^n = \sigma^n_+$ on $K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!]$ via the isomorphism

$$K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!] \xrightarrow{\sim} K_0(\mathsf{Var}_{\mathbb{N}}) \tag{2.1.9}$$

defined by sending $\sum_{n\geqslant 0}[Y_n]t^n \mapsto \left[\coprod_{n\in\mathbb{N}} Y_n \to \mathbb{N}\right]$.

2.1.3 Power structures

The main references for power structures are [45, 46].

Definition 2.1.7 ([45])**.** A *power structure* on a (semi)ring R is a map

$$\begin{aligned}(1+tR[\![t]\!]) \times R &\to 1+tR[\![t]\!]\\ (A(t), m) &\mapsto A(t)^m\end{aligned}$$

satisfying the following conditions:

1. $A(t)^0 = 1$,
2. $A(t)^1 = A(t)$,
3. $(A(t)\cdot B(t))^m = A(t)^m \cdot B(t)^m$,
4. $A(t)^{m+m'} = A(t)^m \cdot A(t)^{m'}$,
5. $A(t)^{mm'} = (A(t)^m)^{m'}$,
6. $(1+t)^m = 1 + mt + O(t^2)$,
7. $A(t)^m\big|_{t\to t^e} = A(t^e)^m$.

Throughout we use the following:

Notation 2.1.8. Partitions $\alpha \vdash n$ are written as $\alpha = (1^{\alpha_1}\cdots i^{\alpha_i}\cdots n^{\alpha_n})$, meaning that there are α_i parts of size i. In particular we recover $n = \sum_i i\alpha_i$. The *automorphism group* of α is the product of symmetric groups $G_\alpha = \prod_i \mathfrak{S}_{\alpha_i}$. We set $||\alpha|| = \sum_i \alpha_i$. We denote by $\mathcal{P}$ the set of all partitions (of varying length).

Example 2.1.9. If $R = \mathbb{Z}$, $A(t) = 1 + \sum_{n>0} A_n t^n \in \mathbb{Z}[\![t]\!]$ and $m \in \mathbb{N}$, the known formula [86, p. 40]

$$A(t)^m = \sum_{\alpha\in\mathcal{P}} \left(\prod_{i=0}^{||\alpha||-1}(m-i)\cdot\frac{\prod_i A_i^{\alpha_i}}{\prod_i \alpha_i!}\right)t^{|\alpha|} = 1 + \sum_{n>0}\sum_{\alpha\vdash n}\left(\prod_{i=0}^{||\alpha||-1}(m-i)\cdot\frac{\prod_i A_i^{\alpha_i}}{\prod_i \alpha_i!}\right)t^n$$

defines a power structure on $\mathbb{Z}$. For $m \in \mathbb{Z}$, say $m = k - l$ with $k, l \in \mathbb{N}$ it is enough to define $A(t)^m = A(t)^k \cdot (A(t)^l)^{-1}$.

Gusein-Zade, Luengo and Melle-Hernández have proved [45, Thm. 2] that there is a unique power structure

$$(A(t), m) \mapsto A(t)^m$$

on $K_0(\mathsf{Var}_{\mathbb{C}})$ extending the one defined in *loc. cit.* on the semiring $S_0(\mathsf{Var}_{\mathbb{C}})$ of effective motives. The latter is given by the formula

$$A(t)^{[X]} = 1 + \sum_{n>0} \sum_{\alpha \vdash n} \pi_{G_\alpha}\left(\left[\prod_i X^{\alpha_i} \setminus \Delta\right] \cdot \prod_i A_i^{\otimes \alpha_i}\right) t^n. \tag{2.1.10}$$

Here, $\Delta \subset \prod_i X^{\alpha_i}$ is the "big diagonal" (the locus in the product where at least two entries are equal), and the product in big round brackets is a G_α-equivariant motive in $\widetilde{K}_0^{G_\alpha}(\mathsf{Var}_{\mathbb{C}})$, thanks to the power map (2.1.7).

An easy but somewhat crucial remark is that, if $A(t) = (1-t)^{-1}$, one has

$$A(t)^{[X]} = (1-t)^{-[X]} = \zeta_X(t),$$

where ζ_X is the Kapranov zeta function introduced in (2.1.2).

Remark 2.1.10. We will not encounter non-effective coefficients here, so we will have direct access to Formula (2.1.10).

2.1.4 Motivic exponential

The *motivic exponential* is a group isomorphism

$$\mathsf{Exp}\colon t K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!] \xrightarrow{\sim} 1 + t K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!], \tag{2.1.11}$$

converting sums into products and preserving effectiveness. If $A = \sum_{n>0} A_n t^n$ is an effective power series, one has by definition

$$\mathsf{Exp}\left(\sum_{n>0} A_n t^n\right) = \prod_{n>0} \left(1 - t^n\right)^{-A_n},$$

and if A and B are effective, one sets

$$\mathsf{Exp}(A - B) = \prod_{n>0} \left(1 - t^n\right)^{-A_n} \cdot \left(\prod_{n>0} \left(1 - t^n\right)^{-B_n}\right)^{-1}. \tag{2.1.12}$$

More generally, if $(S, \nu\colon S \times S \to S)$ is a commutative monoid in the category of schemes, with a submonoid $S_+ \subset S$ such that the induced map $\coprod_{n \geqslant 1} S_+^{\times n} \to S$ is of finite type, we similarly define

$$\mathsf{Exp}_\nu(A) = \sum_{n \geqslant 0} \sigma_\nu^n(A)$$

on effective classes, and for A and B two effective classes, we define $\mathsf{Exp}_\nu(A - B)$ by the analogue of (2.1.12), i.e. by $\mathsf{Exp}_\nu(A) \cdot \mathsf{Exp}_\nu(B)^{-1}$. If $(S, \nu) = (\mathbb{N}, +)$, we simply write Exp instead of Exp_+, as we recover (2.1.11).

Example 2.1.11. For any variety X, we have

$$\mathsf{Exp}[X]t = (1-t)^{-[X]} = \zeta_X(t).$$

Caution 2.1.12. Note that Exp does not behave well with respect to variable substitution, e.g. $\mathsf{Exp}(t) = (1-t)^{-1}$ but $\mathsf{Exp}(-t) = 1 - t \neq \mathsf{Exp}(t)|_{t \to -t} = (1+t)^{-1}$.

2.2 The Denef–Loeser motivic zeta function

Denote by J_n and J_∞ the functors of n-jets and the arc space respectively. Let Y be a smooth variety of dimension $d \geqslant 1$, and let $f\colon Y \to \mathbb{A}^1$ be a nonzero regular function with associated hypersurface $Y_0 = f^{-1}(0) \subset Y$. We have an induced morphism $f_n\colon J_n Y \to J_n\mathbb{A}^1 = \mathbb{A}^{n+1}$. Consider the 'order function'

$$J_n\mathbb{A}^1 \xrightarrow{\mathsf{ord}_t} \mathbb{N} \cup \{\infty\}$$

sending an arc $\gamma \in J_n\mathbb{A}^1$ (viewed as an element $\gamma = \sum_{0\leqslant e\leqslant n} \gamma_e t^e \in \mathbb{C}[t]/t^{n+1}$) to

$$\mathsf{ord}_t(\gamma) = \begin{cases} \min\{\, e \in \mathbb{N} \mid \gamma_e \neq 0 \,\} & \text{if } \gamma \neq 0 \\ \infty & \text{if } \gamma = 0. \end{cases} \tag{2.2.1}$$

Consider the locally closed subsets

$$\mathcal{X}_{f,n} = \{\, \gamma \in J_n Y \mid \mathsf{ord}_t f_n(\gamma) = n \,\} \subset J_n Y$$

for $n \geqslant 1$. These are naturally Y_0-varieties via the projection $\tau_0^n\colon J_n Y \to Y$ sending an arc to its base point, so we have well-defined relative motives $[\mathcal{X}_{f,n} \to Y_0] \in K_0(\mathsf{Var}_{Y_0})$.

Consider, for $n \geqslant 1$, the function

$$\begin{aligned} \mathcal{X}_{f,n} &\xrightarrow{\overline{f}_n} \mathbb{G}_m \\ \gamma &\longmapsto a_n \end{aligned}$$

defined by sending an arc γ such that $\mathsf{ord}_t f_n(\gamma) = a_n t^n$ to the scalar $a_n \in \mathbb{G}_m$. Form the fibre

$$\begin{tikzcd} \mathcal{X}_{f,n,1} \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\square"] & \mathcal{X}_{f,n} \arrow[d, "\overline{f}_n"] \\ 1 \arrow[r, hook] & \mathbb{G}_m \end{tikzcd}$$

so that, set-theoretically, $\mathcal{X}_{f,n,1}$ consists of arcs γ such that $f_n(\gamma) = t^n$. Now, there are actions

$$\begin{tikzcd} \mathbb{G}_m \times \mathcal{X}_{f,n} \arrow[r, "\cdot"] & \mathcal{X}_{f,n} \\ \mu_n \times \mathcal{X}_{f,n,1} \arrow[r, "\cdot"] \arrow[u, hook] & \mathcal{X}_{f,n,1} \arrow[u, hook] \end{tikzcd}$$

defined as follows. Take $\lambda \in \mathbb{G}_m$ and $\gamma \in \mathcal{X}_{f,n}$, so that $f_n(\gamma) = a t^n$ for some $a \in \mathbb{G}_m$ — in other words, $a = \overline{f}_n(\gamma)$. Define the arc $\lambda \cdot \gamma\colon \operatorname{Spec}\mathbb{C}[t]/t^{n+1} \to Y$ as

$$\operatorname{Spec}\mathbb{C}[t]/t^{n+1} \xrightarrow{\lambda\cdot} \operatorname{Spec}\mathbb{C}[t]/t^{n+1} \xrightarrow{\gamma} Y,$$

where the first isomorphism sends $t \mapsto \lambda t$. This way, we have that $f \circ (\lambda\cdot\gamma)\colon \operatorname{Spec}\mathbb{C}[t]/t^{n+1} \to \mathbb{A}^1 = \operatorname{Spec}\mathbb{C}[t]$ corresponds to

$$\mathbb{C}[t]/t^{n+1} \ni f \circ (\lambda\cdot\gamma)(t) = f \circ \gamma(\lambda t) = a(\lambda t)^n = \lambda^n (a t^n) = \lambda^n (f \circ \gamma(t)).$$

In other words,

$$\overline{f}_n(\lambda\cdot\gamma) = \lambda^n a,$$

which confirms at once that $\lambda \cdot \gamma$ has order n and that $\mu_n = \operatorname{Spec}\mathbb{C}[t]/(t^n - 1)$ acts on $\mathcal{X}_{f,n,1}$ by restriction. We obtain equivariant classes

$$[\mathcal{X}_{f,n,1} \longrightarrow Y_0, \hat{\mu}] \in \mathcal{M}_{Y_0}^{\hat{\mu}}$$

where $\hat{\mu} = \varprojlim \mu_n$. Define the *motivic zeta function* attached to the pair (Y, f) as the power series

$$\mathsf{Z}_f(T) = \sum_{n \geqslant 1} [\mathcal{X}_{f,n,1} \to Y_0, \hat{\mu}] \mathbb{L}^{-nd} T^n \in \mathcal{M}_{Y_0}^{\hat{\mu}}.$$

This power series is rational, as we now explain. The key result is the principalisation theorem.

THEOREM 2.2.1 ([12, Thm. 1.10]). *Let Y be a smooth variety, $\mathscr{I} \subset \mathscr{O}_Y$ an ideal sheaf with associated closed subscheme $Z \subset Y$. There exists a proper birational morphism*

$$\widetilde{Y} \xrightarrow{\pi} Y$$

such that

(1) *π is a composition of blowups with smooth centres, and*

(2) *the ideal sheaf $\widetilde{\mathscr{I}} = \pi^* \mathscr{I} \cdot \mathscr{O}_{\widetilde{Y}} \subset \mathscr{O}_{\widetilde{Y}}$ defines a simple normal crossing divisor $\pi^{-1} Z \subset \widetilde{Y}$.*

Consider any embedded resolution $\pi \colon \widetilde{Y} \to Y$ of (Y, f) as in Theorem 2.2.1. Letting Y_0 taking the role of Z, suppose $\pi^{-1} Y_0$ is of the form $\sum_{i \in J} N_i E_i$ when viewed as a simple normal crossing divisor. Also define integers ν_i by the identity $K_\pi = \sum_{i \in J} (\nu_i - 1) E_i$. For a subset $I \subset J$, set

$$m_I = \gcd(N_i \mid i \in I).$$

Form the diagram

$$\begin{tikzcd}
\widetilde{Y}_I^{\nu} \arrow[r] & \widetilde{Y} \times_{\mathbb{A}^1_u} \mathbb{A}^1_t \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & \widetilde{Y} \arrow[d, "f \circ \pi"] \\
& \mathbb{A}^1_t \arrow[r] & \mathbb{A}^1_u
\end{tikzcd}$$

where $\mathbb{A}^1_t \to \mathbb{A}^1_u$ sends $u \mapsto t^{m_I}$ and $\widetilde{Y}_I^{\nu} \to \widetilde{Y} \times_{\mathbb{A}^1_u} \mathbb{A}^1_t$ is the normalisation map. For $i \in J$ and a nonempty subset $\emptyset \neq I \subset J$, consider the (smooth) subvarieties of $\widetilde{Y}$ defined as

$$\begin{aligned}
E_i^\circ &= E_i \setminus \bigcup_{j \neq i} E_j \\
E_I &= \bigcap_{i \in I} E_i \\
E_I^\circ &= E_I \setminus \bigcup_{j \in J \setminus I} E_j.
\end{aligned}$$

The map g_I in the pullback diagram

$$\begin{tikzcd}
\widetilde{E}_I^\circ \arrow[r] \arrow[d, "g_I"'] \arrow[dr, phantom, "\square"] & \widetilde{Y}_I^{\nu} \arrow[d] \\
E_I^\circ \arrow[r, hook] & Y
\end{tikzcd}$$

is an unramified μ_{m_I}-Galois cover, so that in particular we get a good $\hat{\mu}$-action on $\widetilde{E}_I^\circ$. We thus obtain equivariant classes

$$[\widetilde{E}_I^\circ \longrightarrow Y_0, \hat{\mu}] \in \mathcal{M}_{Y_0}^{\hat{\mu}}$$

THEOREM 2.2.2 ([28, 64]). *In $\mathcal{M}_{Y_0}^{\hat{\mu}}[\![T]\!]$ there is an identity*

$$\mathsf{Z}_f(T) = \sum_{\emptyset \neq I \subset J} (\mathbb{L}-1)^{|I|-1} [\widetilde{E}_I^\circ \to Y_0, \hat{\mu}] \prod_{i \in I} \frac{\mathbb{L}^{-\nu_i} T^{N_i}}{1-\mathbb{L}^{-\nu_i} T^{N_i}}.$$

For every point $x \in Y_0$, there is a "fibre map"

$$\mathcal{M}_{Y_0}^{\hat{\mu}} \xrightarrow{\mathrm{Fib}_x} \mathcal{M}_{\mathbb{C}}^{\hat{\mu}}$$

defined on generators by $[U \to Y_0, \hat{\mu}] \mapsto [U \times_{Y_0} \kappa(x), \hat{\mu}]$.

Definition 2.2.3 ([27, §3])**.** Given $f\colon Y \to \mathbb{A}^1$ as above,

1. $\psi_f = -\lim_{T\to\infty} \mathsf{Z}_f(T) \in \mathcal{M}_{Y_0}^{\hat{\mu}}$ is called the *motivic nearby fibre*,
2. $\mathsf{MF}_{f,x} = \mathrm{Fib}_x(\psi_f) \in \mathcal{M}_{\mathbb{C}}^{\hat{\mu}}$ is called the *motivic Milnor fibre* of f at x, and
3. $\phi_f = \psi_f - 1 = \psi_f - [\mathrm{id}\colon Y_0 \to Y_0] \in \mathcal{M}_{Y_0}^{\hat{\mu}}$ is called the *motivic vanishing cycle* of f.

Note that the limit of $\mathsf{Z}_f(T)$ makes sense because the motivic zeta function is rational by Theorem 2.2.2. More precisely, to obtain ψ_f one has to expand $\mathsf{Z}_f(T)$ in T^{-1} and take minus its constant term. In fact, one has

$$\psi_f = \sum_{\emptyset \neq I \subset J} (1-\mathbb{L})^{|I|-1} [\widetilde{E}_I^\circ \to Y_0, \hat{\mu}] \in \mathcal{M}_{Y_0}^{\hat{\mu}}.$$

2.3 Białynicki-Birula decomposition

We recall the classical result of Białynicki-Birula. Recall that a smooth $\mathbb{G}_m$-variety X is *semiprojective* (with respect to the considered action) if the following conditions are satisfied:

(1) the fixed locus $X^{\mathbb{G}_m}$ is proper, and

(2) for every $x \in X$, the limit $\lim_{t\to 0} t \cdot x$ exists in X.

In particular, if X is a projective $\mathbb{G}_m$-variety, then it is automatically semiprojective.

The second condition means the following. Fix $x \in X$. To say that 'the limit exists' means that the family

$$\begin{aligned} \mathbb{G}_m &\xrightarrow{t\cdot} X \\ t &\longmapsto t \cdot x \end{aligned}$$

extends over the origin to a morphism $\mathbb{A}^1 \to X$. For X a moduli space such as the Hilbert scheme, this is clearly a flatness condition: for instance, and this will be the key example for us later, given $[I] \in \mathsf{Hilb}^n(\mathbb{A}^2)$, we say that the limit $\lim_{t\to 0} t \cdot [I]$ exists in $\mathsf{Hilb}^n(\mathbb{A}^2)$ if there is a commutative diagram

$$\begin{tikzcd} \mathbb{G}_m \arrow[r, "t\cdot"] \arrow[d, hook] & \mathsf{Hilb}^n(\mathbb{A}^2) \\ \mathbb{A}^1 \arrow[ur, dotted] & \end{tikzcd}$$

Note that the morphism $\mathbb{G}_m \to \mathsf{Hilb}^n(\mathbb{A}^2)$ *is there* by default: it is the restriction of the action to the slice $\mathbb{G}_m \times [I]$. The morphism $\mathbb{A}^1 \to \mathsf{Hilb}^n(\mathbb{A}^2)$ exists if and only if the $\mathbb{G}_m$-family extends *flatly* to $\mathbb{A}^1$. Finally, it is worth noting that such condition depends on the $\mathbb{G}_m$-action: being semiprojective is a property of the $\mathbb{G}$-action on X, not of X alone!

THEOREM 2.3.1 (Białynicki-Birula [11, Sec. 4]). *Let X be a smooth semiprojective $\mathbb{G}_m$-variety. Let*

$$X^{\mathbb{G}_m} = \coprod_{1\leqslant i\leqslant r} X_i$$

be the decomposition of the fixed locus $X^{\mathbb{G}_m} \subset X$ in its connected components. Then,

(1) *for each $i = 1,\dots,r$ there exists a unique smooth locally closed $\mathbb{G}_m$-invariant subscheme $X_i^+ \subset X$ along with a morphism $\gamma_i : X_i^+ \to X_i$ such that*

- *X_i is a closed subscheme of X_i^+,*
- *γ_i is an affine fibre bundle.*

(2) *for every closed point $x \in X_i$, one has*

$$T_x(X_i^+) = T_x(X)^{\mathsf{fix}} \oplus T_x(X)^+,$$

where $T_x(X)^{\mathsf{fix}}$, resp. $T_x(X)^+$, denotes the $\mathbb{G}_m$-fixed, resp. positive part of $T_x(X)$. In particular, since $T_x(X)^{\mathsf{fix}} = T_x X_i$, the rank of $\gamma_i : X_i^+ \to X_i$ is equal to $\dim_{\mathbb{C}} T_x(X)^+$ for $x \in X_i$.

(3) *The natural morphism*

$$\coprod_{1\leqslant i\leqslant r} X_i^+ \longrightarrow X$$

is bijective. In particular, there is an identity

$$[X] = \sum_{1\leqslant i\leqslant r} [X_i]\mathbb{L}^{\mathrm{d}_i^+} \tag{2.3.1}$$

in $K_0(\mathsf{Var}_{\mathbb{C}})$, where

$$\mathrm{d}_i^+ = \operatorname{rank}\gamma_i = \dim_{\mathbb{C}} T_x(X)^+$$

for an arbitrary closed point $x \in X_i$.

The proof of this theorem is constructive: one defines

$$X_i^+ = \left\{ x \in X \;\middle|\; \lim_{t\to 0} t\cdot x \in X_i \right\} \subset X.$$

Example 2.3.2. Let $X = \mathbb{P}^2 = \operatorname{Proj}\mathbb{C}[x,y,z]$. Consider the action of $\mathbb{G}_m$ with weights $(2,1,0)$. Then $p_1 = (1:0:0)$, $p_2 = (0:1:0)$ and $p_3 = (0:0:1)$ are the fixed points. We have

$$X_i^+ = \left\{ p \in \mathbb{P}^2 \;\middle|\; \lim_{t\to 0} t\cdot p = p_i \right\}$$

for $i = 1,2,3$. Therefore we have 3 cells which are affine spaces, as the fixed locus is isolated. But $\lim_{t\to 0} t\cdot(a:b:c) = \lim_{t\to 0}(t^2a : tb : c)$, so the cells X_i^+ are as follows.

1. $i = 3$: Any point with $c \neq 0$ flows to p_3, as

$$(t^2a : tb : c) \xrightarrow{t\to 0} (0:0:c) = (0:0:1).$$

Therefore $X_3^+ = \mathrm{D}_+(z) = \mathbb{A}^2$.

2. $i = 1$: the only $p = (a:b:c)$ that flows to $p_1 = (1:0:0)$ is p_1 itself, so $X_1^+ = \mathrm{pt}$.

3. $i = 2$: here we look at points $p = (a:b:c)$ with $c = 0$ (otherwise we automatically flow to p_3, see item above) and $b \neq 0$ (otherwise we flow to p_1), and we want $\lim_{t\to 0}(t^2a : tb : 0) = \lim_{t\to 0}(ta : b : 0) = (0:1:0)$, so $X_2^+ = \mathrm{V}_+(z) \cap \mathrm{D}_+(y) = \operatorname{Spec}\mathbb{C}[x/y] \cong \mathbb{A}^1$.

All in all, the Białynicki-Birula decomposition of $\mathbb{P}^2$ with the above action[1] is the classical cell decomposition $\mathbb{P}^2 = \mathbb{A}^2 \amalg \mathbb{A}^1 \amalg \mathrm{pt}$.

Example 2.3.3. The smooth variety $X = \mathbb{A}^2 = \operatorname{Spec}\mathbb{C}[x,y]$ with the action $t\cdot(a,b) = (ta, t^{-1}b)$ is *not* semiprojective. Indeed, for $p = (a,b) \in \mathrm{D}(y) \subset \mathbb{A}^2$, one has

$$\lim_{t\to 0} t\cdot p = \lim_{t\to 0}(ta, t^{-1}b) = (0,\infty) \notin \mathbb{A}^2.$$

One can still construct the (unique) BB cell

$$X_0^+ = \left\{ p \in \mathbb{A}^2 \,\middle|\, \lim_{t\to 0} t\cdot p = 0 \right\}$$

corresponding to the unique fixed point $0 \in \mathbb{A}^2$, but X_0^+ is isomorphic to $\operatorname{Spec}\mathbb{C}[x,y]/y = \mathbb{A}^1$. Therefore the (unique) BB cell does not cover X.

The previous example can be remedied as follows.

Example 2.3.4. The smooth variety $X = \mathbb{A}^2 = \operatorname{Spec}\mathbb{C}[x,y]$ with the action $t\cdot(a,b) = (t^N a, tb)$ such that $N \gg 0$ *is* semiprojective. Indeed, given the unique fixed point $0 \in \mathbb{A}^2$, we have

$$X_0^+ = \left\{ (a,b) \in \mathbb{A}^2 \,\middle|\, \lim_{t\to 0}(t^N a, tb) = 0 \right\} = \mathbb{A}^2.$$

In other words, every point flows to the origin and no point 'escapes at infinity' as in the previous example.

2.4 The motive of the Quot scheme of points on a curve

PROPOSITION 2.4.1. *Let C be a smooth projective curve over $\mathbb{C}$. Fix a locally free sheaf E of rank r on C. Then $\mathsf{Quot}_C(E,n)$ is a smooth projective variety of dimension rn.*

Proof. Let $E \twoheadrightarrow Q$ be a surjection with kernel K. To confirm smooothness, it is enough to show that $\operatorname{Ext}^1(K,Q) = 0$. But $\operatorname{Ext}^1(K,Q)^\vee = \operatorname{Hom}(Q, K\otimes\omega_C)$, which vanishes since Q is 0-dimensional and $K\otimes\omega_C$ is torsion-free. The dimension is computed as $\dim\operatorname{Hom}(K,Q) = h^0(C, K^\vee\otimes Q) = h^0(C, Q^{\oplus r}) = h^0(C,Q)^{\oplus r} = rn$. □

We wish to apply Theorem 2.3.1 to $\mathsf{Quot}_C(E,n)$. The first step is to endow the Quot scheme with a torus action. For this, we need E to be split.

LEMMA 2.4.2 (Rank reduction). *Let X be an algebraic variety. Let $E = L_1 \oplus \cdots \oplus L_r$ be a split vector bundle over X. Then the algebraic torus $\mathbb{T} = \mathbb{G}_m^r$ acts on $\mathsf{Quot}_X(E,n)$ and*

$$\mathsf{Quot}_X(E,n)^{\mathbb{T}} = \coprod_{n_1+\cdots+n_r=n} \prod_{i=1}^{r} \mathsf{Hilb}^{n_i}(X).$$

Proof. This is a special case of the main result in [13]. □

Let $X = C$ be a curve from now on, and let $E = L_1 \oplus \cdots \oplus L_r$ be a split vector bundle. The connected component of $\mathsf{Quot}_C(E,n)^{\mathbb{T}}$ indexed by the r-tuple $\boldsymbol{n} = (n_1,\ldots,n_r)$ will be denoted $Q_{\boldsymbol{n}}$. Now we act on the Quot scheme with the 1-parameter subgroup

$$\mathbb{G}_m = \left\{ (w, w^2, \ldots, w^r) \,\middle|\, w \in \mathbb{C}^\times \right\} \subset \mathbb{T}.$$

[1] You will get the same result if you take arbitrary decreasing weights $\alpha > \beta > 0$.

This 1-parameter subgroup is generic enough to ensure

$$\mathsf{Quot}_X(E,n)^{\mathbb{T}} = \mathsf{Quot}_X(E,n)^{\mathbb{G}_m}.$$

Then Theorem 2.3.1 tells us that we have a stratification

$$\mathsf{Quot}_C(E,n) = \coprod_{\boldsymbol{n}} Q_{\boldsymbol{n}}^+$$

where $Q_{\boldsymbol{n}}^+ \to Q_{\boldsymbol{n}}$ is an affine bundle. Since

$$Q_{\boldsymbol{n}} \cong \prod_{\alpha=1}^{r} \mathsf{Sym}^{n_\alpha}(C),$$

the motive of the base is known. We only need to compute the dimension of the fibre of $Q_{\boldsymbol{n}}^+ \to Q_{\boldsymbol{n}}$. Let $p \in Q_{\boldsymbol{n}} \subset \mathsf{Quot}_C(E,n)^{\mathbb{T}}$ be a fixed point, so that

$$T_p \mathsf{Quot}_C(E,n) = \mathrm{Hom}_C(K,T)$$

is a $\mathbb{T}$-representation. By Lemma 2.4.2, p has the form

$$p = (K_\alpha \hookrightarrow L_\alpha \twoheadrightarrow T_\alpha)_{\alpha=1,\dots,r}$$

where of course $T = \bigoplus_\alpha T_\alpha$ and $\chi(T_\alpha) = n_\alpha$. Then

$$\mathrm{Hom}_C(K,T) = \bigoplus_{1\leqslant\alpha,\beta\leqslant r} \mathrm{Hom}_C(K_\alpha, T_\beta).$$

As a $\mathbb{T}$-representation, we have

$$\begin{aligned} T_p \mathsf{Quot}_C(E,n) &= \bigoplus_{1\leqslant\alpha,\beta\leqslant r} \mathrm{Hom}(K_\alpha \otimes w_\alpha, T_\beta \otimes w_\beta) \\ &= \bigoplus_{1\leqslant\alpha,\beta\leqslant r} \mathrm{Hom}_C(K_\alpha, T_\beta) \otimes w_\beta w_\alpha^{-1}. \end{aligned}$$

The positive part of the tangent space, then, namely the part with positive weights, is computed as

$$T_p^+ \mathsf{Quot}_C(E,n) = \bigoplus_{\alpha<\beta} \mathrm{Hom}_C(K_\alpha, T_\beta) \otimes w_\beta w_\alpha^{-1},$$

where $w_1, \dots, w_r$ are the coordinates of the full torus $\mathbb{T}$. We are interested in the dimension of this vector space. We compute

$$\begin{aligned} \dim T_p^+ \mathsf{Quot}_C(E,n) &= \sum_{\alpha<\beta} \dim \mathrm{Hom}_C(K_\alpha, T_\beta) \\ &= \sum_{\alpha<\beta} \dim \mathrm{H}^0(C, K_\alpha^* \otimes_{\mathcal{O}_C} T_\beta) \\ &= \sum_{\alpha<\beta} n_\beta \\ &= \sum_{1\leqslant\beta\leqslant r} (\beta-1) n_\beta. \end{aligned}$$

It follows from Equation (2.3.1) that

$$\begin{aligned} [Q_{\boldsymbol{n}}^+] &= \prod_{\alpha=1}^{r} [\mathsf{Sym}^{n_\alpha}(C)] \cdot \mathbb{L}^{\sum_{1\leqslant\beta\leqslant r}(\beta-1)n_\beta} \\ &= \prod_{\alpha=1}^{r} [\mathsf{Sym}^{n_\alpha}(C)] \cdot \prod_{\alpha=1}^{r} \mathbb{L}^{(\alpha-1)n_\alpha} \qquad (2.4.1) \\ &= \prod_{\alpha=1}^{r} [\mathsf{Sym}^{n_\alpha}(C)] \mathbb{L}^{(\alpha-1)n_\alpha}. \end{aligned}$$

Form the generating function

$$\mathsf{Quot}_E(t) = \sum_{n\geqslant 0} [\mathsf{Quot}_C(E,n)]t^n.$$

We can now prove the following.

THEOREM 2.4.3. *There is an identity*

$$\mathsf{Quot}_E(t) = \prod_{\alpha=1}^{r} \zeta_C(\mathbb{L}^{\alpha-1}t).$$

Proof. We compute

$$\begin{aligned}
\mathsf{Quot}_E(t) &= \sum_{n\geqslant 0} t^n \sum_{n_1+\cdots+n_r=n} [Q_{\boldsymbol{n}}^+] \\
&= \sum_{n\geqslant 0} t^n \sum_{n_1+\cdots+n_r=n} \prod_{\alpha=1}^{r} [\mathsf{Sym}^{n_\alpha}(C)]\mathbb{L}^{(\alpha-1)n_\alpha} \\
&= \sum_{n_1,\ldots,n_r\geqslant 0} \prod_{\alpha=1}^{r} [\mathsf{Sym}^{n_\alpha}(C)]\mathbb{L}^{(\alpha-1)n_\alpha} t^{n_\alpha} \\
&= \sum_{n_1,\ldots,n_r\geqslant 0} \prod_{\alpha=1}^{r} [\mathsf{Sym}^{n_\alpha}(C)](\mathbb{L}^{\alpha-1}t)^{n_\alpha} \\
&= \prod_{\alpha=1}^{r} \zeta_C(\mathbb{L}^{\alpha-1}t).
\end{aligned}$$

□

In terms of the motivic exponential, since we have $\mathsf{Exp}([X]t) = (1-t)^{-[X]} = \zeta_X(t)$ for any variety X and we have the transformation rule

$$\mathsf{Exp}(\mathbb{L}^i[X]t) = (1-t)^{-\mathbb{L}^i[X]} = (1-\mathbb{L}^i t)^{-[X]} = \zeta_X(\mathbb{L}^i t),$$

we may rewrite

$$\mathsf{Quot}_E(t) = \prod_{\alpha=1}^{r} \mathsf{Exp}(\mathbb{L}^{\alpha-1}[C]t) = \mathsf{Exp}\left(\sum_{\alpha=1}^{r} \mathbb{L}^{\alpha-1}[C]t\right) = \mathsf{Exp}\big([\mathbb{P}^{r-1}\times C]t\big).$$

With very little effort, one can show via power structures that this formula holds for all smooth *quasiprojective* curves and for all locally free sheaves E, not necessarily split.

2.5 The motive of the Hilbert scheme of points on $\mathbb{A}^2$

Let S be a smooth complex quasiprojective surface. Consider the generating function

$$\mathsf{Hilb}_S(t) = \sum_{n\geqslant 0} [\mathsf{Hilb}^n(S)]t^n \in K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!],$$

In this section we shall find a formula for $\mathsf{Hilb}_S(t)$ in the case $S = \mathbb{A}^2$. The case of an arbitrary surface will be handled in Section 2.6.2 thanks to the power structure machinery.

We know that $\mathsf{Hilb}^n(\mathbb{A}^2)$ is acted on by $\mathbb{T} = \mathbb{G}_m^2$ with isolated fixed locus. Each fixed point corresponds to a partition of n, i.e. a Young diagram λ of size $|\lambda| = n$. Explicitly,

$$I_\lambda = \left\langle x^i y^j \;\middle|\; (i,j)\notin\lambda \right\rangle \subset \mathbb{C}[x,y]$$

is the ideal corresponding to $\lambda \vdash n$. The first step is to use a subtorus $\mathbb{G}_m \subset \mathbb{T}$ with the same fixed locus. This can be achieved by acting with *algebraic* (on functions) weights $(N,1)$ such that $N \gg 1$, namely setting up the action

$$t \cdot (x,y) = (t^N x, t y),$$

so that $t \in \mathbb{G}_m$ acts on an ideal $[I] \in \mathsf{Hilb}^n(\mathbb{A}^2)$ via

$$t \cdot I = \left\{ f(t^N x, t y) \,\middle|\, f(x,y) \in I \right\} \subset \mathbb{C}[x,y]. \tag{2.5.1}$$

We already know that the action with weights $(1,-1)$ does not make $\mathbb{A}^2 = \mathsf{Hilb}^1(\mathbb{A}^2)$ semiprojective, cf. Example 2.3.4, and one can show the same for $n > 1$ as well. However, we have the following result.

LEMMA 2.5.1. *The smooth scheme* $\mathsf{Hilb}^n(\mathbb{A}^2)$ *along with the* $\mathbb{G}_m$*-action* (2.5.1) *is semiprojective for all* $n \geqslant 1$.

Proof. Clearly there is an identity $\mathsf{Hilb}^n(\mathbb{A}^2)^{\mathbb{T}} = \mathsf{Hilb}^n(\mathbb{A}^2)^{\mathbb{G}_m}$, so properness of the fixed locus is immediate. We need to confirm that limits exist. Let $[I] \in \mathsf{Hilb}^n(\mathbb{A}^2)$ be a point. We have to extend the map

$$\begin{aligned} \mathbb{G}_m &\longrightarrow \mathsf{Hilb}^n(\mathbb{A}^2) \\ t &\longmapsto t \cdot [I] \end{aligned}$$

to $\mathbb{A}^1$.

For a monomial $x^a y^b$, set

$$\mathsf{wt}(x^a y^b) = aN + b.$$

Let $f = \sum_{a,b} c_{ab} x^a y^b \in R = \mathbb{C}[x,y]$. Define

$$\mathsf{m}_f = \min \left\{ \mathsf{wt}(x^a y^b) \,\middle|\, c_{ab} \neq 0 \right\},$$

so that

$$\begin{aligned} t \cdot f &= \sum_{a,b} c_{ab} t^{aN+b} x^a y^b = \sum_{a,b} c_{ab} t^{\mathsf{wt}(x^a y^b)} x^a y^b \\ &= t^{\mathsf{m}_f} \left(\sum_{\mathsf{wt}(x^a y^b) = \mathsf{m}_f} c_{ab} x^a y^b + \text{higher weight terms} \right). \end{aligned}$$

Define

$$\mathsf{in}_{(N,1)}(f) = \sum_{\mathsf{wt}(x^a y^b) = \mathsf{m}_f} c_{ab} x^a y^b$$

to be the *initial form* of f with respect to the weights $(N,1)$. Let now $[I] \in \mathsf{Hilb}^n(\mathbb{A}^2)$ be a closed point. Define

$$\mathsf{in}_{(N,1)}(I) = \left\langle \mathsf{in}_{(N,1)}(f) \,\middle|\, f \in I \right\rangle \subset R.$$

This is the *initial ideal* of I with respect to the weights $(N,1)$. It is this ideal that will fill in the $\mathbb{G}_m$-family we started with at the origin.

We wish to construct an ideal $\widetilde{I} \subset R[t] = R \otimes_{\mathbb{C}} \mathbb{C}[t]$ (this corresponds to a closed subscheme $\widetilde{Z} \subset \mathbb{A}^2 \times \operatorname{Spec} \mathbb{C}[t]$) such that

(1) over $t \neq 0$, the ideal $\widetilde{I}_t$ corresponds to the ideal $t \cdot I$, and

(2) $R[t]/\widetilde{I}$ is $\mathbb{C}[t]$-flat.

Consider the (exhaustive, bounded below) filtration on R given by

$$R_{\leqslant d} = \mathsf{Span}\left\{ x^a y^b \;\middle|\; aN + b \leqslant d \right\} \subset R.$$

It satisfies $R_{\leqslant 0} = \mathbb{C}$ and $R_{\leqslant d} \cdot R_{\leqslant e} \subset R_{\leqslant d+e}$. We obtain an induced filtration

$$I_{\leqslant d} = I \cap R_{\leqslant d},$$

which we use to define the *Rees ideal*

$$\widetilde{I} = \bigoplus_{d \geqslant 0} I_{\leqslant d}\, t^d \subset R[t].$$

We leave it to the reader to verify that this ideal has the desired properties. To check flatness, it is in fact enough – after having confirmed (1) – that the colength over the origin is equal to n. □

We are now ready to apply Theorem 2.3.1. Fix $n \geqslant 2$. Define the attracting cells

$$H_\lambda^+ = \left\{ I \in \mathsf{Hilb}^n(\mathbb{A}^2) \;\middle|\; \lim_{t \to 0} t \cdot I = I_\lambda \right\} \subset \mathsf{Hilb}^n(\mathbb{A}^2).$$

Combining Theorem 2.3.1 and Lemma 2.5.1, we obtain a locally closed stratification

$$\mathsf{Hilb}^n(\mathbb{A}^2) = \coprod_{\lambda \vdash n} H_\lambda^+$$

where each stratum H_λ^+ is an affine space of dimension

$$\mathrm{d}_\lambda^+ = \dim_{[I_\lambda]} T^+ \mathsf{Hilb}^n(\mathbb{A}^2).$$

In particular, we have an identity of motives

$$[\mathsf{Hilb}^n(\mathbb{A}^2)] = \sum_{\lambda \vdash n} \mathbb{L}^{\mathrm{d}_\lambda^+} \in K_0(\mathsf{Var}_{\mathbb{C}}) \tag{2.5.2}$$

for each $n \geqslant 2$.

PROPOSITION 2.5.2. *For a partition $\lambda \in \mathcal{P}$ of size at least 2, there is an identity*

$$\mathrm{d}_\lambda^+ = |\lambda| + \lambda_1,$$

where λ_1 is the number of columns in the Young diagram attached to λ (cf. Notation 1.3.20).

Proof. Let λ be a partition of n, and let $I_\lambda \subset \mathbb{C}[x, y]$ be the corresponding monomial ideal. Denote by $a(\square)$ and $l(\square)$ the arm-length and the leg-length of a box $\square \in \lambda$. The tangent space

$$T_\lambda \mathsf{Hilb}^n(\mathbb{A}^2)$$

is a $\mathbb{G}_m^2$-representation, and by Exercise 1.3.33 it admits the expression

$$T_\lambda \mathsf{Hilb}^n(\mathbb{A}^2) = \sum_{\square \in \lambda} t_1^{-l(\square)} t_2^{a(\square)+1} + t_1^{l(\square)+1} t_2^{-a(\square)}.$$

Now we restrict to the subtorus

$$\mathbb{G}_m = \left\{ (t^N, t) \;\middle|\; t \in \mathbb{C}^\times \right\} \subset \mathbb{G}_m^2$$

and we count positive tangents. Make the formal substitution $t_1 \mapsto t^N$ and $t_2 \mapsto t$. This way, we see that the first bit of the sum gives a contribution whenever $l(\square) = 0$. But the boxes with trivial leg-length are as many as the columns of the Young diagram corresponding to λ, so this bit contributes precisely λ_1. On the other hand, in the second bit of the sum every box contributes because t_1 approaches 0 much faster than t_2 and so no restriction on $a(\square)$ is present. This proves the desired formula. □

Note that, if λ is the unique partition of size 1, then $|\lambda|+\lambda_1=2$, therefore Equation (2.5.2) also works in the case $n=1$ if one *defines* d^+_λ by the relation in Proposition 2.5.2. The *actual* definition of d^+_λ would give 1! Keeping this in mind, we can now finish the calculation.

COROLLARY 2.5.3. *There are identities*

$$\mathsf{Hilb}_{\mathbb{A}^2}(t)=\prod_{m\geqslant 1}\frac{1}{1-\mathbb{L}^{m+1}t^m}=\mathsf{Exp}\left(\frac{\mathbb{L}^2 t}{1-\mathbb{L}t}\right). \tag{2.5.3}$$

Proof. We have

$$\mathsf{Hilb}_{\mathbb{A}^2}(t)=\sum_{n\geqslant 0}[\mathsf{Hilb}^n(\mathbb{A}^2)]t^n=\sum_{\lambda\in\mathcal{P}}t^{|\lambda|}\mathbb{L}^{|\lambda|+\lambda_1}$$

so the first identity reduces to the next lemma, taken with $a=b=1$. The second identity is obvious, by the very definition of Exp. □

LEMMA 2.5.4. *For every pair $(a,b)\in\mathbb{N}\times\mathbb{N}$, there is an identity*

$$\prod_{m\geqslant 1}\frac{1}{1-\mathbb{L}^{am+b}t^m}=\sum_{\lambda\in\mathcal{P}}t^{|\lambda|}\mathbb{L}^{a|\lambda|+b\lambda_1} \tag{2.5.4}$$

in the subring $A=\mathbb{Z}[\mathbb{L}][\![t]\!]\subset K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!]$.

Proof. Write $\mathsf{F}_m(t)=(1-\mathbb{L}^{am+b}t^m)^{-1}$, so that

$$\mathsf{F}(t)=\prod_{m\geqslant 1}\frac{1}{1-\mathbb{L}^{am+b}t^m}=\prod_{m\geqslant 1}\mathsf{F}_m(t).$$

Note that

$$\mathsf{F}_m(t)=\sum_{k\geqslant 0}\mathbb{L}^{k(am+b)}t^{mk}\in 1+t^mA,$$

so that $\mathsf{Coeff}_{t^N}\mathsf{F}(t)=\mathsf{Coeff}_{t^N}\mathsf{F}_1(t)\mathsf{F}_2(t)\cdots\mathsf{F}_N(t)$ for every $N\geqslant 1$. We have

$$\begin{aligned}\mathsf{Coeff}_{t^N}\mathsf{F}_1(t)\mathsf{F}_2(t)\cdots\mathsf{F}_N(t)&=\mathsf{Coeff}_{t^N}\prod_{1\leqslant m\leqslant N}\sum_{k_m\geqslant 0}\mathbb{L}^{k_m(am+b)}t^{mk_m}\\&=\sum_{\substack{k_1,\dots,k_N\geqslant 0\\ \sum_m mk_m=N}}\prod_{1\leqslant m\leqslant N}\mathbb{L}^{k_m(am+b)}\\&=\sum_{\lambda\vdash N}\mathbb{L}^{\sum_m k_m(am+b)},\end{aligned}$$

where the last identity identifies the tuple $k_\bullet$ with the partition $\lambda=(1^{k_1}\cdots m^{k_m}\cdots N^{k_N})\vdash N$. We may rewrite

$$\sum_m k_m(am+b)=a\sum_m mk_m+b\sum_m k_m=a|\lambda|+b||\lambda||,$$

and hence

$$\mathsf{F}(t)=\sum_{\lambda\in\mathcal{P}}t^{|\lambda|}\mathbb{L}^{a|\lambda|+b||\lambda||}.$$

But we are summing over all partitions, and on this set there is the involution sending a partition to its transpose λ^t (exchanging rows and columns, and preserving the size). Therefore the formula is equivalent to the sought after one. □

Example 2.5.5. Let us do a sanity check explicit example with $n=3$. Expanding (2.5.3) to order 4 we find

$$\mathsf{Hilb}_{\mathbb{A}^2}(t)=1+\mathbb{L}^2+\mathbb{L}^3(1+\mathbb{L})t^2+\mathbb{L}^4(1+\mathbb{L}+\mathbb{L}^2)t^3+\cdots$$

so we should try and confirm that

$$[\mathsf{Hilb}^3(\mathbb{A}^2)] = \mathbb{L}^4(1+\mathbb{L}+\mathbb{L}^2) = \mathbb{L}^3(\mathbb{L}+\mathbb{L}^2+\mathbb{L}^3)$$

using the tangent representation. There are 3 partitions to consider, namely

$$\ydiagram{3} \qquad \ydiagram{1,1,1} \qquad \ydiagram{2,1}$$

Let us compute $T_\lambda = T_\lambda \mathsf{Hilb}^3(\mathbb{A}^2)$ for all 3 partitions λ. For instance, the first partition has

$$\begin{aligned} a(0,0) &= 2, & l(0,0) &= 0 \\ a(1,0) &= 1, & l(1,0) &= 0 \\ a(2,0) &= 0, & l(2,0) &= 0. \end{aligned}$$

Recall from Equation (2.5.5) that we computed

$$\begin{aligned} T_{\ydiagram{3}} &= t_1 + t_2 + t_2^2 + t_2^3 + t_1 t_2^{-1} + t_1 t_2^{-2} \\ T_{\ydiagram{1,1,1}} &= t_1 + t_2 + t_1^2 + t_1^3 + t_1^{-1} t_2 + t_1^{-2} t_2 \\ T_{\ydiagram{2,1}} &= 2t_1 + 2t_2 + t_1^2 t_2^{-1} + t_1^{-1} t_2^2. \end{aligned} \tag{2.5.5}$$

Sending $t_1 \mapsto t^N$ and $t_2 \mapsto t$, we find

$$\begin{aligned} T_{\ydiagram{3}} &= (t^3 + t^{N-2}) + (t^2 + t^{N-1}) + (t + t^N) && \to \quad \mathbb{L}^6 \\ T_{\ydiagram{1,1,1}} &= (t^{1-2N} + t^{3N}) + (t^{1-N} + t^{2N}) + (t + t^N) && \to \quad \mathbb{L}^4 \\ T_{\ydiagram{2,1}} &= (t^{2-N} + t^{2N-1}) + (t + t^N) + (t + t^N) && \to \quad \mathbb{L}^5 \end{aligned}$$

which confirms the formula (the negative tangents are highlighted in red).

2.6 The motive of the Quot scheme of points

2.6.1 Power structure formula for Quot schemes

Let X be a smooth quasiprojective variety of dimension d. Let E be a rank r locally free sheaf on X. For a given integer $n \geqslant 0$, the Quot scheme

$$\mathsf{Quot}_X(E,n)$$

parametrises isomorphism classes of quotients $E \twoheadrightarrow Q$ such that

$$\dim(\operatorname{Supp} Q) = 0, \quad \chi(Q) = n.$$

The Quot-to-Chow map

$$\mathsf{Quot}_X(E,n) \xrightarrow{\sigma_n} \mathsf{Sym}^n X$$

introduced in (1.3.1) and constructed in [82, Cor. 7.15] takes a quotient $E \twoheadrightarrow Q$ to the zero-cycle (with multiplicities) determined by the set-theoretic support of Q. We define the *punctual Quot scheme* to be the preimage

$$\mathsf{Quot}_X(E,n)_p = \sigma_n^{-1}(n \cdot p)$$

of the cycle $n \cdot p \in \mathsf{Sym}^n X$, where $p \in X$ is a point. This is easily seen to only depend on a formal neighborhood of $p \in X$ (but not on p, X or E). One has isomorphisms

$$\mathsf{Quot}_X(E,n)_p \cong \mathsf{Quot}_X(\mathscr{O}_X^{\oplus r},n)_p \cong \mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}_{\mathbb{A}^d}^{\oplus r},n)_0 \tag{2.6.1}$$

where 0 is the origin in $\mathbb{A}^d$. This scheme will be denoted $P_{r,n}$ from now on, and we set

$$\mathsf{P}_{r,n} = \left[P_{r,n}\right] \in K_0(\mathsf{Var}_{\mathbb{C}}).$$

Remark 2.6.1. The *punctual motives* $\mathsf{P}_{r,n}$ clearly depend on the dimension $d = \dim X$, but we omit d from the notation.

Define the generating functions

$$\mathsf{P}_r(t) = \sum_{n \geqslant 0} \mathsf{P}_{r,n} t^n,$$
$$\mathsf{Quot}_E(t) = \sum_{n \geqslant 0} \left[\mathsf{Quot}_X(E,n)\right] t^n$$

in the power series ring $K_0(\mathsf{Var}_{\mathbb{C}})[\![t]\!]$. The following result is the higher rank analogue of the corresponding statement for the Hilbert scheme of points [46, Thm. 1], obtained by setting $r = 1$.

THEOREM 2.6.2. *Let X be a smooth quasiprojective variety. Let E be a rank r locally free sheaf on X. There is an identity*

$$\mathsf{Quot}_E(t) = \mathsf{P}_r(t)^{[X]}. \tag{2.6.2}$$

Proof. For α a partition of n, let $\mathsf{Sym}^\alpha X \subset \mathsf{Sym}^n X$ be the locally closed subvariety parameterising zero-cycles whose support is distributed according to α. We get a motivic decomposition

$$\left[\mathsf{Quot}_X(E,n)\right] = \sum_{\alpha \vdash n} \left[\mathsf{Quot}_X(E,n)_\alpha\right], \tag{2.6.3}$$

where we have set $\mathsf{Quot}_X(E,n)_\alpha = \sigma_n^{-1}(\mathsf{Sym}^\alpha X)$. By standard arguments (see e.g. [10, Sec. 4] and [79, Sec. 3]), one sees that the deepest stratum of the Quot-to-Chow map

$$\sigma_{(n)} \colon \mathsf{Quot}_X(E,n)_{(n)} \to X$$

is a Zariski locally trivial fibration with fibre $P_{r,n}$. This relies on the local case $X = \mathbb{A}^d$, where one has a global decomposition

$$\mathsf{Quot}_{\mathbb{A}^d}(\mathscr{O}^{\oplus r},n)_{(n)} \cong \mathbb{A}^d \times P_{r,n}$$

under which $\sigma_{(n)}$ is identified with the first projection.

For a fixed partition $\alpha \vdash n$, let

$$V_\alpha \hookrightarrow \prod_i \mathsf{Quot}_X(E,i)^{\alpha_i}$$

be the open subscheme parameterising finite quotients with disjoint supports. By [7, Prop. A.3], taking the union of points gives an étale map

$$u_\alpha \colon V_\alpha \longrightarrow \mathsf{Quot}_X(E,n)$$

and we let U_α denote its image. The stratum $\mathsf{Quot}_X(E,n)_\alpha$ sits inside U_α as a closed subscheme. We let the cartesian diagram

$$\begin{tikzcd} Z_\alpha \arrow[r, hook] \arrow[d, "\widetilde{u}_\alpha"'] \arrow[dr, phantom, "\square"] & V_\alpha \arrow[d, "u_\alpha"] \\ \mathsf{Quot}_X(E,n)_\alpha \arrow[r, hook] & U_\alpha \end{tikzcd} \tag{2.6.4}$$

define the scheme Z_α. The map $\widetilde{u}_\alpha$ is a finite étale cover with Galois group G_α, in particular we have

$$\mathsf{Quot}_X(E,n)_\alpha = Z_\alpha / G_\alpha. \tag{2.6.5}$$

In fact, Z_α can also be realised as the fibre product

$$\begin{tikzcd} Z_\alpha \arrow[r, hook] \arrow[d, "f_\alpha"'] \arrow[dr, phantom, "\square"] & \prod_i \mathsf{Quot}_X(E,i)_{(i)}^{\alpha_i} \arrow[d] \\ \prod_i X^{\alpha_i} \setminus \Delta \arrow[r, hook] & \prod_i X^{\alpha_i} \end{tikzcd} \tag{2.6.6}$$

where the top open immersion is the complement of the big diagonal and the map f_α is a G_α-equivariant piecewise trivial fibration with fibre $\prod_i P_{r,i}^{\alpha_i}$. This implies the identity

$$[Z_\alpha] = \left[\prod_i X^{\alpha_i} \setminus \Delta\right] \cdot \prod_i \mathsf{P}_{r,i}^{\otimes \alpha_i}$$

in $K_0^{G_\alpha}(\mathsf{Var}_{\mathbb{C}})$. Using (2.6.5), it follows that

$$\begin{aligned} \left[\mathsf{Quot}_X(E,n)_\alpha\right] &= \pi_{G_\alpha}[Z_\alpha] \\ &= \pi_{G_\alpha}\left(\left[\prod_i X^{\alpha_i} \setminus \Delta\right] \cdot \prod_i \mathsf{P}_{r,i}^{\otimes \alpha_i}\right), \end{aligned}$$

where $\pi_{G_\alpha} \colon K_0^{G_\alpha}(\mathsf{Var}_{\mathbb{C}}) \to K_0(\mathsf{Var}_{\mathbb{C}})$ is the quotient map extending (2.1.6). Since the classes $\mathsf{P}_{r,i}$ are effective, combining the decomposition (2.6.3) with the power structure formula (2.1.10) and summing over n proves the result. □

COROLLARY 2.6.3. *The series* $\mathsf{Quot}_E(t)$ *does not depend on* E*. In particular, the identity*

$$\left[\mathsf{Quot}_X(E,n)\right] = \left[\mathsf{Quot}_X(\mathscr{O}_X^{\oplus r},n)\right]$$

holds in $K_0(\mathsf{Var}_{\mathbb{C}})$ *for all locally free sheaves* E *of rank* r *on* X*.*

Definition 2.6.4. Define absolute classes $\Omega_{r,n} \in K_0(\mathsf{Var}_{\mathbb{C}})$ via

$$\mathsf{Exp}\left(\sum_{n>0} \Omega_{r,n} t^n\right) = \mathsf{P}_r(t). \tag{2.6.7}$$

Remark 2.6.5. In terms of the motivic exponential, we can rephrase Equation (2.6.2) as

$$\mathsf{Quot}_E(t) = \mathsf{Exp}\left([X] \sum_{n>0} \Omega_{r,n} t^n\right). \tag{2.6.8}$$

It is then clear that to determine the series Quot_E one has to compute the fully punctual classes $\Omega_{r,n}$.

Euler characteristics

Since the Euler characteristic

$$K_0(\mathsf{Var}_{\mathbb{C}}) \xrightarrow{\chi} \mathbb{Z}$$

is a morphism of rings with power structure, Theorem 2.6.2 yields the relation

$$\sum_{n\geqslant 0}\chi(\mathsf{Quot}_X(F,n))q^n = \left(\sum_{n\geqslant 0}\chi(\mathsf{Quot}_X(F,n)_0)q^n\right)^{\chi(X)}, \tag{2.6.9}$$

where $\mathsf{Quot}_X(F,n)_0$ is the *punctual Quot scheme*. It is easy to verify that the right hand side can be rewritten in terms of the rank 1 Quot schemes (i.e. Hilbert schemes).

Exercise 2.6.6. Let F be a locally free sheaf of rank r on a smooth quasiprojective variety X of dimension d. Show the identity

$$\sum_{n\geqslant 0}\chi(\mathsf{Quot}_X(F,n)_0)q^n = \left(\sum_{n\geqslant 0}\chi(\mathsf{Hilb}^n(X)_0)q^n\right)^r.$$

(**Hint**: first prove that $\mathsf{Quot}_X(F,n)_0 \cong \mathsf{Quot}_X(\mathscr{O}^{\oplus r},n)_0$, then use the $\mathbb{G}_m^r$-action on the latter punctual Quot scheme. You are allowed to use Bifet's theorem [13]

$$\mathsf{Quot}_X(\mathscr{O}^{\oplus r},n)_0^{\mathbb{G}_m^r} \cong \coprod_{n_1+\cdots+n_r=n}\prod_{i=1}^r \mathsf{Hilb}^{n_i}(X)_0$$

to conclude).

Combining Exercise 2.6.6 and Equation (2.6.9) with one another, we find the following result.

PROPOSITION 2.6.7. *Let F be a locally free sheaf of rank r on a smooth quasiprojective variety X of dimension d. Then*

$$\sum_{n\geqslant 0}\chi(\mathsf{Quot}_X(F,n))q^n = \left(\sum_{n\geqslant 0}\mathsf{p}_{d-1}(n)q^n\right)^{r\chi(X)}.$$

Example 2.6.8. If $\dim X \leqslant 3$, we have

$$\sum_{n\geqslant 0}\chi(\mathsf{Quot}_X(F,n))q^n = \begin{cases} (1-q)^{-r\chi(X)} & \text{if } d=1 \\ \prod_{m\geqslant 1}(1-q^m)^{-r\chi(X)} & \text{if } d=2 \\ \mathsf{M}(q)^{r\chi(X)} & \text{if } d=3 \end{cases}$$

where

$$\mathsf{M}(q) = \prod_{m\geqslant 1}(1-q^m)^{-m}$$

is the MacMahon function.

Exercise 2.6.9. Let Y be an abelian variety. Show directly (i.e. without using Example 2.6.8) that $\chi(\mathsf{Hilb}^n Y) = 0$ for all $n > 0$.

Fact 2.6.10. Assume $r = 1$. The relation

$$\mathsf{Hilb}_X(q) = \mathsf{Hilb}_{\mathbb{A}^d}(q)^{\chi(X)}$$

holds true for every smooth quasiprojective variety of dimension d, but there is no infinite product formula available for $\mathsf{Hilb}_{\mathbb{A}^d}(q)$ if $d \geqslant 4$. There also is no closed formula for the number $\mathsf{p}_1(n)$ of Young diagrams of size n, or for the number $\mathsf{p}_2(n)$ of plane partitions of size n. However, for $d = 1,2$ there are recursive formulae

$$n\mathsf{p}_d(n) = \sum_{k=0}^{n} \sigma_d(k)\mathsf{p}_d(n-k),$$

where $\sigma_d(m) = \sum_{\ell|m} \ell^d$. The formula for $d = 1$ was proved by Hardy–Ramanujan [47] in 1918, and by Erdös [34] in 1942. For the $d = 2$ case, see Example 2 in [4, Chapter 6].

2.6.2 Computation of Hilb_S

Let S be a smooth quasiprojective surface, and set $r = 1$, so that $\mathsf{Quot}_S(L,n) = \mathsf{Hilb}^n S$ for every line bundle L. We know by Formula (2.6.8) that

$$\mathsf{Hilb}_S(t) = \mathsf{Exp}\left([S]\sum_{n>0}\Omega_{1,n}t^n\right) = \prod_{n>0}(1-t^n)^{-\Omega_{1,n}[S]}.$$

On the other hand, we know that

$$\begin{aligned}\prod_{n>0}(1-t^n)^{-\Omega_{1,n}\mathbb{L}^2} &= \mathsf{Hilb}_{\mathbb{A}^2}(t)\\ &= \prod_{n>0}\left(1-\mathbb{L}^{n+1}t^n\right)^{-1}\\ &= \prod_{n>0}(1-t^n)^{-\mathbb{L}^{n+1}}.\end{aligned}$$

We deduce that

$$\Omega_{1,n} = \mathbb{L}^{n-1}, \quad n > 0. \tag{2.6.10}$$

In particular, the generating function of the motives of punctual Hilbert schemes is given by the formula

$$\mathsf{P}_1(t) = \sum_{n\geqslant 0}\left[\mathsf{Hilb}^n(\mathbb{A}^2)_0\right]t^n = \prod_{n\geqslant 1}\left(1-\mathbb{L}^{n-1}t^n\right)^{-1}, \tag{2.6.11}$$

and for an arbitrary surface S one has the relation

$$\mathsf{Hilb}_S(t) = \prod_{n>0}\left(1-\mathbb{L}^{n-1}t^n\right)^{-[S]}.$$

This is entirely computable as soon as one "knows" the motivic class of S.

2.7 Motivic DT invariants of $\mathbb{A}^3$

Let $K_0(\mathsf{St}_{\mathbb{C}})$ be the Grothendieck ring of algebraic stacks (with affine geometric stabilisers). It can be defined as the localisation of the ordinary Grothendieck ring of varieties $K_0(\mathsf{Var}_{\mathbb{C}})$ at the classes $[\mathrm{GL}_k]$ of general linear groups [18]. The invariants we want to study will live in the extended ring

$$\mathcal{M}_{\mathbb{C}} = K_0(\mathsf{St}_{\mathbb{C}})\left[\mathbb{L}^{-\frac{1}{2}}\right],$$

where $\mathbb{L} = [\mathbb{A}^1] \in K_0(\mathsf{Var}_{\mathbb{C}}) \to K_0(\mathsf{St}_{\mathbb{C}})$ is the Lefschetz motive. The main character in this section is the singular scheme

$$Q_{r,n} = \mathsf{Quot}_{\mathbb{A}^3}(\mathscr{O}^{\oplus r}, n).$$

Its description as a critical locus

$$\begin{tikzcd} Q_{r,n} \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & \mathsf{ncQuot}_{r,n} \arrow[d, "\mathrm{d}f_{r,n}"] \\ \mathsf{ncQuot}_{r,n} \arrow[r, "0"'] & T^*\mathsf{ncQuot}_{r,n} \end{tikzcd}$$

gives us the motivic Donaldson–Thomas invariant

$$\mathsf{DT}_{r,n} = -\mathbb{L}^{-(2n^2+rn)/2}\phi_{f_{r,n}} \in \mathcal{M}_{\mathbb{C}}.$$

Here, ϕ_f is the motivic vanishing cycle class attached to a regular function f (on a smooth scheme), introduced by Denef–Loeser [27].

Define the generating function

$$\mathsf{DT}_r(\mathbb{A}^3, q) = \sum_{n\geqslant 0} \mathsf{DT}_{r,n}\, q^n.$$

Our goal is to prove the following theorem.

THEOREM 2.7.1. *In $\mathcal{M}_{\mathbb{C}}[\![q]\!]$, there are identities*

$$\mathsf{DT}_r(\mathbb{A}^3, q) = \prod_{m\geqslant 1}\prod_{k=0}^{rm-1}\left(1-\mathbb{L}^{2+k-\frac{rm}{2}}q^m\right)^{-1} = \prod_{i=1}^{r}\mathsf{DT}_1(q\mathbb{L}^{\frac{-r-1}{2}+i}).$$

To prove Theorem 2.7.1, we exploit an r-framed version of motivic wall-crossing. This technique, inspired by [76, 74, 75], is also applied to small crepant resolutions of affine toric Calabi–Yau 3-folds in [1].

2.7.1 The virtual motive of a critical locus

Let U be a smooth d-dimensional $\mathbb{C}$-scheme, $f : U \to \mathbb{A}^1$ a regular function. The *virtual motive* of the critical locus $\mathsf{Crit}\, f = Z(\mathrm{d}f) \subset U$, depending on the pair (U, f), is defined in [9] as the motivic class

$$\left[\mathsf{Crit}\, f\right]_{\mathsf{vir}} = -\mathbb{L}^{-\frac{d}{2}} \cdot \phi_f \in \mathcal{M}_{\mathbb{C}}^{\hat{\mu}},$$

where $\phi_f \in K_0^{\hat{\mu}}(\mathsf{Var}_{\mathbb{C}})$ is the (absolute) motivic vanishing cycle class defined by Denef and Loeser [27]. The '$\hat{\mu}$' decoration means that we are considering $\hat{\mu}$-equivariant motives, where $\hat{\mu}$ is the group of all roots of unity. However, the motivic invariants studied here will live in the subring $\mathcal{M}_{\mathbb{C}} \subset \mathcal{M}_{\mathbb{C}}^{\hat{\mu}}$ of classes carrying the trivial $\hat{\mu}$-action.

Example 2.7.2. Set $f = 0$. Then $\mathsf{Crit}\, f = U$ and $[U]_{\mathsf{vir}} = \mathbb{L}^{-(\dim U)/2} \cdot [U]$. For instance, $[\mathrm{GL}_k]_{\mathsf{vir}} = \mathbb{L}^{-k^2/2} \cdot [\mathrm{GL}_k]$.

Remark 2.7.3. We use lambda-ring conventions on $\mathcal{M}_{\mathbb{C}}$ from [9, 24]. In particular we use the definition of $[\mathsf{Crit}\, f]_{\mathsf{vir}}$ from [9, §2.8], which differs (slightly) from the one in [74]. The difference amounts to the substitution $\mathbb{L}^{1/2} \to -\mathbb{L}^{1/2}$. The Euler number specialisation with our conventions is $\mathbb{L}^{1/2} \to -1$.

2.7.2 Quivers: framings, and motivic quantum torus

A quiver Q is a finite directed graph, determined by its sets Q_0 and Q_1 of vertices and edges, respectively, along with the maps $h, t : Q_1 \to Q_0$ specifying where an edge starts or ends. We use the notation

$$t(a)\bullet \xrightarrow{\quad a \quad} \bullet\, h(a)$$

to denote the *tail* and the *head* of an edge $a \in Q_1$.

All quivers will be assumed to be connected. The *path algebra* $\mathbb{C}Q$ of a quiver Q is defined, as a $\mathbb{C}$-vector space, by using as a $\mathbb{C}$-basis the set of all paths in the quiver, including a trivial path ϵ_i for each $i \in Q_0$. The product is defined by concatenation of paths whenever the operation is possible, and is set to be 0 otherwise. The identity element is $\sum_{i\in Q_0} \epsilon_i \in \mathbb{C}Q$.

On a quiver Q one can define the *Euler–Ringel form* $\chi_Q(-,-)\colon \mathbb{Z}^{Q_0} \times \mathbb{Z}^{Q_0} \to \mathbb{Z}$ by

$$\chi_Q(\alpha,\beta) = \sum_{i\in Q_0} \alpha_i\beta_i - \sum_{a\in Q_1} \alpha_{t(a)}\beta_{h(a)},$$

as well as the skew-symmetric form

$$\langle \alpha,\beta\rangle_Q = \chi_Q(\alpha,\beta) - \chi_Q(\beta,\alpha).$$

Definition 2.7.4 (r-framing)**.** Let Q be a quiver with a distinguished vertex $0 \in Q_0$, and let r be a positive integer. We define the quiver $\widetilde{Q}$ by adding one vertex, labelled ∞, to the original vertices in Q_0, and r edges $\infty \to 0$. We refer to $\widetilde{Q}$ as the *r-framed* quiver obtained out of $(Q,0)$.

Let Q be a quiver. Define its *motivic quantum torus* (or *twisted motivic algebra*) as

$$\mathcal{T}_Q = \prod_{\alpha\in\mathbb{N}^{Q_0}} \mathcal{M}_{\mathbb{C}} \cdot y^\alpha$$

with product

$$y^\alpha \cdot y^\beta = \mathbb{L}^{\frac{1}{2}\langle\alpha,\beta\rangle_Q}\, y^{\alpha+\beta}. \tag{2.7.1}$$

If $\widetilde{Q}$ is the r-framed quiver associated to $(Q,0)$, one has a decomposition

$$\mathcal{T}_{\widetilde{Q}} = \mathcal{T}_Q \oplus \prod_{d\geqslant 0} \mathcal{M}_{\mathbb{C}} \cdot y_\infty^d,$$

where we have set $y_\infty = y^{(1,\mathbf{0})}$. A generator $y^\alpha \in \mathcal{T}_Q$ will be identified with its image $y^{(0,\alpha)} \in \mathcal{T}_{\widetilde{Q}}$.

2.7.3 Quiver representations and their stability

Let Q be a quiver. A *representation* ρ of Q is the datum of a finite dimensional $\mathbb{C}$-vector space ρ_i for every vertex $i \in Q_0$, and a linear map $\rho(a)\colon \rho_i \to \rho_j$ for every edge $a\colon i \to j$ in Q_1. The *dimension vector* of ρ is $\underline{\dim}\,\rho = (\dim_{\mathbb{C}} \rho_i)_i \in \mathbb{N}^{Q_0}$, where $\mathbb{N} = \mathbb{Z}_{\geqslant 0}$.

Convention 2.7.5. Let Q be a quiver, $\widetilde{Q}$ its r-framing. The dimension vector of a representation $\widetilde{\rho}$ of $\widetilde{Q}$ will be denoted by (d,α), where $d = \dim_{\mathbb{C}} \widetilde{\rho}_\infty \in \mathbb{N}$ and $\alpha \in \mathbb{N}^{Q_0}$.

The space of all representations of Q with a fixed dimension vector $\alpha \in \mathbb{N}^{Q_0}$ is the affine space

$$\mathsf{Rep}(Q,\alpha) = \prod_{a\in Q_1} \mathrm{Hom}_{\mathbb{C}}(\mathbb{C}^{\alpha_{t(a)}}, \mathbb{C}^{\alpha_{h(a)}}).$$

The gauge group $\mathrm{GL}_\alpha = \prod_{i\in Q_0} \mathrm{GL}_{\alpha_i}$ acts on $\mathsf{Rep}(Q,\alpha)$ by the rule

$$(g_i)_i \cdot (\rho(a))_{a\in Q_1} = (g_{h(a)} \circ \rho(a) \circ g_{t(a)}^{-1})_{a\in Q_1}.$$

Following [74], we recall the notion of (semi)stability of a representation.

Definition 2.7.6. A *central charge* is a group homomorphism $\mathrm{Z}\colon \mathbb{Z}^{Q_0} \to \mathbb{C}$ such that the image of $\mathbb{N}^{Q_0} \setminus 0$ lies inside $\mathbb{H}_+ = \{ re^{\sqrt{-1}\pi\varphi} \mid r > 0, 0 < \varphi \leqslant 1 \}$. For every $\alpha \in \mathbb{N}^{Q_0} \setminus 0$, we denote by $\varphi(\alpha)$ the real number φ such that $\mathrm{Z}(\alpha) = re^{\sqrt{-1}\pi\varphi}$. It is called the *phase* of α with respect to Z.

Note that every vector $\zeta \in \mathbb{R}^{Q_0}$ induces a central charge Z_ζ if we set $\mathrm{Z}_\zeta(\alpha) = -\zeta \cdot \alpha + |\alpha|\sqrt{-1}$, where $|\alpha| = \sum_{i \in Q_0} \alpha_i$. We denote by φ_ζ the induced phase function, and we set $\varphi_\zeta(\rho) = \varphi_\zeta(\underline{\dim}\,\rho)$ for every representation ρ of Q.

Definition 2.7.7. Fix $\zeta \in \mathbb{R}^{Q_0}$. A representation ρ of Q is called *ζ-semistable* if

$$\varphi_\zeta(\rho') \leqslant \varphi_\zeta(\rho)$$

for every nonzero proper subrepresentation $0 \neq \rho' \subsetneq \rho$. If '$\leqslant$' can be replaced by '$<$', we say that ρ is *ζ-stable*. Vectors $\zeta \in \mathbb{R}^{Q_0}$ are referred to as *stability parameters*.

Definition 2.7.8. Let $\alpha \in \mathbb{N}^{Q_0}$ be a dimension vector. A stability parameter ζ is called *α-generic* if for any $0 < \beta < \alpha$ one has $\varphi_\zeta(\beta) \neq \varphi_\zeta(\alpha)$.

The sets of ζ-stable and ζ-semistable representations with given dimension vector α form a chain of open subsets

$$\mathsf{Rep}^{\zeta\text{-}\mathsf{st}}(Q,\alpha) \subset \mathsf{Rep}^{\zeta\text{-}\mathsf{ss}}(Q,\alpha) \subset \mathsf{Rep}(Q,\alpha).$$

If ζ is α-generic, one has $\mathsf{Rep}^{\zeta\text{-}\mathsf{st}}(Q,\alpha) = \mathsf{Rep}^{\zeta\text{-}\mathsf{ss}}(Q,\alpha)$.

2.7.4 Quivers with potential

Let Q be a quiver. Consider the quotient $\mathbb{C}Q/[\mathbb{C}Q,\mathbb{C}Q]$ of the path algebra by the commutator ideal. A finite linear combination of cyclic paths $W \in \mathbb{C}Q/[\mathbb{C}Q,\mathbb{C}Q]$ is called a *potential*. Given a cyclic path w and an arrow $a \in Q_1$, one defines the noncommutative derivative

$$\frac{\partial w}{\partial a} = \sum_{\substack{w = cac' \\ c,c' \text{ paths in } Q}} c'c \in \mathbb{C}Q.$$

This rule extends to an operator $\partial/\partial a$ acting on every potential. The *Jacobi algebra* $J = J_{Q,W}$ of (Q,W) is the quotient of $\mathbb{C}Q$ by the two-sided ideal generated by $\partial W/\partial a$ for all edges $a \in Q_1$. For every $\alpha \in \mathbb{N}^{Q_0}$, the potential $W = \sum_c a_c c$ determines a regular function

$$f_\alpha\colon \mathsf{Rep}(Q,\alpha) \to \mathbb{A}^1, \quad \rho \mapsto \sum_{c \text{ cycle in } Q} a_c \operatorname{Tr}\big(\rho(c)\big).$$

The points in the critical locus $\mathsf{Crit}\, f_\alpha \subset \mathsf{Rep}(Q,\alpha)$ correspond to α-dimensional *J-modules*.

Fix an α-generic stability parameter $\zeta \in \mathbb{R}^{Q_0}$. If $f_{\zeta,\alpha}\colon \mathsf{Rep}^{\zeta\text{-}\mathsf{st}}(Q,\alpha) \to \mathbb{A}^1$ is the restriction of f_α, then

$$\mathfrak{M}(J,\alpha) = [\mathsf{Crit}\, f_\alpha / G_\alpha], \quad \mathfrak{M}_\zeta(J,\alpha) = [\mathsf{Crit}\, f_{\zeta,\alpha} / \mathrm{GL}_\alpha]$$

are, by definition, the stacks of α-dimensional J-modules and ζ-stable J-modules.

Definition 2.7.9. A quiver with potential (Q,W) admits a *cut* if there is a subset $I \subset Q_1$ such that every cyclic monomial appearing in W contains exactly one edge in I.

From now on we assume (Q,W) admits a cut. This condition ensures that the motive $[\mathfrak{M}(J,\alpha)]_{\mathsf{vir}}$ introduced in the next definition lives in $\mathcal{M}_{\mathbb{C}}$. The 3-loop quiver L_3 with the cubic potential $W = A_3[A_1,A_2]$ has $I = \{A_3\}$ as a cut. We go back to this in Example 2.7.15.

Definition 2.7.10 ([74])**.** We define motivic Donaldson–Thomas invariants

$$[\mathfrak{M}(J,\alpha)]_{\mathsf{vir}} = \frac{[\mathsf{Crit}\, f_\alpha]_{\mathsf{vir}}}{[\mathrm{GL}_\alpha]_{\mathsf{vir}}}, \qquad [\mathfrak{M}_\zeta(J,\alpha)]_{\mathsf{vir}} = \mathbb{L}^{\frac{1}{2}\chi(\alpha,\alpha)} \frac{[f_{\zeta,\alpha}^{-1}(0)]-[f_{\zeta,\alpha}^{-1}(1)]}{[\mathrm{GL}_\alpha]}$$

in $\mathcal{M}_{\mathbb{C}}$, where $[\mathrm{GL}_\alpha]_{\mathsf{vir}}$ is taken as in Example 2.7.2. The generating function

$$A_U = \sum_{\alpha\in\mathbb{N}^{Q_0}} [\mathfrak{M}(J,\alpha)]_{\mathsf{vir}} \cdot y^\alpha \in \mathcal{T}_Q \tag{2.7.2}$$

is called the *universal series* of (Q,W).

Definition 2.7.11. A stability parameter $\zeta\in\mathbb{R}^{Q_0}$ is called *generic* if $\zeta\cdot\underline{\dim}\,\rho\neq 0$ for every nontrivial ζ-stable J-module ρ.

2.7.5 Framed motivic DT invariants

Let Q be a quiver, $r\geqslant 1$ be an integer, and consider its r-framing $\widetilde{Q}$ with respect to a vertex $0\in Q_0$ (Definition 2.7.4). A representation $\widetilde{\rho}$ of $\widetilde{Q}$ can be uniquely written as a pair (u,ρ), where ρ is a representation of Q and $u=(u_1,\dots,u_r)$ is an r-tuple of linear maps $u_i\colon\widetilde{\rho}_\infty\to\rho_0$. From now on, we assume our framed representations to satisfy $\dim_{\mathbb{C}}\widetilde{\rho}_\infty=1$, so that according to Convention 2.7.5 we can write $\underline{\dim}\,\widetilde{\rho}=(1,\underline{\dim}\,\rho)$. We also view ρ as a subrepresentation of $\widetilde{\rho}$ of dimension $(0,\underline{\dim}\,\rho)$, based at the vertex $0\in Q_0$.

Definition 2.7.12. Fix $\zeta\in\mathbb{R}^{Q_0}$. A representation (u,ρ) of $\widetilde{Q}$ (resp. a $\widetilde{J}$-module) with $\dim_{\mathbb{C}}\widetilde{\rho}_\infty=1$ is said to be ζ-*(semi)stable* if it is (ζ_∞,ζ)-(semi)stable in the sense of Definition 2.7.7, where $\zeta_\infty=-\zeta\cdot\underline{\dim}\,\rho$.

We now define motivic DT invariants for moduli stacks of r-framed representations of a given quiver Q. Fix a potential W on Q. Let $\widetilde{Q}$ be the r-framing of Q at a given vertex $0\in Q_0$, and let $\widetilde{J}$ be the Jacobi algebra $J_{\widetilde{Q},W}$, where W is viewed as a potential on $\widetilde{Q}$ in the obvious way. For a generic stability parameter $\zeta\in\mathbb{R}^{Q_0}$, and an arbitrary dimension vector $\alpha\in\mathbb{N}^{Q_0}$, set

$$\zeta_\infty=-\zeta\cdot\alpha, \quad \widetilde{\zeta}=(\zeta_\infty,\zeta), \quad \widetilde{\alpha}=(1,\alpha).$$

As in §2.7.4, consider the trace map $f_{\widetilde{\alpha}}\colon\mathsf{Rep}(\widetilde{Q},\widetilde{\alpha})\to\mathbb{A}^1$, induced by W, and its restriction to the framed-stable locus $f_{\widetilde{\zeta},\widetilde{\alpha}}\colon\mathsf{Rep}^{\widetilde{\zeta}\text{-st}}(\widetilde{Q},\widetilde{\alpha})\to\mathbb{A}^1$. Define the moduli stacks

$$\mathfrak{M}(\widetilde{J},\alpha)=\big[\mathsf{Crit}\, f_{\widetilde{\alpha}}\big/\mathrm{GL}_\alpha\big], \quad \mathfrak{M}_\zeta(\widetilde{J},\alpha)=\big[\mathsf{Crit}\, f_{\widetilde{\zeta},\widetilde{\alpha}}\big/\mathrm{GL}_\alpha\big].$$

Note that we are not quotienting by $\mathrm{GL}_{\widetilde{\alpha}}=\mathrm{GL}_\alpha\times\mathbb{C}^\times$, but only by GL_α.

Definition 2.7.13. We define r-framed motivic Donaldson–Thomas invariants

$$[\mathfrak{M}(\widetilde{J},\alpha)]_{\mathsf{vir}} = \frac{[\mathsf{Crit}\, f_{\widetilde{\alpha}}]_{\mathsf{vir}}}{[\mathrm{GL}_\alpha]_{\mathsf{vir}}}, \qquad [\mathfrak{M}_\zeta(\widetilde{J},\alpha)]_{\mathsf{vir}} = \frac{[\mathsf{Crit}\, f_{\widetilde{\zeta},\widetilde{\alpha}}]_{\mathsf{vir}}}{[\mathrm{GL}_\alpha]_{\mathsf{vir}}} \tag{2.7.3}$$

in $\mathcal{M}_{\mathbb{C}}$, and the associated motivic generating functions

$$\widetilde{A}_U = \sum_{\alpha\in\mathbb{N}^{Q_0}} [\mathfrak{M}(\widetilde{J},\alpha)]_{\mathsf{vir}} \cdot y^{\widetilde{\alpha}} \in \mathcal{T}_{\widetilde{Q}}$$
$$\mathsf{Z}_\zeta = \sum_{\alpha\in\mathbb{N}^{Q_0}} [\mathfrak{M}_\zeta(\widetilde{J},\alpha)]_{\mathsf{vir}} \cdot y^{\widetilde{\alpha}} \in \mathcal{T}_{\widetilde{Q}}.$$

The fact that the r-framed invariants (2.7.3) live in $\mathcal{M}_{\mathbb{C}}$ (i.e. have no monodromy) follows from [74, Lemma 1.10]. The reason is that the dimension vector $\widetilde{\alpha}=(1,\alpha)$ contains '1' as a component.

2.7.6 Dimensional reduction

If I is a cut for (Q,W), one can define a new quiver $Q_I=(Q_0,Q_1\setminus I)$. Let $J_{W,I}$ be the quotient of $\mathbb{C}Q_I$ by the two-sided ideal generated by the noncommutative derivatives $\partial W/\partial a$ for $a\in I$. Let $\mathsf{Rep}(J_{W,I},\alpha)\subset\mathsf{Rep}(Q_I,\alpha)$ be the space of $J_{W,I}$-modules of dimension vector α. Then one has the following dimensional reduction principle.

PROPOSITION 2.7.14 ([74, Prop. 1.15]). *Suppose I is a cut for (Q,W). Set $\mathrm{d}_I(\alpha)=\sum_{a\in I}\alpha_{t(a)}\alpha_{h(a)}$. Then*

$$A_U=\sum_{\alpha\in\mathbb{N}^{Q_0}}\mathbb{L}^{\frac{1}{2}\chi_Q(\alpha,\alpha)+\mathrm{d}_I(\alpha)}\frac{[\mathsf{Rep}(J_{W,I},\alpha)]}{[\mathrm{GL}_\alpha]}\cdot y^\alpha.$$

Example 2.7.15. Let $Q=L_3$ be the 3-loop quiver (see Figure 2.1, and remove the framing vertex to obtain a picture of this quiver) with the potential $W=A_3[A_1,A_2]$. Notice that $J=J_{L_3,W}=\mathbb{C}[x,y,z]$, and $I=\{A_3\}$ is a cut for (L_3,W). The quiver Q_I is the 2-loop quiver and $J_{W,I}=\mathbb{C}[x,y]$. We have $\mathrm{d}_I(n)=n^2$ and $\chi_Q(n,n)=-2n^2$. Therefore Proposition 2.7.14 yields an identity

$$\sum_{n\geqslant 0}[\mathfrak{M}(J,n)]_{\mathsf{vir}}\cdot y^n=\sum_{n\geqslant 0}\frac{[C_n]}{[\mathrm{GL}_n]}\cdot y^n=\prod_{m\geqslant 1}\prod_{k\geqslant 1}\left(1-\mathbb{L}^{2-k}y^m\right)^{-1}, \tag{2.7.4}$$

where $\mathsf{Rep}(J_{W,I},n)$ is identified with the *commuting variety*

$$C_n=\left\{(A_1,A_2)\in\mathrm{End}_{\mathbb{C}}(\mathbb{C}^n)^{\oplus 2}\,\middle|\,[A_1,A_2]=0\right\}\subset\mathrm{End}_{\mathbb{C}}(\mathbb{C}^n)^{\oplus 2},$$

and the second identity in (2.7.4) is the Feit–Fine formula [40, 9, 20].

2.7.7 Stability on the framed 3-loop quiver

The main character in this section is the framed quiver $\widetilde{L}_3$ of Figure 2.1, which we equip with the potential $W=A_3[A_1,A_2]$.

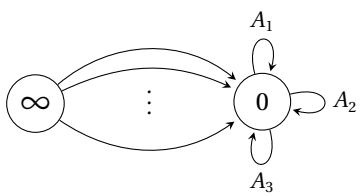


Figure 2.1: The r-framed 3-loop quiver $\widetilde{L}_3$.

Definition 2.7.16. Let $\widetilde{\rho}=(u,\rho)$ be a representation of $\widetilde{L}_3$ of dimension $(1,n)$. We denote by $\langle u,\rho\rangle\subset\widetilde{\rho}$ the smallest subrepresentation of $\widetilde{\rho}$ containing $u(\rho_\infty)$. More precisely, if $\rho=(A_1,A_2,A_3)\in\mathsf{Rep}(L_3,n)$, then $\langle u,\rho\rangle$ is the subrepresentation of $\widetilde{\rho}$ with $\langle u,\rho\rangle_\infty=\widetilde{\rho}_\infty=\mathbb{C}$ and

$$\langle u,\rho\rangle_0=\mathrm{span}_{\mathbb{C}}\left\{A_1^{a_1}A_2^{a_2}A_3^{a_3}\cdot u_\ell(1)\,\middle|\,a_i\geqslant 0,1\leqslant\ell\leqslant r\right\}\subset\widetilde{\rho}_0,$$

and with linear maps induced naturally by those defined by $\widetilde{\rho}$.

From now on we identify the space of stability parameters for L_3 with $\mathbb{R}$.

LEMMA 2.7.17. *Let $\zeta\in\mathbb{R}$ be a stability parameter, and let $\widetilde{\rho}=(u,\rho)$ be a representation of $\widetilde{L}_3$ of dimension $(1,n)$. Set $\widetilde{\zeta}=(-n\zeta,\zeta)$. Then:*

1. *if $\zeta < 0$, $\widetilde{\rho}$ is ζ-semistable if and only if it is ζ-stable if and only if $\widetilde{\rho} = \langle u, \rho \rangle$;*

2. *if $\zeta = 0$, $\widetilde{\rho}$ is ζ-semistable;*

3. *if $\zeta > 0$, $\widetilde{\rho}$ is ζ-semistable if and only if it is ζ-stable if and only if $n = 0$.*

Proof. For the case $\zeta < 0$ we refer to [7, Prop. 2.4]. Consider the case $\zeta > 0$. If we had $n = \dim_{\mathbb{C}} \widetilde{\rho}_0 > 0$, then $\rho \subset \widetilde{\rho}$ would be destabilising, for $\mathrm{Z}_{\widetilde{\zeta}}(0, n) = -n\zeta + n\sqrt{-1}$ implies $\varphi_{\widetilde{\zeta}}(\rho) > 1/2 = \varphi_{\widetilde{\zeta}}(\widetilde{\rho})$, since $\mathrm{Z}_{\widetilde{\zeta}}(1, n) = (n+1)\sqrt{-1}$ has vanishing real part. On the other hand, if $n = 0$ then $\widetilde{\rho}$ is simple and hence ζ-stable. In the case $\zeta = 0$ there is nothing to prove, as all representations have phase $1/2$. □

Consider the following regions of the space of stability parameters $\mathbb{R}$:

- $\Omega_+ = \{\zeta \in \mathbb{R} \mid \zeta < 0\}$,
- $\Omega_0 = \{\zeta \in \mathbb{R} \mid \zeta = 0\}$,
- $\Omega_- = \{\zeta \in \mathbb{R} \mid \zeta > 0\}$.

By Lemma 2.7.17 the space of stability parameters on $\widetilde{L}_3$ admits a particularly simple wall-and-chamber decomposition $\mathbb{R} = \Omega_+ \amalg \Omega_0 \amalg \Omega_-$ with one wall (the origin) and two chambers.

2.7.8 The virtual motive of the Quot scheme of points

Let $\widetilde{L}_3$ be the r-framed 3-loop quiver (Figure 2.1), and fix the potential $W = A_3[A_1, A_2]$. Fix a stability parameter $\zeta^+ \in \Omega_+ = \mathbb{R}_{<0}$. Fixing $n \geqslant 0$ and setting $\widetilde{\zeta}^+ = (-n\zeta, \zeta)$, $\widetilde{n} = (1, n)$, the quotient stack

$$\mathfrak{M}_{\zeta^+}(\widetilde{L}_3, n) = \left[\mathsf{Rep}^{\widetilde{\zeta}^+\text{-st}}(\widetilde{L}_3, \widetilde{n}) \big/ \mathrm{GL}_n\right]$$

is a smooth quasi-projective *variety* of dimension $2n^2 + rn$, called the *noncommutative Quot scheme* in [7]. The regular function $f_n \colon \mathsf{Rep}(\widetilde{L}_3, \widetilde{n}) \to \mathbb{A}^1$ given by taking the trace of W descends to a regular function on $\mathfrak{M}_{\zeta^+}(\widetilde{L}_3, n)$, still denoted f_n. We have the following description of the Quot scheme of length n quotients of $\mathscr{O}_{\mathbb{A}^3}^{\oplus r}$.

PROPOSITION 2.7.18 ([7, Thm. 2.6]). *There is an identity of closed subschemes*

$$\mathsf{Quot}_{\mathbb{A}^3}(\mathscr{O}^{\oplus r}, n) = \mathsf{Crit}(f_n) \subset \mathfrak{M}_{\zeta^+}(\widetilde{L}_3, n).$$

Thanks to Proposition 2.7.18, we can form the virtual motives of the Quot scheme, as in §2.7.1, and define their generating function

$$\mathsf{DT}_r(\mathbb{A}^3, q) = \sum_{n \geqslant 0} \left[\mathsf{Quot}_{\mathbb{A}^3}(\mathscr{O}^{\oplus r}, n)\right]_{\mathsf{vir}} \cdot q^n \in \mathcal{M}_{\mathbb{C}}[\![q]\!]. \tag{2.7.5}$$

Remark 2.7.19. The main result of [9] is the formula

$$\mathsf{DT}_1(\mathbb{A}^3, q) = \prod_{m \geqslant 1} \prod_{k=0}^{m-1} \left(1 - \mathbb{L}^{2+k-\frac{m}{2}} q^m\right)^{-1}.$$

The series $\mathsf{DT}_1(Y, q)$ studied in [9] for an arbitrary smooth 3-fold Y also appeared in [24] as the wall-crossing factor in the motivic DT/PT correspondence based at a fixed smooth curve $C \subset Y$ in Y. This correspondence refined its enumerative counterpart [79, 78]. The same phenomenon occurred in [74, 75] in the context of framed motivic DT invariants. See [80, § 4] for a generalisation $\mathsf{DT}_r(Y, q)$ of (2.7.5) to an arbitrary smooth 3-fold Y.

The following consideration will constitute the final step in proving Theorem 2.7.1, so we anticipate it here.

LEMMA 2.7.20. *There is an identity*

$$\prod_{m\geqslant 1}\prod_{k=0}^{rm-1}\left(1-\mathbb{L}^{2+k-\frac{rm}{2}}q^m\right)^{-1}=\prod_{i=1}^{r}\mathsf{DT}_1\left(\mathbb{A}^3,q\mathbb{L}^{\frac{-r-1}{2}+i}\right). \tag{2.7.6}$$

Proof. The claimed identity follows from a simple manipulation:

$$\begin{aligned}\prod_{i=1}^{r}\mathsf{DT}_1(\mathbb{A}^3,q\mathbb{L}^{\frac{-r-1}{2}+i})&=\prod_{i=1}^{r}\prod_{m\geqslant 1}\prod_{k=0}^{m-1}\left(1-\mathbb{L}^{2+k-\frac{m}{2}}\mathbb{L}^{\frac{-r-1}{2}m+im}q^m\right)^{-1}\\&=\prod_{m\geqslant 1}\prod_{k=0}^{m-1}\prod_{i=1}^{r}\left(1-\mathbb{L}^{2+k+(i-1)m-\frac{rm}{2}}q^m\right)^{-1}\\&=\prod_{m\geqslant 1}\prod_{k=0}^{rm-1}\left(1-\mathbb{L}^{2+k-\frac{rm}{2}}q^m\right)^{-1}.\end{aligned}$$

□

2.7.9 Calculation via wall-crossing

In this subsection we prove Theorem 2.7.1. Consider the universal generating function

$$\widetilde{A}_U=\sum_{n\geqslant 0}\left[\mathfrak{M}(\widetilde{J},n)\right]_{\mathsf{vir}}\cdot y^{(1,n)}\in\mathcal{T}_{\widetilde{L}_3}$$

as an element of the motivic quantum torus. To (generic) stability parameters $\zeta^{\pm}\in\Omega_{\pm}$ we associate elements (cf. Definition 2.7.13)

$$\mathsf{Z}_{\zeta^{\pm}}=\sum_{n\geqslant 0}\left[\mathfrak{M}_{\zeta^{\pm}}(\widetilde{J},n)\right]_{\mathsf{vir}}\cdot y^{(1,n)}\in\mathcal{T}_{\widetilde{L}_3}.$$

By Lemma 2.7.17(3), we have an identity

$$\mathsf{Z}_{\zeta^-}=y_\infty=y^{(1,0)}, \tag{2.7.7}$$

whereas the series Z_{ζ^+} is, essentially, a "shift" of the generating function $\mathsf{DT}_r(\mathbb{A}^3,y^{(0,1)})$. More precisely, by Proposition 2.7.18 we have an identification

$$\mathfrak{M}_{\zeta^+}(\widetilde{J},n)=\mathsf{Quot}_{\mathbb{A}^3}(\mathcal{O}^{\oplus r},n)\subset\mathfrak{M}_{\zeta^+}(\widetilde{L}_3,n)$$

of critical loci sitting inside the noncommutative Quot scheme; in particular, the associated virtual motives are the same. The shift is intended in the following sense: since $\langle(0,n),(1,0)\rangle_{\widetilde{L}_3}=rn$, the product rule (2.7.1) yields the identity

$$y^{(1,n)}=\mathbb{L}^{-\frac{rn}{2}}y^{(0,n)}\cdot y_\infty\in\mathcal{T}_{\widetilde{L}_3}. \tag{2.7.8}$$

Since we can express $y^{(0,n)}$ as the n-fold product of $y^{(0,1)}$ with itself, we obtain

$$\mathsf{Z}_{\zeta^+}=\mathsf{DT}_r\left(\mathbb{A}^3,\mathbb{L}^{-\frac{r}{2}}y^{(0,1)}\right)\cdot y_\infty. \tag{2.7.9}$$

The last generating function we need to analyse is

$$A_U=\sum_{n\geqslant 0}\left[\mathfrak{M}(J,n)\right]_{\mathsf{vir}}\cdot y^{(0,n)}\in\mathcal{T}_{L_3}\subset\mathcal{T}_{\widetilde{L}_3},$$

whose n-th coefficient is the virtual motive of the stack of 0-dimensional $\mathbb{C}[x,y,z]$-modules of length n. This was already computed in Example 2.7.15:

$$A_U(y) = \prod_{m\geqslant 1}\prod_{k\geqslant 1}\left(1-\mathbb{L}^{2-k}y^m\right)^{-1}. \tag{2.7.10}$$

The next ingredient of the proof is a particular instance of Mozgovoy's motivic wall-crossing formula [76].

PROPOSITION 2.7.21. *In $\mathcal{T}_{\widetilde{L}_3}$, there are identities*

$$\mathsf{Z}_{\zeta^+}\cdot A_U = \widetilde{A}_U = A_U\cdot \mathsf{Z}_{\zeta^-}. \tag{2.7.11}$$

Proof. Let $\widetilde{\rho}=(u,\rho)$ be a $\widetilde{J}$-module. Consider $\zeta^+\in\Omega_+$, and let $\langle u,\rho\rangle\subset\widetilde{\rho}$ be the submodule introduced in Definition 2.7.16. We have that $\langle u,\rho\rangle$ is ζ^+-stable by Lemma 2.7.17(1) and the quotient $\widetilde{\rho}/\langle u,\rho\rangle$ is supported at the vertex 0. From this we deduce the decomposition $\widetilde{A}_U=\mathsf{Z}_{\zeta^+}\cdot A_U$. Consider now $\zeta^-\in\Omega_-$. The quotient of $\widetilde{\rho}$ by the submodule ρ based at the vertex 0 is the simple module supported at the framing vertex ∞. By Lemma 2.7.17(3) this is the unique ζ^--stable module for the current choice of ζ^-, so we obtain the decomposition $\widetilde{A}_U = A_U\cdot\mathsf{Z}_{\zeta^-}$. □

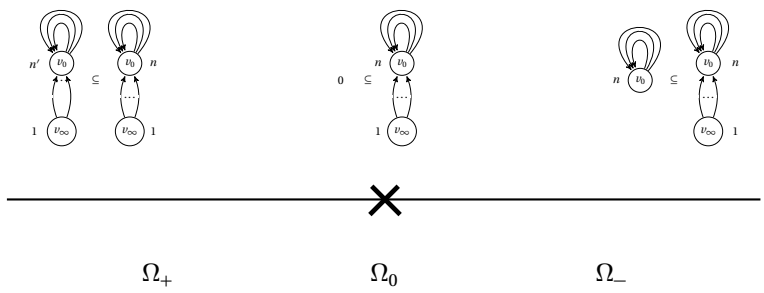


Figure 2.2: An illustration of Proposition 2.7.21.

Note that we have an identity

$$y_\infty\cdot y^{(0,n)} = \mathbb{L}^{-\frac{rn}{2}}\cdot y^{(1,n)} = \mathbb{L}^{-rn}\cdot y^{(0,n)}\cdot y_\infty,$$

where we have used (2.7.8) for the second equality. By Formula (2.7.9), the left-hand term of Formula (2.7.11) can then be rewritten as

$$\begin{aligned}\mathsf{DT}_r\left(\mathbb{A}^3,\mathbb{L}^{-\frac{r}{2}}y^{(0,1)}\right)\cdot\sum_{n\geqslant 0}\left[\mathfrak{M}(J,n)\right]_{\mathsf{vir}}\cdot y_\infty\cdot y^{(0,n)}&\\ = \mathsf{DT}_r\left(\mathbb{A}^3,\mathbb{L}^{-\frac{r}{2}}y^{(0,1)}\right)\cdot\sum_{n\geqslant 0}\left[\mathfrak{M}(J,n)\right]_{\mathsf{vir}}\mathbb{L}^{-rn}\cdot y^{(0,n)}\cdot y_\infty.&\end{aligned}$$

Therefore, by Equation (2.7.7), identifying the two expressions for $\widetilde{A}_U$ in Equation (2.7.11) yields

$$\mathsf{DT}_r\left(\mathbb{A}^3,\mathbb{L}^{-\frac{r}{2}}y^{(0,1)}\right)\cdot A_U\left(\mathbb{L}^{-r}y^{(0,1)}\right)\cdot y_\infty = A_U\left(y^{(0,1)}\right)\cdot y_\infty,$$

which is equivalent to

$$\mathsf{DT}_r\left(\mathbb{A}^3,\mathbb{L}^{-\frac{r}{2}}y^{(0,1)}\right) = \frac{A_U\left(y^{(0,1)}\right)}{A_U\left(\mathbb{L}^{-r}y^{(0,1)}\right)}.$$

Setting $q = \mathbb{L}^{-\frac{r}{2}} y^{(0,1)}$, and using Equation (2.7.10), a simple substitution yields

$$\begin{aligned}
\mathsf{DT}_r(\mathbb{A}^3, q) &= \frac{A_U\left(\mathbb{L}^{\frac{r}{2}} q\right)}{A_U\left(\mathbb{L}^{-\frac{r}{2}} q\right)} \\
&= \prod_{m\geqslant 1}\prod_{j\geqslant 0} \frac{\left(1-\mathbb{L}^{1-j+\frac{rm}{2}} q^m\right)^{-1}}{\left(1-\mathbb{L}^{1-j-\frac{rm}{2}} q^m\right)^{-1}} \\
&= \prod_{m\geqslant 1}\prod_{j=0}^{rm-1} \left(1-\mathbb{L}^{1-j+\frac{rm}{2}} q^m\right)^{-1} \\
&= \prod_{m\geqslant 1}\prod_{k=0}^{rm-1} \left(1-\mathbb{L}^{2+k-\frac{rm}{2}} q^m\right)^{-1}.
\end{aligned}$$

This is what we wanted. By Lemma 2.7.20, Theorem 2.7.1 is fully proved.

Remark 2.7.22. The generating function $\mathsf{DT}_r(\mathbb{A}^3, (-1)^r q)$ admits the plethystic expression

$$\mathsf{DT}_r(\mathbb{A}^3, (-1)^r q) = \mathsf{Exp}\left(\frac{(-1)^r q \mathbb{L}^{\frac{3}{2}}}{\left(1-(-\mathbb{L}^{-\frac{1}{2}})^r q\right)\left(1-(-\mathbb{L}^{\frac{1}{2}})^r q\right)} \frac{\mathbb{L}^{-\frac{r}{2}} - \mathbb{L}^{\frac{r}{2}}}{\mathbb{L}^{-\frac{1}{2}} - \mathbb{L}^{\frac{1}{2}}}\right).$$

This is exploited in [80, § 4] to define $[\mathsf{Quot}_Y(F, n)]_{\mathsf{vir}}$ for all 3-folds Y and locally free sheaves F over Y.

Remark 2.7.23. The Euler number specialisation $\mathbb{L}^{1/2} \to -1$ applied to Formula (**??**) yields a formula for the generating function of *virtual Euler characteristics* of Quot schemes,

$$\sum_{n\geqslant 0} \chi_{\mathsf{vir}}(\mathsf{Quot}_{\mathbb{A}^3}(\mathscr{O}^{\oplus r}, n)) \cdot q^n = \mathsf{M}((-1)^r q)^r,$$

where $\mathsf{M}(q) = \prod_{m\geqslant 1}(1-q^m)^{-m}$ is the MacMahon function, the generating function of plane partitions, and where $\chi_{\mathsf{vir}}(-) = \chi(-, \nu)$ is the Euler characteristic weighted by Behrend's microlocal function [8]. The above identity can be seen as the Calabi–Yau specialisation (i.e. the specialisation $s_1 + s_2 + s_3 = 0$) of the generating function of *cohomological Donaldson–Thomas invariants* of $\mathbb{A}^3$,

$$\sum_{n\geqslant 0} \mathsf{DT}_r^{\mathsf{coh}}(\mathbb{A}^3, n) \cdot q^n = \mathsf{M}((-1)^r q)^{-r\frac{(s_1+s_2)(s_1+s_3)(s_2+s_3)}{s_1 s_2 s_3}} \in \mathbb{Q}(\!(s_1, s_2, s_3)\!)[\![q]\!],$$

obtained in [39, Thm. B] (as a higher rank version of [66, Thm. 1]), where s_1, s_2 and s_3 are the equivariant parameters of the torus $\mathbb{T} = \mathbb{G}_m^3$ acting on the Quot scheme.

3 | Localisation techniques

3.1 Equivariant cohomology

Equivariant cohomology was introduced by Borel in his seminar on trasformation groups [15]. The goal of this section is to introduce the framework necessary to state the Atiyah–Bott *localisation formula* [16, 6]. This will be done in Section 3.2. We shall see how the formula works via concrete examples in Section 3.3. See [3, 68] for the theory of localisation in the context of algebraic geometry, [89, 30] for extensions from cohomology to Chow, and see also [33] for the first appearance of equivariant cohomology in enumerative geometry.

3.1.1 Universal principal bundles and classifying spaces

Classifying spaces in topology

We work, for now, in the category Top of Hausdorff and paracompact topological spaces. A topological group is a group object G in Top. Given a topological group G, there is a category of (left, or right) G-*spaces*, i.e. pairs (X,σ) where $X \in \mathsf{Top}$ and σ is a (left, or right) continuous G-action. Morphisms in this category are the G-equivariant maps between the underlying topological spaces. For instance, a morphism of left G-spaces $(X,\sigma_X) \to (Y,\sigma_Y)$ is a continuous map $\alpha\colon X \to Y$ such that the diagram

$$\begin{tikzcd} G \times X \arrow[r, "\sigma_X"] \arrow[d, "\mathrm{id}\times\alpha"'] & X \arrow[d, "\alpha"] \\ G \times Y \arrow[r, "\sigma_Y"] & Y \end{tikzcd}$$

commutes. A G-*homotopy* between G-equivariant maps $\alpha,\beta\colon X \rightrightarrows Y$ is defined to be a G-equivariant homotopy $f\colon [0,1]\times X \to Y$, where G acts trivially on the interval.

Let S be a topological space, E another topological space equipped with a (fixed) right G-action and with a G-invariant map $p\colon E \to S$ (i.e. p is G-equivariant, with respect to the trivial G-action on S). We say that (E,p) is a *principal G-bundle* over S if p is furthermore G-equivariantly locally trivial, i.e. S has an open covering $\{S_i\}_i$ such that on each S_i there are G-equivariant homeomorphisms

$$p^{-1}(S_i) \xrightarrow{\sim} S_i \times G$$

where G acts on $S_i \times G$ by $(s,h)\cdot g = (s,hg)$. This implies that the initial right G-action on E is *free*, and moreover there is a canonical homeomorphism

$$E/G \xrightarrow{\sim} S,$$

where E/G is the topological orbit space induced by the G-action $E \times G \to E$. Every fibre $p^{-1}(s)$ of a principal G-bundle $p\colon E \to S$ is in particular homeomorphic to the group G.

A morphism between principal G-bundles $E \to S$ and $F \to S$ is, by definition, a G-equivariant map between the total spaces, commuting with the projections down to S. Any morphism of principal G-bundles over the same base S is in fact an isomorphism (one way to rephrase this is by saying: the category of principal G-bundles over S is a groupoid), and a principal G-bundle $E \to S$ is isomorphic to the trivial principal G-bundle $\mathrm{pr}_1 \colon S \times G \to S$ if and only if $E \to S$ has a section (i.e. a continuous right inverse).

If $p \colon E \to S$ is a principal G-bundle and $\alpha \colon S' \to S$ is a continuous map, then the pullback

$$\begin{tikzcd} \alpha^* E \arrow[r] \arrow[d, "p'"'] \arrow[dr, phantom, "\square"] & E \arrow[d, "p"] \\ S' \arrow[r, "\alpha"'] & S \end{tikzcd}$$

defines a principal G-bundle $(\alpha^* E, p')$. Note that the pullback of (E, p) along p itself defines a trivial principal G-bundle $E \times_S E \to E$, since the diagonal provides a section. Given two homotopic maps $\alpha, \beta \colon S' \rightrightarrows S$, the two principal G-bundles $\alpha^* E \to S'$ and $\beta^* E \to S'$ are isomorphic. This fact has the following use. Let HTop denote the *homotopy category* of Hausdorff, paracompact topological spaces, and let $G \in \mathsf{HTop}$ be a topological group. Then, sending a space $S \in \mathsf{HTop}$ to the set of isomorphism classes of principal G-bundles over S defines a *functor* $k_G \colon \mathsf{HTop}^{\mathrm{op}} \to \mathsf{Sets}$. This functor is representable, as proved for instance in [55, Ch. 4.12], i.e. there exists an object $\mathrm{B}G \in \mathsf{HTop}$, unique up to unique isomorphism in HTop, such that

$$k_G(S) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{HTop}}(S, \mathrm{B}G)$$

functorially in S.

Definition 3.1.1 (Homotopy equivalences)**.** Let X and Y be two topological spaces.

- A continuous map $h \colon X \to Y$ is called a *weak homotopy equivalence* if it induces isomorphisms on all homotopy groups.
- A *homotopy equivalence* between X and Y is a pair of maps $h \colon X \to Y$ and $l \colon Y \to X$ such that $l \circ h$ is homotopy to id_X and $h \circ l$ is homotopic to id_Y. We say that X and Y are *homotopy equivalent*, or have the same *homotopy type*. We sometimes use the notation $X \approx Y$ to denote the existence of a homotopy equivalence between X and Y.
- A topological space E is called *weakly contractible* if the constant map $E \to \mathrm{pt}$ is a weak homotopy equivalence, which is equivalent to all the homotopy groups of E being trivial. It is called *contractible* if it is homotopy equivalent to a point.

Remark 3.1.2. If X and Y have the same homotopy type, then their cohomology groups are isomorphic.

Definition 3.1.3 (Universal principal bundle)**.** A principal G-bundle $p \colon E \to S$ is called *universal* if its total space E is contractible.

THEOREM 3.1.4 (Whitehead [91]). *Let $h \colon X \to Y$ be a weak homotopy equivalence of CW complexes. Then h induces a homotopy equivalence between X and Y. In particular, every weakly contractible CW complex is contractible.*

Definition 3.1.5 (Classifying space)**.** If $p \colon E \to S$ is a principal G-bundle over a topological space S, such that E is a weakly contractible CW complex, we say that S is a *classifying space* for G.

If $p\colon E \to S$ is as in Definition 3.1.5, then it is universal by Whitehead's theorem. Moreover, S can be taken to be a CW complex (by exploiting a weak equivalence $S' \to S$ where S' is a CW complex, cf. [51, Proposition 4.13]), and the classifying space S (resp. the total space E) is unique up to homotopy equivalence (resp. up to G-homotopy equivalence). In fact, such an S represents the functor k_G, with

$$\eta_G = \big[p\colon E \to S\big] \in k_G(S)$$

the tautological object. The existence of a principal G-bundle $p\colon E \to S$ as in Definition 3.1.5, for every topological group G, was proved by Milnor [70]. We stress that the total space E of such a bundle is contractible, by Whitehead's theorem. Therefore the equivalence class η_G contains a representative with contractible total space, that we will denote

$$\mathrm{E}G \longrightarrow \mathrm{B}G.$$

The tautological object $\eta_G = [\mathrm{E}G \to \mathrm{B}G] \in k_G(\mathrm{B}G)$ corresponds to the identity $\mathrm{id} \in \mathrm{Hom}_{\mathsf{HTop}}(\mathrm{B}G, \mathrm{B}G)$ under the isomorphism of functors $k_G \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{HTop}}(-, \mathrm{B}G)$. For every CW complex S and for every principal G-bundle (F, p) over S, there exists a unique morphism

$$\varphi_{(F,p)} \in \mathrm{Hom}_{\mathsf{HTop}}(S, \mathrm{B}G)$$

such that $\varphi^*_{(F,p)}\mathrm{E}G$ is isomorphic to $p\colon F \to S$.

Milnor's proof is usually referred to as the *join construction*, for in [71] he constructs $\mathrm{E}G$ as

$$\mathrm{E}G = \varinjlim_n G * G * \cdots * G$$

where $*$ denotes the topological join. If we identify G, up to homotopy, with a CW complex W, then $\mathrm{E}G$ is homotopy equivalent to a colimit of finite joins of W, and is therefore weakly contractible (which implies the above universal property). The fact that $\mathrm{E}G$ is contractible will be crucial, since then the product space $\mathrm{E}G \times X$ will have the same homotopy type as X, for every topological space X.

Finally, the reader should keep in mind that $\mathrm{E}G$ and its free quotient $\mathrm{B}G = \mathrm{E}G/G$ are only determined up to homotopy.

First examples of classifying spaces

We now list a few classical examples of universal principal G-bundles $\mathrm{E}G \to \mathrm{B}G$. We insist that $\mathrm{E}G$ should be contractible.

(1) If $G = 1$, one can take $\mathsf{pt} \to \mathsf{pt}$ as $\mathrm{E}G \to \mathrm{B}G$.

(2) If G and K are two groups, then one can always take $\mathrm{E}(G \times K) = \mathrm{E}G \times \mathrm{E}K$.

(3) If $G = \mathbb{R}$, then one can take $\mathrm{E}\mathbb{R} = \mathbb{R}$ with the constant map $\mathbb{R} \to \mathsf{pt}$.

(4) If $G = \mathbb{Z}^n$, one can take $\mathbb{R}^n \to (S^1)^n$ to be the map

$$(y_1, \dots, y_n) \mapsto \big(\exp(i\pi y_1), \dots, \exp(i\pi y_n)\big).$$

(5) If $G = \mathbb{Z}/2$, one can take the double cover $S^\infty \to \mathbb{RP}^\infty$.

(6) If $G = \mathbb{C}^\times$, one can take $\mathbb{C}^\infty \setminus 0 \to \mathbb{P}^\infty$.

(7) If $G = S^1 \subset \mathbb{C}^\times$ is the circle, we can restrict the bundle from the previous example to get the (infinite) *Hopf bundle* $S^\infty \to \mathbb{P}^\infty$.

(8) If $K \subset G$ is closed subgroup, one can always take

$$\mathrm{E}K = \mathrm{E}G, \quad \mathrm{B}K = \mathrm{E}G/K,$$

with $\mathrm{E}K = \mathrm{E}G \to \mathrm{E}G/K$ the quotient map.

Notation 3.1.6. In the algebraic context, explored in later sections, we shall denote the group $\mathbb{C}^\times$ by $\mathbb{G}_m$, as customary in algebraic geometry. This is to emphasise the *multiplicative group* structure on the variety $\mathbb{C}^\times = \operatorname{Spec}\mathbb{C}[x]_x = \operatorname{Spec}\mathbb{C}[x,y]/(xy-1)$. Similarly, $\mathbb{C}^n$ will be denoted $\mathbb{A}^n$ as customary in the algebraic category.

Remark 3.1.7. Let us compare Items (7) and (8). The infinite Hopf bundle certainly works, as S^∞ is contractible, and the quotient $S^\infty \to S^\infty/S^1 = \mathbb{P}^\infty$ is a principal bundle. It seems like we 'kept' the classifying space and 'changed' the universal space over it, which looks like the opposite of what is happening in Item (8). But there is no contradiction: Item (8) tells us we can equally well take

$$\mathrm{E}S^1 = \mathrm{E}\mathbb{C}^\times = \mathbb{C}^\infty \setminus 0 \longrightarrow (\mathbb{C}^\infty \setminus 0)/S^1 = \mathbb{P}^\infty \times \mathbb{R}.$$

The target $\mathbb{P}^\infty \times \mathbb{R}$ still works as $\mathrm{B}S^1$, as it is homotopy equivalent to $\mathbb{P}^\infty$. This remark is good to see that the spaces $\mathrm{E}G$ and $\mathrm{B}G$ are only defined up to homotopy.

Remark 3.1.8. Example (6) generalises in two different ways:

(6′) If $G = (\mathbb{C}^\times)^n$, one has $(\mathbb{C}^\infty \setminus 0)^n \to (\mathbb{P}^\infty)^n$ by (2). Similarly, replacing $\mathbb{C}^\times$ by S^1, example (7) becomes $(S^\infty)^n \to (\mathbb{P}^\infty)^n$, the direct limit of the $(S^1)^n$-bundles $(S^{2m+1})^n \to (\mathbb{P}^m)^n$.

(6″) If $G = \mathrm{GL}_n(\mathbb{C})$, one has $F(n,\mathbb{C}^\infty) \to G(n,\mathbb{C}^\infty)$ where $F(n,\mathbb{C}^\infty)$ denotes the space of ordered n-frames in an infinite dimensional vector space and the map sends an n-frame to its span, viewed as a point of the infinite Grassmannian.

Remark 3.1.9. The infinite dimensional complex Grassmannian

$$G(n,\mathbb{C}^\infty) = \varinjlim_m G(n,\mathbb{C}^m)$$

is also the classifying space for the unitary group $\mathrm{U}(n)$, for which one can take

$$\mathrm{EU}(n) = F^{\mathrm{orth}}(n,\mathbb{C}^\infty) = \varinjlim_m F^{\mathrm{orth}}(n,\mathbb{C}^m)$$

as universal principal bundle, where $F^{\mathrm{orth}}(n,\mathbb{C}^m)$ denotes the space of orthonormal n-frames in $\mathbb{C}^m$. Similarly, if $G = \mathrm{O}(n)$ is the real orthogonal group, one can take $F^{\mathrm{orth}}(n,\mathbb{R}^\infty) \to G(n,\mathbb{R}^\infty)$ as universal principal G-bundle, and if $G = \mathrm{GL}_n(\mathbb{R})$, one can take all n-frames, and end up with $F(n,\mathbb{R}^\infty) \to G(n,\mathbb{R}^\infty)$ as universal principal bundle.

Before defining equivariant cohomology, we pause for a second. The reader may have observed that in most examples above, the spaces involved are infinite dimensional. We want to answer the following questions:

(A) Why is that the case?

(B) Why is it not a problem for algebraic geometers?

Consider example (6), say. We have that $G = \mathbb{C}^\times$ acts on $\mathbb{C}^n$ by

$$\lambda \cdot (x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n),$$

and removing the origin we get a nice principal $\mathbb{C}^\times$-bundle

$$\mathbb{C}^n \setminus 0 \longrightarrow \mathbb{P}^{n-1}.$$

But we cannot stop here and declare this to be our $\mathrm{E}G \to \mathrm{B}G$, since we want the total space to be contractible, and unfortunately

$$\pi_n(\mathbb{C}^n \setminus 0) \neq 0.$$

It turns out that taking the limit is what we need in order to kill all higher homotopy groups. Formally, we can use the $\mathbb{C}^\times$-equivariant maps

$$\mathbb{C}^n \setminus 0 \hookrightarrow \mathbb{C}^{n+1} \setminus 0, \quad (x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, 0)$$

to get an inductive system of inclusions $\mathbb{P}^{n-1} \hookrightarrow \mathbb{P}^n$. The result of the limit process

$$\begin{tikzcd} \mathbb{C}^n \setminus 0 \arrow[r, hook] \arrow[d] & \mathbb{C}^{n+1} \setminus 0 \arrow[d] & & \mathbb{C}^\infty \setminus 0 \arrow[d] \\ \mathbb{P}^{n-1} \arrow[r, hook] & \mathbb{P}^n & \arrow[r, dashed, "n \to \infty"] & \mathbb{P}^\infty \end{tikzcd}$$

has the desired property $\pi_n(\mathbb{C}^\infty \setminus 0) = 0$ for all n. This answers Question (A). As for Question (B), we will see in Section 3.1.3 that computations can be performed at the level of certain finite dimensional *approximation spaces*.

3.1.2 Definition of equivariant cohomology

Let X be any object in Top which is (homotopy equivalent to) a CW complex, and suppose that X comes with a left G-action. Choose a universal principal G-bundle $\mathrm{E}G \to \mathrm{B}G$, with a free *right* G-action on $\mathrm{E}G$. Form the quotient space

$$\mathrm{E}G \times^G X = \frac{\mathrm{E}G \times X}{(e \cdot g, x) \sim (e, g \cdot x)}. \tag{3.1.1}$$

This quotient exists as an 'infinite-dimensional smooth manifold' when the initial G-space X is a smooth manifold, since the diagonal action of G on $\mathrm{E}G \times X$ is *free*. Note that the product $\mathrm{E}G \times X$ is homotopy equivalent to X since $\mathrm{E}G$ is contractible. The quotient (3.1.1) is the right notion of orbit space (from a homotopy theory point of view), for it is much better behaved than the (topological) orbit space X/G.

Definition 3.1.10 (Equivariant cohomology)**.** The *G-equivariant cohomology of X* is defined to be the singular cohomology of the quotient (3.1.1). The notation will be

$$\mathrm{H}^*_G(X) = \mathrm{H}^*(\mathrm{E}G \times^G X). \tag{3.1.2}$$

The definition works also for real and complex algebraic varieties, since any real or complex algebraic variety is homotopy equivalent to a CW complex [54]. The above definition is upgraded from cohomology to Chow in [30] for algebraic varieties.

Convention 3.1.11. We work with $\mathbb{Q}$ coefficients, so $\mathrm{H}^*(V) = \mathrm{H}^*_{\mathrm{sing}}(V, \mathbb{Q})$ for any space V. Moreover, in the algebraic context (i.e. when X is a complex algebraic variety and G is an algebraic group), we implicitly assume all representations to be *complex representations*, and vector bundles to be *algebraic vector bundles*.

Example 3.1.12. When $X = \mathrm{pt}$, we set

$$\mathrm{H}^*_G = \mathrm{H}^*_G(\mathrm{pt}) = \mathrm{H}^*(\mathrm{B}G) = \mathrm{H}^*_{\mathrm{sing}}(\mathrm{B}G, \mathbb{Q}).$$

This can be viewed as the ring of characteristic classes of principal G-bundles. See [89] for the definition of the G-equivariant *Chow ring* $A^*(\mathrm{B}G)$, a special case of [30].

Remark 3.1.13. A G-space X can have nonzero equivariant cohomology in degrees higher than its dimension, unlike ordinary cohomology. This is already true for $X = \mathrm{pt}$.

Exercise 3.1.14. Let E be a right G-space, and let G act on itself by left multiplication. Then there is an isomorphism $E \times^G G \xrightarrow{\sim} E$.

Exercise 3.1.15. Let E be a right G-space, $K \subset G$ a closed subgroup. Then there is an isomorphism $E \times^G G/K \cong E/K$.

Remark 3.1.16. Even though ordinary cohomology is a special case of equivariant cohomology (it is enough to set $G = 1$), it is not true that one can reconstruct ordinary cohomology knowing equivariant cohomology. For instance, setting $X = G = \mathbb{C}^\times$, with G acting on itself by left complex multiplication, one finds (using Exercise 3.1.14 for the second equality)

$$\mathrm{H}^1_G(X) = \mathrm{H}^1(\mathbb{C}^\infty \setminus 0 \times^{\mathbb{C}^\times} \mathbb{C}^\times) = \mathrm{H}^1(\mathbb{C}^\infty \setminus 0) = 0,$$

whereas $\mathrm{H}^1(\mathbb{C}^\times) = \mathrm{H}^1(S^1) = \mathbb{Q}$.

Remark 3.1.17. The cohomology of a topological orbit space X/G does not agree with the G-equivariant cohomology of X, unless the G-action on X is free (cf. Equation (3.1.8)). One example is $G = S^1$ acting on $X = S^2 \subset \mathbb{R}^3$ by rotation along the z-axis. There are two fixed points so the action is not free. The quotient X/G can be identified with a closed interval, which is contractible. Therefore the cohomology of the quotient is the cohomology of a point, and the action has manifestly been forgotten. But $\mathrm{H}^*_{S^1}(S^2)$ is more interesting (cf. Exercise 3.1.27).

LEMMA 3.1.18. *Let X be a CW complex with a left G-action. Then the definition of $\mathrm{H}^*_G(X)$ does not depend upon the choice of universal principal G-bundle.*

Proof. Let $\mathrm{E}G \to \mathrm{B}G$ and $\mathrm{F}G \to \mathrm{B}G$ be two universal principal G-bundles. Consider the space $Y = (\mathrm{E}G \times \mathrm{F}G \times X)/G$. Then we have fibrations

$$\begin{tikzcd} & Y \arrow[dl, "p_E"'] \arrow[dr, "p_F"] & \\ \mathrm{E}G \times^G X & & \mathrm{F}G \times^G X \end{tikzcd}$$

with contractible fibers $\mathrm{F}G$ and $\mathrm{E}G$, respectively. Let us focus on p_E. For every positive integer n, we have the fibration long exact sequence

$$\cdots \longrightarrow \pi_n(\mathrm{F}G) \longrightarrow \pi_n(Y) \longrightarrow \pi_n(\mathrm{E}G \times^G X) \longrightarrow \pi_{n-1}(\mathrm{F}G) \longrightarrow \cdots$$

but the homotopy groups of the fiber $\mathrm{F}G$ are trivial, therefore p_E induces a family of isomorphisms

$$\pi_n(Y) \xrightarrow{\sim} \pi_n(\mathrm{E}G \times^G X).$$

By Whitehead's theorem (cf. Theorem 3.1.4), p_E is in fact a homotopy equivalence, so it induces isomorphisms between cohomology groups. Repeating the process with p_F gives the result. □

Remark 3.1.19. There is a fibre bundle

$$\begin{tikzcd} X \arrow[r, hook] & \mathrm{E}G \times^G X \arrow[d, "p"] \\ & \mathrm{B}G \end{tikzcd} \tag{3.1.3}$$

with fibre X, induced by the G-equivariant projection $\mathrm{E}G \times X \to \mathrm{E}G$. The standard pullback in cohomology induces a canonical ring homomorphism

$$\mathrm{H}^*_G \xrightarrow{p^*} \mathrm{H}^*_G(X), \tag{3.1.4}$$

making $\mathrm{H}^*_G(X)$ into a H^*_G-module.

Example 3.1.20 (Equivariant formality)**.** When G acts trivially on X we have $\mathrm{E}G \times^G X = \mathrm{B}G \times X$, so that

$$\mathrm{H}^*_G(X) \cong \mathrm{H}^*_G \otimes \mathrm{H}^*(X). \tag{3.1.5}$$

For instance, given any action of G on X and letting X^G be the fixed locus, we have $\mathrm{H}^*_G(X^G) \cong \mathrm{H}^*_G \otimes \mathrm{H}^*(X^G)$. There are G-spaces X with nontrivial action such that the isomorphism (3.1.5) holds — see Exercise 3.1.27. When one has an isomorphism (3.1.5), we say that X (or its G-action) is *equivariantly formal*.

Example 3.1.21 (Contractible G-space)**.** Let X be a contractible space with a left G-action. Then (3.1.4) is an isomorphism.

Example 3.1.22 (Equivariant cohomology of a subgroup)**.** Let $K \subset G$ be a closed subgroup. Then $\mathrm{E}G/K$ exists and in fact $\mathrm{E}K = \mathrm{E}G \to \mathrm{E}G/K = \mathrm{B}K$ is a classifying space for K. Therefore,

$$\mathrm{H}^*_G(G/K) = \mathrm{H}^*(\mathrm{E}G \times^G (G/K)) = \mathrm{H}^*(\mathrm{E}G/K) = \mathrm{H}^*(\mathrm{B}K) = \mathrm{H}^*_K,$$

where the second equality follows by Exercise 3.1.15.

Example 3.1.23. Generalising the previous example, let $K \subset G$ be again a closed subgroup, acting on a space X on the left. We can also consider the diagonal action of K upon $G \times X$. The quotient

$$G \times^K X = (G \times X)/(gk, x) \sim (g, k \cdot x)$$

makes sense and we have

$$\mathrm{H}^*_G(G \times^K X) = \mathrm{H}^*(\mathrm{E}G \times^G G \times^K X) = \mathrm{H}^*(\mathrm{E}G \times^K X) = \mathrm{H}^*_K(X).$$

Again, we have used that $\mathrm{E}G \to \mathrm{E}G/K$ is a classifying space for $K \subset G$.

THEOREM 3.1.24 ([84, I.5]). *Let $f : X \to S$ be a topological fibre bundle with contractible fibre. Then f is a homotopy equivalence. In particular, f^* induces an isomorphism*

$$\mathrm{H}^*(X, \mathbb{Z}) \xrightarrow{\sim} \mathrm{H}^*(S, \mathbb{Z}).$$

Exercise 3.1.25. Let $K \subset G$ be a closed subgroup of a topological group G such that G/K is contractible, and let X be a G-space. Then

$$\mathrm{H}^*_K(X) \cong \mathrm{H}^*_G(X).$$

(**Hint**: use the fact that one can take $\mathrm{E}K = \mathrm{E}G$ to construct a map $\mathrm{E}K \times^K X = \mathrm{E}G \times^K X \to \mathrm{E}G \times^G X$. Show that it is a fibre bundle with fibre G/K. Conclude by Theorem 3.1.24).

Example 3.1.26. Let us revisit the examples (6′) and (6″) discussed in Section 3.1.1.

(A) In the case $\mathbb{T} = (\mathbb{C}^\times)^n$, we find

$$\mathrm{H}^*_{\mathbb{T}} = \mathrm{H}^*((\mathbb{P}^\infty)^n, \mathbb{Q}) = \mathbb{Q}[s_1, \dots, s_n] \tag{3.1.6}$$

where $s_i \in \mathrm{H}^2_{\mathbb{T}}$ are the Chern classes obtained by pulling back to $\mathrm{B}\mathbb{T}$ the universal line bundles living over the individual spaces $\mathbb{P}^\infty$. In other words, if $\pi_i \colon \mathrm{B}\mathbb{T} \to \mathbb{P}^\infty$ is the i-th projection, the generators of $\mathrm{H}^*_{\mathbb{T}}$ are

$$s_i = c_1(\pi_i^* \mathscr{O}(-1)).$$

(B) In the case $G = \mathrm{GL}_n(\mathbb{C})$, we find

$$\mathrm{H}^*_G = \mathrm{H}^*(G(n, \mathbb{C}^\infty), \mathbb{Q}) = \mathbb{Q}[e_1, \dots, e_n] \tag{3.1.7}$$

where $e_i = c_i(\mathscr{S})$ are the Chern classes of the universal rank n bundle $\mathscr{S}$ living over the Grassmannian.

Exercise 3.1.27. Let $G = S^1$ act on the sphere $S^2 \subset \mathbb{R}^3$ by rotation along the z-axis. Show that S^2 is equivariantly formal for this S^1-action. In other words, show that

$$\mathrm{H}^*_{S^1}(S^2) \cong \mathrm{H}^*(\mathrm{B}S^1) \otimes \mathrm{H}^*(S^2),$$

even though the action has two fixed points. (**Hint**: Find a suitable open cover $S^2 = X_1 \cup X_2$, inducing an open cover of $\mathrm{E}S^1 \times^{S^1} S^2$. Apply Mayer–Vietoris to this open cover, and use that $\mathrm{H}^*_{S^1} = \mathbb{Q}[c]$ is one-dimensional, resp. zero-dimensional, in even degree, resp. odd degree). The conclusion of this exercise also holds with integral coefficients.

Preview: how to calculate via equivariant cohomology

Let X be a smooth manifold acted on (smoothly) by a Lie group G. There is a commutative diagram

$$\begin{tikzcd}
\mathrm{E}G \arrow[d] & \mathrm{E}G \times X \arrow[l] \arrow[r] \arrow[d] & X \arrow[d] \\
\mathrm{B}G & \mathrm{E}G \times^G X \arrow[l, "p"'] \arrow[r, "\sigma"] & X/G
\end{tikzcd}$$

where σ, unlike p, is in general not a fiber bundle, but has the property that

$$\sigma^{-1}(\mathrm{Orb}\, x) = \mathrm{E}G/G_x \cong \mathrm{B}G_x$$

for all $x \in X$, where $G_x \subset G$ is the stabiliser at x. When G is a compact Lie group acting (smoothly and) *freely* on X, the map σ induces a homotopy equivalence $\mathrm{E}G \times^G X \approx X/G$ yielding a natural isomorphism

$$\sigma^* \colon \mathrm{H}^*(X/G) \xrightarrow{\sim} \mathrm{H}^*_G(X). \tag{3.1.8}$$

Let $\iota\colon X \hookrightarrow \mathrm{E}G \times^G X$ denote the inclusion of X as a fibre of $p\colon \mathrm{E}G \times^G X \to \mathrm{B}G$ (after choosing a base point $\mathsf{pt} \in \mathrm{B}G$). Then ι induces a natural map (surjective in sufficiently nice situations)

$$\mathrm{H}^*_G(X) \xrightarrow{\iota^*} \mathrm{H}^*(X)$$

from equivariant to ordinary cohomology.

There is a pullback diagram (on the left) inducing, whenever X is compact and oriented, a commutative diagram of cohomology rings:

$$\begin{tikzcd} X \arrow[r, "q"] \arrow[d, "\iota"'] \arrow[dr, phantom, "\square"] & \mathsf{pt} \arrow[d, "b"] \\ \mathrm{E}G \times^G X \arrow[r, "p"] & \mathrm{B}G \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} \mathrm{H}^*(X) \arrow[r, "q_*"] & \mathbb{Z} \\ \mathrm{H}^*_G(X) \arrow[r, "p_*"] \arrow[u, "\iota^*"'] & \mathrm{H}^*_G \arrow[u, "b^*"'] \end{tikzcd}$$

In algebraic geometry, this compatibility makes sense and is useful when X is a proper variety, which ensures the existence of q_*. The *equivariant pushforward* p_*, on the other hand, will be defined in Section 3.2.2. Note that the above commutativity implies in particular that

$$\boxed{\textit{ordinary integrals can be performed in equivariant cohomology.}}$$

Indeed, if one wanted to compute

$$\int_X \alpha = q_*\alpha \in \mathbb{Z},$$

then one could proceed as follows: suppose one can pick a lift $\widetilde{\alpha}$ of α along ι^*. Then one can compute the above integral as

$$b^* p_* \widetilde{\alpha},$$

i.e. via equivariant cohomology. This strategy will be crucial for the applications discussed in Section 3.3.

Remark 3.1.28. We want to stress that, even though we often want to compute numbers, an equivariant integral (still undefined for now) will take values in H^*_G.

3.1.3 Moving to algebraic geometry: approximation spaces

Let us now assume that X is a complex algebraic variety and G is an algebraic group. The fact that the spaces involved, like $\mathrm{E}G$ and $\mathrm{B}G$, are infinite-dimensional, is not quite an obstacle to the computation of the equivariant cohomology groups. This is the case because of the following "approximation" result.

THEOREM 3.1.29. *Let $(E_m)_{m\geqslant 0}$ be a family of path-connected spaces on which G acts freely on the right. Let $\mathsf{k}\colon \mathbb{N} \to \mathbb{N}$ be a function such that $\pi_i(E_m) = 0$ for $0 < i < \mathsf{k}(m)$ and such that $\lim_{m\to\infty} \mathsf{k}(m) = +\infty$. Then, for any left G-space X, there are natural isomorphisms*

$$\mathrm{H}^i_G(X) \cong \mathrm{H}^i(E_m \times^G X), \quad 0 < i < \mathsf{k}(m).$$

Proof. Set $E = \mathrm{E}G$. The diagonal action of G on $E \times E_m$ induces a commutative diagram

$$\begin{tikzcd} E_m \times X \arrow[d] & E \times E_m \times X \arrow[l, "E"'] \arrow[r, "E_m"] \arrow[d] & E \times X \arrow[d] \\ E_m \times^G X & (E \times E_m) \times^G X \arrow[l, "E"'] \arrow[r, "E_m"] & E \times^G X \end{tikzcd} \tag{3.1.9}$$

where the vertical maps are the (free) quotient maps, and the horizontal maps are locally trivial fibre bundles with fibre indicated on top of the corresponding arrow. As a consequence of the Leray–Hirsch Lemma, if a fibre bundle $Y \to B$ has $\mathrm{H}^i(\mathsf{fibre}) = 0$ for $0 < i < m$, then the pullback induces a natural isomorphism $\mathrm{H}^i(B,\mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^i(Y,\mathbb{Q})$ for $0 < i < m$. (See also Theorem 3.1.24). Since $\mathsf{k}(m)$ goes to infinity as m grows, we can apply the previous statement to $\mathsf{k}(m)$ directly: granting for a second that $\mathrm{H}^i(E_m,\mathbb{Q}) = 0$ for all $0 < i < \mathsf{k}(m)$, we have by Leray–Hirsch (using also that $\mathrm{H}^*(E,\mathbb{Q}) = 0$), isomorphisms

$$\mathrm{H}^i(E_m \times^G X) \xrightarrow{\sim} \mathrm{H}^i((E \times E_m) \times^G X) \xleftarrow{\sim} \mathrm{H}^i_G(X)$$

for all $0 < i < \mathsf{k}(m)$, induced by the lower row of (3.1.9). All that is left to do is to confirm that $\mathrm{H}^i(E_m,\mathbb{Q}) = 0$ for all $0 < i < \mathsf{k}(m)$. But because E_m is path-connected and $(\mathsf{k}(m)-1)$-connected by assumption, Hurewicz's theorem says precisely that $\mathrm{H}_i(E,\mathbb{Z}) = 0$ for $0 < i < \mathsf{k}(m)$. Tensoring with $\mathbb{Q}$ and dualising, we get $\mathrm{H}^i(E_m,\mathbb{Q}) = 0$ for all $0 < i < \mathsf{k}(m)$, as required. □

We will refer to the E_m spaces as *approximation spaces*. The idea behind their existence can be traced back to [15, Remark XII.3.7].

Remark 3.1.30. In the smooth category, if G is a compact Lie group, then $\mathrm{E}G \to \mathrm{B}G$ is a colimit of smooth principal G-bundles

$$E_m \longrightarrow B_m,$$

where E_m is m-connected, i.e. $\pi_i(E_m) = 0$ for $1 \leqslant i \leqslant m$, i.e. if S^k is a sphere of dimension $k \leqslant m$, then any continuous map $S^k \to E_m$ is homotopic to the constant map.

Example 3.1.31. Let us revisit once more the examples $\mathbb{T} = (\mathbb{C}^\times)^n$ and $G = \mathrm{GL}_n(\mathbb{C})$. For the case of the torus, one can take $E_m = (\mathbb{C}^m \setminus 0)^n \to B_m = (\mathbb{P}^{m-1})^n$ as approximations of $\mathrm{E}\mathbb{T} \to \mathrm{B}\mathbb{T}$. Since $\mathbb{C}^m \setminus 0$ has the same homotopy type as S^{2m-1}, the function $\mathsf{k}(m) = n(2m-1)$ satisfies the assumption $\pi_i(E_m) = 0$ of Theorem 3.1.29, where $0 < i < \mathsf{k}(m)$. Setting for instance $X = \mathsf{pt}$, we find isomorphisms

$$\mathrm{H}^i_{\mathbb{T}} \cong \mathrm{H}^i((\mathbb{P}^{m-1})^n), \quad i < n(2m-1).$$

When $G = \mathrm{GL}_n(\mathbb{C})$, we can take E_m, for $m > n$, to be the set of $m \times n$ rank n matrices. The approximation spaces then look like the free quotients $E_m \to G(n,\mathbb{C}^m)$. The function $\mathsf{k}(m) = 2(m-n)$ will do the job (cf. Remark 3.1.32 below) and we would find, again for $X = \mathsf{pt}$, isomorphisms

$$\mathrm{H}^i_{\mathrm{GL}_n} \cong \mathrm{H}^i(G(n,\mathbb{C}^m)), \quad i < 2(m-n).$$

Remark 3.1.32. If $Y \subset \mathbb{A}^N$ is a subvariety of codimension d, then

$$\pi_i(\mathbb{A}^N \setminus Y) = 0, \quad 0 < i \leqslant 2d-2,$$

as proved in [44, Section A.4]. Assume $n < m$. By [5, Section II.2], the locus of matrices of rank at most k is irreducible and of codimension $(m-k)(n-k)$ in $\mathbb{A}^{mn}$ for every $k \leqslant n$. Thus, setting $k = n-1$, we see that the open subset

$$U = \{\text{full rank matrices}\} \subset \mathsf{Mat}_{m\times n}(\mathbb{C}) = \mathbb{A}^{mn}$$

has complement $Y = \mathbb{A}^{mn} \setminus U$ of codimension $m-n+1$, and the choice $\mathsf{k}(m) = 2(m-n)$ works.

Remark 3.1.33. It is clear that the method of approximation spaces allows one to compute H^*_G in all degrees, since $\mathsf{k}(m) \to \infty$ for $m \to \infty$.

3.1.4 Equivariant vector bundles

Let G be a group acting on X and H a group acting on Y. Suppose there are maps $\phi: G \to H$ and $f: X \to Y$. The condition

$$f(g \cdot x) = \phi(g) \cdot f(x) \tag{3.1.10}$$

for all $x \in X$ and $g \in G$ is the condition under which one can construct a natural map

$$\mathrm{E}G \times^G X \longrightarrow \mathrm{E}H \times^H Y. \tag{3.1.11}$$

When (3.1.10) is satisfied, taking cohomology of the map (3.1.11) yields the *equivariant pullback* homomorphism

$$\mathrm{H}^*_H(Y) \xrightarrow{f^*} \mathrm{H}^*_G(X). \tag{3.1.12}$$

When $\phi = \mathrm{id}_G$ we say that f is G-*equivariant*, and when moreover G acts trivially on Y we say that f is G-*invariant*.

A crucial example is that of a G-equivariant vector bundle $\pi: E \to X$. Such an object is a geometric vector bundle together with a lift of the G-action on X to a G-action on E, as the diagram

$$\begin{tikzcd} G \times E \arrow[r] \arrow[d, "\mathrm{id}\times\pi"'] & E \arrow[d, "\pi"] \\ G \times X \arrow[r, "\text{action}"] & X \end{tikzcd}$$

illustrates. A choice of lift of the G-action is also called a G-*linearisation* on E. Given an equivariant vector bundle one can define a new vector bundle

$$V_E = \mathrm{E}G \times^G E \longrightarrow \mathrm{E}G \times^G X \tag{3.1.13}$$

of the same rank as E.

Next, we define equivariant Chern classes. The definition is based on ordinary Chern classes, for which the reader is referred to [52, Chapter 3] or [72].

Definition 3.1.34 (Equivariant Chern classes)**.** The *equivariant Chern classes* of a G-equivariant vector bundle $E \to X$ are the characteristic classes

$$c_i^G(E) = c_i(V_E) \in \mathrm{H}^{2i}_G(X).$$

These classes can clearly be computed through approximation spaces. Indeed, the vector bundle (3.1.13) can be approximated by vector bundles

$$V_{E,m} = E_m \times^G E \longrightarrow E_m \times^G X$$

whose Chern classes $c_i(V_{E,m})$ live in $\mathrm{H}^{2i}(E_m \times^G X) \cong \mathrm{H}^{2i}_G(X)$ for $m \gg 0$.

Example 3.1.35 (Equivariant fundamental class)**.** Let X be a smooth algebraic variety acted on by a linear algebraic group G, and let $Y \subset X$ be a G-invariant closed subvariety of codimension d. The space $E_m \times^G X$ is smooth, and

$$E_m \times^G Y \hookrightarrow E_m \times^G X$$

is a closed subvariety of codimension d. Then

$$[Y]^G = [E_m \times^G Y] \in \mathrm{H}^{2d}(E_m \times^G X) \cong \mathrm{H}^{2d}_G(X) \tag{3.1.14}$$

is called the *equivariant fundamental class* of $Y \subset X$.

Exercise 3.1.36. Show that the classes $[Y]^G$ defined in (3.1.14) are compatible when m varies and are independent of the choice of $\mathrm{E}G$ and $(E_m)_{m\geqslant 0}$.

Example 3.1.37 (Weight of a character)**.** When X is a point, an equivariant vector bundle is a representation $\rho\colon G\to \mathrm{GL}(E)$. This still gives rise to a vector bundle

$$\begin{array}{c} V_\rho \\ \downarrow \\ \mathrm{B}G \end{array}$$

whose Chern classes live in H^*_G. When E is one-dimensional, an equivariant line bundle is simply a *character* $\chi\colon G\to\mathbb{C}^\times$ and one defines the *weight* of χ to be the Chern class

$$w_\chi = c_1(V_\chi)\in \mathrm{H}^2_G.$$

When a complex representation $\rho\colon G\to\mathrm{GL}(E)$ of dimension r splits as a direct sum of characters χ_i (for example this is always the case when $G=\mathbb{G}_m^n$ is a torus), the G*-equivariant Euler class* splits as a product of weights,

$$e^G(E) = c^G_{\mathrm{top}}(E) = c_{\mathrm{top}}(V_\rho) = \prod_{i=1}^r w_i \in \mathrm{H}^{2r}_G,$$

where $w_i = c_1(V_{\chi_i})\in \mathrm{H}^2_G$ is the weight of χ_i.

More generally, for a complex G-representation $\rho\colon G\to\mathrm{GL}(E)$ with $\rho=\chi_1\oplus\cdots\oplus\chi_r$, the i-th equivariant Chern class $c_i^G(E)$ is the i-th symmetric function in $w_1,\dots,w_r$. See Example 3.1.39 for a detailed description of this fact.

Example 3.1.38. Let $\mathbb{T}=\mathbb{G}_m=\mathbb{C}^\times$ be the 1-dimensional torus and $\rho_a\colon\mathbb{T}\to\mathbb{C}^\times=\mathrm{GL}(\mathbb{C}_a)$ the character $z\mapsto z^a$ for an integer $a\in\mathbb{Z}$. Here we are viewing the 1-dimensional vector space $\mathbb{C}_a$ as an equivariant vector bundle over $X=\mathrm{pt}$. The line bundle $V_{\rho_a}\to\mathrm{B}\mathbb{T}$ is approximated by line bundles $V_{\rho_a,m}\to\mathbb{P}^{m-1}$. In fact, $V_{\rho_a,m}\cong\mathcal{O}_{\mathbb{P}^{m-1}}(-a)$ so that the weight of ρ_a is

$$c_1^{\mathbb{T}}(\mathbb{C}_a) = c_1(V_{\rho_a,m}) = a\cdot s\in \mathrm{H}^2_{\mathbb{T}}.$$

Notice that s corresponds to the case $a=1$. This is called the *standard action* of $\mathbb{T}=\mathbb{G}_m$ on $\mathbb{C}$. It generalises in the next example.

Example 3.1.39. Let $F=\mathbb{C}^n$ be an n-dimensional vector space and let the torus $\mathbb{T}=\mathbb{G}_m^n$ act on F via the *standard action*

$$t\cdot(v_1,\dots,v_n) = (t_1v_1,\dots,t_nv_n).$$

The induced representation $\rho\colon\mathbb{T}\to\mathrm{GL}(F)$, defined by $\rho_t(v)=t\cdot v$ as above, gives a rank n vector bundle

$$\begin{array}{rcl} V_\rho & = & \mathrm{E}\mathbb{T}\times^{\mathbb{T}} F \\ & & \downarrow \\ \mathrm{B}\mathbb{T} & = & (\mathbb{P}^\infty)^n \end{array}$$

which can of course be approximated by rank n vector bundles

$$\begin{array}{rcl} V_{\rho,m} & = & (\mathbb{C}^m\setminus 0)^n\times^{\mathbb{T}} F \\ & & \downarrow \\ \mathrm{B}\mathbb{T} & = & (\mathbb{P}^{m-1})^n \end{array}$$

Now, there is a splitting $F = F_1 \oplus \cdots \oplus F_n$ where each summand corresponds to a character

$$\begin{aligned} \mathbb{T} &\xrightarrow{\chi_i} \mathbb{C}^\times \\ t &\longmapsto t_i \end{aligned}$$

whose weight is just $s_i \in \mathrm{H}^2_{\mathbb{T}}$ by Example 3.1.38. In other words, for each i, one has that $V_{\chi_i} = \mathrm{E}\mathbb{T} \times^{\mathbb{T}} F_i \to (\mathbb{P}^\infty)^n$ is the total space of the line bundle $\mathscr{O}_i(-1) = \mathrm{pr}_i^* \mathscr{O}(-1)$. We quickly verify that

$$c_i^{\mathbb{T}}(F) = e_i(s_1, \ldots, s_n) \in \mathrm{H}^{2i}_{\mathbb{T}}, \tag{3.1.15}$$

the i-th symmetric function in the Chern classes $s_i = c_1(\mathscr{O}_i(-1))$. We have a decomposition

$$V_\rho = \bigoplus_{i=1}^n \mathrm{E}\mathbb{T} \times^{\mathbb{T}} F_i = \bigoplus_{i=1}^n \mathscr{O}_i(-1).$$

Then V_ρ has the obvious filtration

$$0 \subset \mathscr{O}_1(-1) \subset \mathscr{O}_1(-1) \oplus \mathscr{O}_2(-1) \subset \cdots \subset V_\rho$$

with $\mathscr{O}_i(-1)$ as line bundle quotients. The identity (3.1.15) is then a straightforward property of Chern classes. For instance, the top Chern class of the standard representation is given by

$$e^{\mathbb{T}}(F) = c_n(V_\rho) = s_1 \cdots s_n \in \mathrm{H}^{2n}_{\mathbb{T}}.$$

3.1.5 Two computations on $\mathbb{P}^{n-1}$

In this section we compute the equivariant cohomology of projective space $\mathbb{P}^{n-1}$ and the weights of the tangent representation induced by the standard action of Example 3.1.39. As a convention, we identify

$$\mathbb{P}^{n-1} = \mathbb{P}(F^*) = \{\, 1\text{-dimensional quotients } F^* \twoheadrightarrow \Lambda \,\}$$

where $F = \mathbb{C} \cdot e_1 \oplus \cdots \oplus \mathbb{C} \cdot e_n$ is the standard representation, which means

$$\mathrm{weight}(\mathbb{C} \cdot e_i) = s_i \in \mathrm{H}^2_{\mathbb{T}}, \quad \mathbb{T} = \mathbb{G}_m^n.$$

In other words, we are acting on $\mathbb{P}^{n-1}$ by $t \cdot (a_1, \ldots, a_n) = (t_1 a_1, \ldots, t_n a_n)$, and we have a natural identification $F = \mathrm{H}^0(\mathbb{P}^{n-1}, \mathscr{O}_{\mathbb{P}^{n-1}}(1))$ so that e_i corresponds to the linear form x_i.

Equivariant cohomology of $\mathbb{P}^{n-1}$

We know, thanks to Equation (3.1.15), that

$$c_i^{\mathbb{T}}(F) = c_i(V_{\rho,m}) = e_i(s_1, \ldots, s_n) \in \mathrm{H}^{2i}_{\mathbb{T}} \cong \mathrm{H}^{2i}(B_m) = \mathrm{H}^{2i}((\mathbb{P}^{m-1})^n),$$

where ρ is the standard representation and $E_m = (\mathbb{C}^m \setminus 0)^n \to B_m = (\mathbb{P}^{m-1})^n$ are the finite approximations of $\mathrm{E}\mathbb{T} \to \mathrm{B}\mathbb{T}$. The tautological short exact sequence

$$0 \longrightarrow \mathscr{S} \longrightarrow F^* \otimes_{\mathbb{C}} \mathscr{O}_{\mathbb{P}^{n-1}} \longrightarrow \mathscr{O}_{\mathbb{P}^{n-1}}(1) \longrightarrow 0$$

defines a $\mathbb{T}$-equivariant structure on the line bundle $\mathscr{O}_{\mathbb{P}^{n-1}}(1)$ living on the $\mathbb{T}$-space $\mathbb{P}^{n-1} = \mathbb{P}(F^*)$. Let

$$\begin{array}{c} W = \mathrm{Tot}_{\mathbb{P}^{n-1}}(\mathscr{O}_{\mathbb{P}^{n-1}}(1)) \\ \downarrow \\ \mathbb{P}^{n-1} \end{array}$$

be its total space, and let ζ be the first Chern class of the induced line bundle $E_m \times^{\mathbb{T}} W$ living over the total space of the projective space bundle $\mathbb{P}(V_{\rho,m})$. In other words, given

$$\begin{tikzcd} E_m \times^{\mathbb{T}} W \arrow[d, "\text{line bundle}"] & \\ E_m \times^{\mathbb{T}} \mathbb{P}^{n-1} \arrow[r, "\sim"] & \mathbb{P}(V_{\rho,m}) \end{tikzcd}$$

one has $\zeta = c_1(\mathcal{O}_{\mathbb{P}(V_{\rho,m})}(1)) = c_1^{\mathbb{T}}(\mathcal{O}_{\mathbb{P}^{n-1}}(1)) \in \mathrm{H}^2_{\mathbb{T}}(\mathbb{P}^{n-1})$. We can now compute

$$\begin{aligned} \mathrm{H}^*_{\mathbb{T}}(\mathbb{P}^{n-1}) &\cong \mathrm{H}^*(E_m \times^{\mathbb{T}} \mathbb{P}^{n-1}) \\ &\cong \mathrm{H}^*(\mathbb{P}(V_{\rho,m})) \\ &\cong \mathrm{H}^*_{\mathbb{T}}[\zeta] \Big/ \sum_{0\leqslant i\leqslant n} c_i(V_{\rho,m})\zeta^{n-i} \\ &= \mathrm{H}^*_{\mathbb{T}}[\zeta] \Big/ \prod_{1\leqslant i\leqslant n} (\zeta + s_i). \end{aligned} \tag{3.1.16}$$

The last equality follows by the description $c_i(V_{\rho,m}) = e_i(s_1, \dots, s_n)$. Equation (3.1.16) is the $\mathbb{T}$-equivariant analog of the projective bundle formula for cohomology.

The tangent representation

In the tautological exact sequence

$$0 \longrightarrow \mathcal{S} \longrightarrow F^* \otimes_{\mathbb{C}} \mathcal{O}_{\mathbb{P}^{n-1}} \longrightarrow \mathcal{O}_{\mathbb{P}^{n-1}}(1) \longrightarrow 0$$

we have, for each coordinate point $p_i \in \mathbb{P}^{n-1}$,

$$\mathcal{S}\big|_{p_i} = \bigoplus_{j\neq i} \mathbb{C}\cdot x_j^*, \quad \mathcal{O}_{\mathbb{P}^{n-1}}(1)\big|_{p_i} = \mathbb{C}\cdot x_i^*.$$

The tangent space at the fixed point p_i is then identified with

$$T_{p_i}\mathbb{P}^{n-1} = \mathcal{S}^*\big|_{p_i} \otimes \mathcal{O}_{\mathbb{P}^{n-1}}(1)\big|_{p_i} = \bigoplus_{j\neq i} (x_j \otimes x_i^*)\cdot \mathbb{C}.$$

This says that the weights of $T_{p_i}\mathbb{P}^{n-1}$ are $s_j - s_i$ for $j \neq i$, in particular the Euler class is computed as

$$e^{\mathbb{T}}(T_{p_i}\mathbb{P}^{n-1}) = \prod_{j\neq i} (s_j - s_i).$$

3.2 The Atiyah–Bott localisation formula

In this chapter we prove a version of the Atiyah–Bott localisation formula in equivariant cohomology. We mainly follow [68, 3]. We note that the results hold in equivariant Chow theory [68, 30], but for the purpose of our applications we content ourselves with equivariant cohomology.

3.2.1 A glimpse of the self-intersection formula

Let X be a connected oriented closed manifold (i.e. compact without boundary). The orientation defines a *fundamental class* $[X] \in \mathrm{H}_{\dim X}(X)$, inducing the Poincaré duality isomorphisms $\mathrm{pd}\colon \mathrm{H}^k(X) \xrightarrow{\sim} \mathrm{H}_{\dim X - k}(X)$ for every k. If $E \to X$ is an oriented vector bundle of rank r on X, the orientation

$$\eta \in \mathrm{H}^r(E, E \setminus X)$$

corresponds, by construction, to the identity element

$$\mathbf{1} = \mathsf{pd}^{-1}([X]) \in \mathrm{H}^0(X)$$

under the Thom isomorphism

$$\mathrm{H}^0(X) \xrightarrow{\sim} \mathrm{H}^r(E, E \setminus X),$$

where $X \hookrightarrow E$ is embedded as the zero section. The inclusions

$$(X, \emptyset) \hookrightarrow (E, \emptyset) \hookrightarrow (E, E \setminus X)$$

induce maps in relative cohomology

$$\begin{tikzcd} \mathrm{H}^r(E, E \setminus X) \arrow[r, "u_E"] \arrow[rr, bend left, "\alpha"] & \mathrm{H}^r(E) \arrow[r] & \mathrm{H}^r(X) \end{tikzcd}$$

and the *Euler class* of E is, by definition, the image of η under this composition, i.e.

$$e(E) = \alpha(\eta) \in \mathrm{H}^r(X).$$

Let $f : X \to Y$ be a map of connected compact oriented manifolds. Set $n = \dim X$, $m = \dim Y$ and $d = m - n$. There are Gysin maps

$$\mathrm{H}^p(X) \xrightarrow{f_*} \mathrm{H}^{p+d}(Y),$$

defined via Poincaré duality through the diagram

$$\begin{tikzcd} \mathrm{H}^p(X) \arrow[r, dashed] \arrow[d, "\mathsf{pd}"', "\wr"] & \mathrm{H}^{p+d}(Y) \\ \mathrm{H}_{n-p}(X) \arrow[r, "f_*"] & \mathrm{H}_{n-p}(Y) \arrow[u, "\wr"', "\mathsf{pd}^{-1}"] \end{tikzcd} \tag{3.2.1}$$

where f_* is the pushforward in homology.

When $f = \iota$ is a closed embedding of codimension d, and the normal bundle

$$N_{X/Y} = T_Y|_X / T_X$$

is oriented compatibly with respect to f (this means that the canonical isomorphism of line bundles $\det T_X \otimes \det N_{X/Y} \xrightarrow{\sim} \det T_Y|_X$ preserves orientations), there is a factorisation of ι_* as

$$\mathrm{H}^p(X) \xrightarrow{\sim} \mathrm{H}^{p+d}(N_{X/Y}, N_{X/Y} \setminus X) \xrightarrow{\sim} \mathrm{H}^{p+d}(Y, Y \setminus X) \longrightarrow \mathrm{H}^{p+d}(Y),$$

where the last map comes from Mayer–Vietoris for the inclusion $\iota : X \hookrightarrow Y$, and the middle isomorphism is computed via excision by fixing a tubular neighborhood T of X in Y, so that

$$\mathrm{H}^*(N_{X/Y}, N_{X/Y} \setminus X) \cong \mathrm{H}^*(T, T \setminus X) \cong \mathrm{H}^*(Y, Y \setminus X).$$

Now set $p = 0$. Then there is a diagram

$$\begin{tikzcd} \mathrm{H}^0(X) \arrow[r, "\sim"] \arrow[rrr, bend left, "\iota_*"] & \mathrm{H}^d(N_{X/Y}, N_{X/Y} \setminus X) \arrow[r, "\sim"] \arrow[d, "u_{N_{X/Y}}"] & \mathrm{H}^d(Y, Y \setminus X) \arrow[r] & \mathrm{H}^d(Y) \arrow[d, "\iota^*"] \\ & \mathrm{H}^d(N_{X/Y}) \arrow[rr] & & \mathrm{H}^d(X) \end{tikzcd}$$

whose commutativity implies the content of the *self-intersection formula*.

THEOREM 3.2.1 (Self-intersection formula). *Let $\iota : X \hookrightarrow Y$ be a closed embedding, of codimension d, of connected closed oriented manifolds. The operator $\iota^*\iota_*$ agrees with multiplication by the Euler class of $N_{X/Y}$. In particular, there is an identity*

$$\iota^*\iota_*\mathbf{1} = e(N_{X/Y}) \in \mathrm{H}^d(X).$$

See Fulton [42] for the corresponding algebraic statement.

3.2.2 Equivariant pushforward

Let G be a compact Lie group. Let $f : X \to Y$ be a G-equivariant map of compact manifolds. Set

$$n = \dim X, \quad m = \dim Y, \quad d = m - n.$$

In order to construct an equivariant pushforward $f_*^G : \mathrm{H}_G^*(X) \to \mathrm{H}_G^*(Y)$ we cannot apply the same procedure leading to Diagram 3.2.1, because Poincaré duality is not available on infinite dimensional spaces such as $X_G = \mathrm{E}G \times^G X$ and $Y_G = \mathrm{E}G \times^G Y$.

We use approximation spaces to solve this issue. Fix a directed system of principal G-bundles

$$\{ E_i \to B_i \}_{i \geqslant 0}$$

whose limit recovers the classifying space $\mathrm{E}G \to \mathrm{B}G$ (cf. Remark 3.1.30). Since E_i are compact spaces, so are the Borel spaces

$$X_G^i = E_i \times^G X, \quad Y_G^i = E_i \times^G Y,$$

and the approximation theorem (Theorem 3.1.29) ensures that

$$\mathrm{H}_G^p(X) \cong \mathrm{H}^p(X_G^i), \quad \mathrm{H}_G^p(Y) \cong \mathrm{H}^p(Y_G^i), \quad p \leqslant i. \tag{3.2.2}$$

Recall from (3.1.3) that we have fibrations

$$\begin{tikzcd}
X \arrow[r, hook] & X_G^i \arrow[rr, "f^i"] \arrow[dr, "p_X"'] & & Y_G^i \arrow[dl, "p_Y"] & Y \arrow[l, hook'] \\
& & B_i & &
\end{tikzcd}$$

with fibre X and Y, respectively. It follows that

$$\dim X_G^i = \ell_i + n, \quad \dim Y_G^i = \ell_i + m,$$

where $\ell_i = \dim B_i$. Exploiting Poincaré duality on the compact (finite dimensional) spaces X_G^i and Y_G^i, we can replace the diagram (3.2.1) by a new diagram

$$\begin{tikzcd}
\mathrm{H}^p(X_G^i) \arrow[r, dashed] \arrow[d, "\mathsf{pd}"', "\wr"] & \mathrm{H}^{p+d}(Y_G^i) \\
\mathrm{H}_{\ell_i+n-p}(X_G^i) \arrow[r, "f_*^i"] & \mathrm{H}_{\ell_i+n-p}(Y_G^i) \arrow[u, "\wr"', "\mathsf{pd}^{-1}"]
\end{tikzcd}$$

which we can of course redraw as

$$\begin{tikzcd}
\mathrm{H}_G^p(X) \arrow[r, dashed, "f_*^{G,p}"] \arrow[d, "\mathsf{pd}"', "\wr"] & \mathrm{H}_G^{p+d}(Y) \\
\mathrm{H}_{\ell_i+n-p}(X_G^i) \arrow[r, "f_*^i"] & \mathrm{H}_{\ell_i+n-p}(Y_G^i) \arrow[u, "\wr"', "\mathsf{pd}^{-1}"]
\end{tikzcd}$$

by exploiting (3.2.2). This yields a system of maps

$$f_*^{G,p}\colon \mathrm{H}_G^p(X) \longrightarrow \mathrm{H}_G^{p+d}(Y).$$

Exercise 3.2.2. Prove that the maps $f_*^{G,p}$ are compatible with the structure of inverse system of $\mathrm{H}^p(X_G^i)$ and $\mathrm{H}^p(Y_G^i)$ attached to the directed systems $(X_G^i)_i$ and $(Y_G^i)_i$ respectively.

By Exercise 3.2.2, we can construct the *equivariant pushforward*

$$f_*^G\colon \mathrm{H}_G^*(X) \longrightarrow \mathrm{H}_G^*(Y).$$

The superscript 'G' will often be omitted.

3.2.3 Trivial torus actions

Let $\mathbb{T}$ be a torus acting *trivially* on a smooth complex algebraic variety X. Given a $\mathbb{T}$-equivariant vector bundle $E \to X$ of rank r, we get a canonical decomposition

$$E = \bigoplus_{\chi} E_\chi, \tag{3.2.3}$$

where $\mathbb{T}$ acts by the character χ on E_χ. The characters χ vary in the character group $\widehat{\mathbb{T}} = \mathrm{Hom}(\mathbb{T}, \mathbb{G}_m) = \mathbb{Z}^{\dim \mathbb{T}}$ of the torus. One should expect to be able to express the $\mathbb{T}$-equivariant Chern classes of E in terms of the $\mathbb{T}$-equivariant Chern classes of its subbundles $E_\chi \subset E$.

Let

$$V_E = \mathrm{E}\mathbb{T} \times^{\mathbb{T}} E \longrightarrow \mathrm{E}\mathbb{T} \times^{\mathbb{T}} X = \mathrm{B}\mathbb{T} \times X$$

be the induced rank r bundle, cf. (3.1.13) (and (3.1.5) for the identification on the right hand side).

Exercise 3.2.3. Show that $V_{E_\chi} \cong E_\chi \boxtimes V_\chi$, where the box product refers to the canonical projections from $\mathrm{B}\mathbb{T} \times X$ and V_χ is the line bundle on $\mathrm{B}\mathbb{T}$ introduced in Example 3.1.37.

Exercise 3.2.4. Assume again $\mathbb{T}$ acts trivially on a smooth variety X, and let E be a $\mathbb{T}$-equivariant vector bundle. Show that

$$c_i^{\mathbb{T}}(E_\chi) = \sum_{k=0}^{i} \binom{r_\chi - k}{i-k} c_k(E_\chi)\chi^{i-k} \in \mathrm{H}_{\mathbb{T}}^{2i}(X),$$

where $r_\chi = \mathrm{rk}\, E_\chi$, and we denote simply by χ the weight $w_\chi \in \mathrm{H}_{\mathbb{T}}^2$ of Example 3.1.37. (**Hint**: use the previous exercise and a standard property of Chern classes of tensor products, cf. [42, Example 3.2.2]).

3.2.4 Torus fixed loci

Throughout this section, we let X be a smooth complex algebraic variety acted on by a torus $\mathbb{T}$. The scheme structure of the fixed locus

$$X^{\mathbb{T}} \subset X$$

is discussed in [41, Section 2]. We shall grant the following important result in what follows.

THEOREM 3.2.5 ([57, 41]). *If X is a smooth $\mathbb{T}$-variety, the fixed locus $X^{\mathbb{T}}$ is smooth.*

Exercise 3.2.6. Let Y be a compact normal $\mathbb{T}$-variety. Show that if Y has a singular point, then it has a *torus-fixed* singular point.

Let N be the normal bundle of the inclusion $F \subset X$ of a component $F \subset X^{\mathbb{T}}$. Then N is $\mathbb{T}$-equivariant and for each $x \in F$ one has $(T_x X)^{\mathbb{T}} = T_x F$ by results of [41], so that the action of $\mathbb{T}$ on the normal space

$$N_x = T_x X / T_x F$$

is nontrivial, i.e. N_x has no trivial subrepresentations, i.e. $N_x^{\mathbb{T}} = 0$. It follows that the Euler class of the normal bundle N is nonzero, being a product of nonzero weights. In fact, $e^{\mathbb{T}}(N)$ becomes invertible in the ring $\mathrm{H}^*_{\mathbb{T}}(F)[\chi_i^{-1}]$, where χ_i are the characters that occur in the decomposition of N into eigenbundles.

Let E be a $\mathbb{T}$-equivariant vector bundle of rank r over X. Let $F \subset X^{\mathbb{T}}$ be a component of the fixed locus, so that $\mathrm{H}^*_{\mathbb{T}}(F) = \mathrm{H}^*(F) \otimes \mathrm{H}^*_{\mathbb{T}}$. According to Equation (3.2.3), the vector bundle $E|_F$ on F has a decomposition

$$E\big|_F = \bigoplus_{\chi} E_{F,\chi}$$

into eigen-subbundles. By Exercise 3.2.4, the component of

$$c_i^{\mathbb{T}}(E_{F,\chi}) \in \mathrm{H}^{2i}_{\mathbb{T}}(F) \tag{3.2.4}$$

in $\mathrm{H}^{2i}_{\mathbb{T}}$ is given by

$$\binom{\operatorname{rk} E_{F,\chi}}{i} \chi^i \in \mathrm{H}^{2i}_{\mathbb{T}}. \tag{3.2.5}$$

Here we are denoting simply by χ the weight $w_\chi \in \mathrm{H}^2_{\mathbb{T}}$, cf. Example 3.1.37. Since $\mathrm{H}^{2k}(F) = 0$ for $k > \dim F$, we have that for all $j > 0$, the classes in $\mathrm{H}^{2j}(F)$ are nilpotent in $\mathrm{H}^*_{\mathbb{T}}(F)$. Therefore the element (3.2.4) is invertible if and only if its component (3.2.5) is invertible. It follows that

$$c_i^{\mathbb{T}}(E_{F,\chi}) \text{ is invertible in the localised ring } \mathrm{H}^*_{\mathbb{T}}(F)\big[\chi^{-1}\big]. \tag{3.2.6}$$

PROPOSITION 3.2.7. *Let X be a smooth $\mathbb{T}$-variety, let $F \subset X^{\mathbb{T}}$ be a component of codimension d. Then there are finitely many characters $\chi_1, \dots, \chi_s$ such that the Euler class*

$$e^{\mathbb{T}}(N_{F/X}) \in \mathrm{H}^{2d}_{\mathbb{T}}(F)$$

becomes invertible in the ring

$$\mathrm{H}^*_{\mathbb{T}}(F)\big[\chi_s^{-1}, \dots, \chi_s^{-1}\big].$$

Proof. Set $N = N_{F/X}$. We saw above that the action on the normal space N_x is nontrivial for all $x \in F$, so the characters appearing in the decomposition

$$N = \bigoplus_{i=1}^{s} N_{\chi_i}$$

are all nontrivial, and $e^{\mathbb{T}}(N)$ is a product of nonzero weights,

$$0 \neq e^{\mathbb{T}}(N) = \prod_{i=1}^{s} e^{\mathbb{T}}(N_{\chi_i}),$$

so the observation (3.2.6) implies the result. □

3.2.5 The localisation formula

This subsection introduces the technique of localisation in Algebraic Geometry. Our purpose is to apply this powerful tool to solve enumerative problems, such as finding the number of lines on a general cubic surface $S \subset \mathbb{P}^3$, or on a general quintic 3-fold $Y \subset \mathbb{P}^4$.

Let $\iota\colon X^{\mathbb{T}} \hookrightarrow X$ be the inclusion of the fixed point locus. We have the equivariant pushforward

$$\iota_*\colon \mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}}) \longrightarrow \mathrm{H}^*_{\mathbb{T}}(X),$$

and the localisation theorem states that this map becomes an isomorphism after inverting finitely many nontrivial characters. In particular, it becomes an isomorphism after extending scalars to the field of fractions

$$\mathcal{H}_{\mathbb{T}} = \operatorname{Frac} \mathrm{H}^*_{\mathbb{T}} = \mathbb{Q}(s_1, \dots, s_g), \quad g = \dim \mathbb{T}.$$

Notice that

$$\mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}}) = \bigoplus_{\alpha} \mathrm{H}^*_{\mathbb{T}}(F_\alpha),$$

where F_α are the components of the fixed locus. The crucial point is that, if N_α is the normal bundle of the inclusion $\iota_\alpha\colon F_\alpha \hookrightarrow X$, then

$$\iota_\alpha^* \iota_{\alpha *}(-) = e^{\mathbb{T}}(N_\alpha) \cap -$$

and as we saw this Euler class is nonzero when restricted to any point $x \in F_\alpha$. This is enough for $e^{\mathbb{T}}(N_\alpha)$ to become invertible after a suitable localisation.

The statement of the localisation theorem for compact manifolds with torus action is the following. See also the work of Edidin–Graham [30] for the localisation formula in Chow.

THEOREM 3.2.8 (Atiyah–Bott [6]). *Let M be a compact smooth manifold equipped with an action of a torus $\mathbb{T}$. Then the equivariant pushforward along $\iota\colon M^{\mathbb{T}} \hookrightarrow M$ induces an isomorphism*

$$\iota_*\colon \mathrm{H}^*_{\mathbb{T}}(M^{\mathbb{T}}) \otimes_{\mathrm{H}^*_{\mathbb{T}}} \mathcal{H}_{\mathbb{T}} \xrightarrow{\sim} \mathrm{H}^*_{\mathbb{T}}(M) \otimes_{\mathrm{H}^*_{\mathbb{T}}} \mathcal{H}_{\mathbb{T}}.$$

Its inverse is given by

$$\psi \mapsto \sum_{\alpha} \frac{\iota_\alpha^* \psi}{e^{\mathbb{T}}(N_\alpha)}.$$

Remark 3.2.9. In particular, every class $\psi \in \mathrm{H}^*_{\mathbb{T}}(M) \otimes_{\mathrm{H}^*_{\mathbb{T}}} \mathcal{H}_{\mathbb{T}}$ writes uniquely as

$$\psi = \sum_{\alpha} \iota_{\alpha *} \frac{\iota_\alpha^* \psi}{e^{\mathbb{T}}(N_\alpha)}. \tag{3.2.7}$$

Let M be a compact manifold with a $\mathbb{T}$-action and structure map $q\colon M \to \mathrm{pt}$. We have the equivariant pushforward $q_*\colon \mathrm{H}^*_{\mathbb{T}}(M) \to \mathrm{H}^*_{\mathbb{T}}$, which after tensoring by $\mathcal{H}_{\mathbb{T}}$ yields the integration map

$$\int_M \colon \mathrm{H}^*_{\mathbb{T}}(M) \otimes_{\mathrm{H}^*_{\mathbb{T}}} \mathcal{H}_{\mathbb{T}} \to \mathcal{H}_{\mathbb{T}}.$$

For any component $F_\alpha \subset M^{\mathbb{T}}$, the structure map $q_\alpha\colon F_\alpha \to \mathrm{pt}$ factors as $q \circ \iota_\alpha$ where $\iota_\alpha\colon F_\alpha \hookrightarrow M$ is the inclusion. The condition $q_{\alpha *} = q_* \circ \iota_{\alpha *}$ then allows us, simply by looking at (3.2.7), to deduce the following integration formula: for any equivariant class $\psi \in \mathrm{H}^*_{\mathbb{T}}(M) \otimes_{\mathrm{H}^*_{\mathbb{T}}} \mathcal{H}_{\mathbb{T}}$, one has

$$\int_M \psi = \sum_{\alpha} q_{\alpha *} \frac{\iota_\alpha^* \psi}{e^{\mathbb{T}}(N_\alpha)} = \sum_{\alpha} \int_{F_\alpha} \frac{\iota_\alpha^* \psi}{e^{\mathbb{T}}(N_\alpha)} \in \mathcal{H}_{\mathbb{T}}. \tag{3.2.8}$$

Let us go back to the algebraic setting (in the special case of finitely many fixed points).

PROPOSITION 3.2.10. *Let X be a smooth $\mathbb{T}$-variety with finitely many fixed points. Set*

$$\mathsf{e} = \prod_{p \in X^{\mathbb{T}}} e^{\mathbb{T}}(T_p X) \in \mathrm{H}^*_{\mathbb{T}},$$

and fix a multiplicative subset $S \subseteq \mathrm{H}^*_{\mathbb{T}} \setminus 0$ containing e. Then

(i) *The pullback map $S^{-1}\iota^* \colon S^{-1}\mathrm{H}^*_{\mathbb{T}}(X) \to S^{-1}\mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}})$ is onto, and*

(ii) *if $\mathrm{H}^*_{\mathbb{T}}(X)$ is a free $\mathrm{H}^*_{\mathbb{T}}$-module of rank $r \leqslant |X^{\mathbb{T}}|$, then $r = |X^{\mathbb{T}}|$ and $S^{-1}\iota^*$ is an isomorphism.*

Proof. To prove (i), note that the composition

$$S^{-1}\mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}}) \xrightarrow{S^{-1}\iota_*} S^{-1}\mathrm{H}^*_{\mathbb{T}}(X) \xrightarrow{S^{-1}\iota^*} S^{-1}\mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}})$$

is surjective because it equals $S^{-1}(\iota^* \circ \iota_*)$, and the determinant of $\iota^* \circ \iota_*$ is precisely e because it is a diagonal map, equal to $e^{\mathbb{T}}(T_p X)$ on the component indexed by p. But e becomes invertible after localisation by Proposition 3.2.7, so (i) follows.

By Part (i), we have

$$r = \mathrm{rk}_{S^{-1}\mathrm{H}^*_{\mathbb{T}}} S^{-1}\mathrm{H}^*_{\mathbb{T}}(X) \geqslant \mathrm{rk}_{S^{-1}\mathrm{H}^*_{\mathbb{T}}} S^{-1}\mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}}) = |X^{\mathbb{T}}| \geqslant r,$$

which implies $\mathrm{rk}_{S^{-1}\mathrm{H}^*_{\mathbb{T}}} S^{-1}\mathrm{H}^*_{\mathbb{T}}(X) = |X^{\mathbb{T}}|$. To prove Part (ii), observe that $S^{-1}\mathrm{H}^*_{\mathbb{T}}$ is a noetherian ring (being a localisation of a noetherian ring), hence a surjective $S^{-1}\mathrm{H}^*_{\mathbb{T}}$-linear map of free modules of the same rank, such as $S^{-1}\iota^*$, is necessarily an isomorphism. □

Remark 3.2.11. Let X be a smooth complex projective variety, with an action by a torus $\mathbb{T}$ having finitely many fixed points $p_1, \dots, p_s$. Then the Białynicki-Birula decomposition [11] yields s $\mathbb{T}$-invariant subvarieties

$$Y_1, \dots, Y_s \subset X$$

with the property that the $\mathbb{T}$-equivariant cohomology classes (cf. Example 3.1.35)

$$[Y_\ell]^{\mathbb{T}} \in \mathrm{H}^*_{\mathbb{T}}(X)$$

form a *free* $\mathrm{H}^*_{\mathbb{T}}$*-basis* of the equivariant cohomology ring, restricting to a $\mathbb{Z}$-basis of the ordinary cohomology ring $\mathrm{H}^*(X)$. In other words,

$$\mathrm{H}^*_{\mathbb{T}}(X) \cong \bigoplus_{\ell=1}^{s} \mathrm{H}^*_{\mathbb{T}} \cdot [Y_\ell]^{\mathbb{T}}$$

is a free $\mathrm{H}^*_{\mathbb{T}}$-module of rank s.

The form of the localisation theorem that we will need is the following.

COROLLARY 3.2.12 (Integration Formula). *Let X be a smooth complex projective $\mathbb{T}$-variety with finitely many fixed points. Then for all $\psi \in \mathrm{H}^*_{\mathbb{T}}(X)$ there is an identity*

$$\int_X \psi = \sum_{q \in X^{\mathbb{T}}} \frac{i_q^* \psi}{e^{\mathbb{T}}(T_q X)} \in \mathcal{H}_{\mathbb{T}}. \tag{3.2.9}$$

Proof. Let $S \subseteq \mathrm{H}^*_{\mathbb{T}} \setminus 0$ be a multiplicative subset as in Theorem 3.2.10. By the surjectivity of

$$S^{-1}\iota_* \colon S^{-1}\mathrm{H}^*_{\mathbb{T}}(X^{\mathbb{T}}) \xrightarrow{\sim} S^{-1}\mathrm{H}^*_{\mathbb{T}}(X),$$

due to part (ii) of Proposition 3.2.10 (which we may apply thanks to Remark 3.2.11), we may assume

$$\psi = \iota_{p*}\theta,$$

for $p \in X^{\mathbb{T}}$ and $\theta \in \mathrm{H}^*_{\mathbb{T}}$. Then clearly

$$\int_X \psi = \int_X \iota_{p*}\theta = \theta,$$

because $(\int_X) \circ \iota_{p*}$ is an isomorphism. On the other hand,

$$\sum_{q \in X^{\mathbb{T}}} \frac{i_q^* \iota_{p*}\theta}{e^{\mathbb{T}}(T_q X)} = \frac{\iota_p^* \iota_{p*}\theta}{e^{\mathbb{T}}(T_p X)} = \frac{e^{\mathbb{T}}(T_p X)}{e^{\mathbb{T}}(T_p X)} \cap \theta = \theta,$$

as required. □

Remark 3.2.13. Formula (3.2.8) is true in the algebraic context as well (for X a smooth projective variety) if one sets $S = \mathrm{H}^*_{\mathbb{T}} \setminus 0$, but we will not deal with positive dimensional fixed loci in these notes. See e.g. [68].

LEMMA 3.2.14. *Let M be a smooth oriented compact manifold with a torus action having finitely many fixed points $p_1, \dots, p_s$. Then*

$$\chi(M) = s.$$

Proof. Recall that $\chi(M) = \int_M e(T_M)$ by Poincaré–Hopf. We have

$$\int_M e(T_M) = \int_M e^{\mathbb{T}}(T_M) = \sum_{1 \leqslant i \leqslant s} \frac{e^{\mathbb{T}}(T_M)|_{p_i}}{e^{\mathbb{T}}(N_{p_i/M})} = \sum_{1 \leqslant i \leqslant s} 1 = s,$$

as desired. □

In the algebraic context, we can use the localisation formula to prove the following.

PROPOSITION 3.2.15. *Let $\mathbb{T}$ be a torus acting on a smooth complex projective variety Y. Then*

$$\chi(Y) = \chi(Y^{\mathbb{T}}).$$

Proof. By the ($\mathbb{T}$-equivariant) normal bundle exact sequence

$$0 \longrightarrow T_{Y^{\mathbb{T}}} \longrightarrow T_Y|_{Y^{\mathbb{T}}} \longrightarrow N_{Y^{\mathbb{T}}/Y} \longrightarrow 0,$$

we have an identity

$$e^{\mathbb{T}}(T_Y|_{Y^{\mathbb{T}}}) = e^{\mathbb{T}}(T_{Y^{\mathbb{T}}}) \cdot e^{\mathbb{T}}(N_{Y^{\mathbb{T}}/Y}). \tag{3.2.10}$$

We compute

$$\begin{aligned}
\chi(Y) &= \int_Y e(T_Y) && \text{by Poincaré–Hopf} \\
&= \int_Y e^{\mathbb{T}}(T_Y) && \text{by equivariance of } T_Y \\
&= \int_{Y^{\mathbb{T}}} \frac{e^{\mathbb{T}}(T_Y|_{Y^{\mathbb{T}}})}{e^{\mathbb{T}}(N_{Y^{\mathbb{T}}/Y})} && \text{by localisation} \\
&= \int_{Y^{\mathbb{T}}} e^{\mathbb{T}}(T_{Y^{\mathbb{T}}}) && \text{by Equation (3.2.10)} \\
&= \int_{Y^{\mathbb{T}}} e(T_{Y^{\mathbb{T}}}) && \text{by equivariance of } T_{Y^{\mathbb{T}}} \\
&= \chi(Y^{\mathbb{T}}) && \text{by Poincaré–Hopf,}
\end{aligned}$$

as required. □

The last result holds in greater generality:

LEMMA 3.2.16 ([23, Proposition 2.5.1]). *Let $\mathbb{T}$ be a torus acting on a quasiprojective scheme Y of finite type over $\mathbb{C}$. Then*

$$\chi(Y) = \chi(Y^{\mathbb{T}}).$$

The following result is also very useful in computations.

LEMMA 3.2.17. *Let Y be a variety with a $\mathbb{T}$-action. Suppose the fixed locus $Y^{\mathbb{T}}$ is finite. Then there is a 1-dimensional subtorus $\mathbb{G}_m \subset \mathbb{T}$ such that $Y^{\mathbb{T}} = Y^{\mathbb{G}_m}$.*

Proof. Let $Y^{\mathbb{T}} = \{ y_1, \dots, y_e \}$ be the fixed locus. For each $i = 1, \dots, e$, the tangent space $T_{y_i} Y$, as a representation, splits as a sum of characters $\chi_1^{(i)}, \dots, \chi_{r_i}^{(i)}$. Consider the set $\Upsilon = \{ \chi_j^{(i)} \mid 1 \leqslant i \leqslant e, 1 \leqslant j \leqslant r_i \}$. Then, for every 1-parameter subgroup $\lambda \colon \mathbb{G}_m \to \mathbb{T}$, we have that $Y^{\mathbb{G}_m} = Y^{\mathbb{T}}$ if and only if $\chi \circ \lambda \neq 0 \in \mathbb{Z} = \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m)$ for every $\chi \in \Upsilon$. So it is enough to pick $\lambda \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{T}) \cong \mathbb{Z}^{\dim \mathbb{T}}$ away from $|\Upsilon|$ hyperplanes. □

Exercise 3.2.18. Compute the Euler characteristic of the Grassmannian $G(k, n)$. (**Hint**: Lift the standard torus action of $\mathbb{T} = \mathbb{G}_m^n$ on $\mathbb{C}^n$ to the Grassmannian).

Exercise 3.2.19. Let $n > 0$ be an integer. Show that

$$\sum_{i=1}^{n} \frac{(-s_i)^k}{\prod_{\substack{1 \leqslant j \leqslant n \\ j \neq i}} (s_j - s_i)} = \begin{cases} 0 & \text{if } 0 \leqslant k < n-1 \\ 1 & \text{if } k = n-1. \end{cases}$$

(**Hint**: Apply Equation (3.2.9) to $\psi = c_1^{\mathbb{T}}(\mathcal{O}_{\mathbb{P}^{n-1}}(1))$ having fixed the standard action on $X = \mathbb{P}^{n-1}$).

3.3 Applications of the localisation formula

In this chapter we give four complete examples on how to use Theorem 3.2.8 to solve enumerative problems. The work of Meireles Araújo and Vainsencher [68] was of great inspiration for the first three sections in this chapter, and we take the opportunity to refer the reader to loc. cit. for more examples of application of the localisation formula (upgraded to equivariant Chow theory) in enumerative geometry.

3.3.1 How not to compute the simplest intersection number

In this section we show how to use localisation to compute the number of intersection points between two general lines in $\mathbb{P}^2$. Note that this problem is (equivalent, and) dual to the problem of counting lines through two general points in $\mathbb{P}^2$. This explains why we work with the usual weights on $\mathbb{P}^2$ instead of the dual weights on $\mathbb{P}^{2*} = \mathbb{G}(1,2)$, which would have required us to start out with $\mathrm{H}^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(1))^*$ instead of $\mathrm{H}^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(1))$.

Let $\mathbb{T} = \mathbb{G}_m \subset \mathbb{G}_m^3$ be a one-parameter subgroup acting with weights w_0, w_1, w_2 on the vector space of linear forms

$$V = \mathrm{H}^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(1)) = \mathsf{Span}_{\mathbb{C}} \{ x_0, x_1, x_2 \}.$$

This means that $t \cdot x_i = t^{w_i} x_i$. We might also use the notation

$$V = \bigoplus_{i=0}^{2} \mathbb{C} \cdot x_i \otimes t^{w_i}$$

if we were to emphasise the equivariant splitting. Choosing the w_i's distinct from one another ensures that the induced action on $\mathbb{P}^2$ has the three fixed points $p_0 = (1,0,0), p_1 = (0,1,0), p_2 = (0,0,1)$. The intersection number we want to compute is

$$\int_{\mathbb{P}^2} c_1(\mathscr{O}_{\mathbb{P}^2}(1))^2 = \int_{\mathbb{P}^2} c_1^{\mathbb{T}}(\mathscr{O}_{\mathbb{P}^2}(1))^2 = \sum_{i=0}^{2} \frac{c_1^{\mathbb{T}}(\mathscr{O}_{\mathbb{P}^2}(1)|_{p_i})^2}{e^{\mathbb{T}}(T_{p_i}\mathbb{P}^2)}, \tag{3.3.1}$$

where the last identity is the application of the localisation formula. The universal exact sequence

$$0 \to \mathscr{S} \to V \otimes_{\mathbb{C}} \mathscr{O}_{\mathbb{P}^2} \to \mathscr{O}_{\mathbb{P}^2}(1) \to 0$$

restricts, at p_i, to a short exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & V_{jk} & \longrightarrow & V & \longrightarrow & V/V_{jk} & \longrightarrow & 0 \\ & & & & & & \| & & \\ & & & & & & \mathbb{C} \cdot x_i & & \end{array}$$

where $V_{jk} \subset V$ is the span of x_j, x_k, i.e. the space of linear forms vanishing at p_i. The weights of the tangent representations

$$T_{p_i}\mathbb{P}^2 = V_{jk}^* \otimes V/V_{jk} = \mathsf{Span}_{\mathbb{C}}\{x_j^* \otimes x_i, x_k^* \otimes x_i\}$$

are simply $w_i - w_j, w_i - w_k$, where $\{i,j,k\} = \{0,1,2\}$. The line bundle $\mathscr{O}_{\mathbb{P}^2}(1)$ is the universal quotient bundle and clearly

$$\mathscr{O}_{\mathbb{P}^2}(1)|_{p_i} = V/V_{jk} = \mathbb{C} \cdot x_i$$

has weight $c_1^{\mathbb{T}}(\mathscr{O}_{\mathbb{P}^2}(1)|_{p_i}) = w_i$. The sum in the right hand side of (3.3.1) then equals

$$\frac{w_0^2}{(w_0-w_1)(w_0-w_2)} + \frac{w_1^2}{(w_1-w_0)(w_1-w_2)} + \frac{w_2^2}{(w_2-w_0)(w_2-w_1)}. \tag{3.3.2}$$

The latter equals 1 for *every* choice of (pairwise distinct) w_0, w_1, w_2.

Exercise 3.3.1. Compare the latter calculation to the one of Exercise 3.2.19.

Remark 3.3.2. The fact that the sum (3.3.2) equals 1 has the following interpretation. The three fractions in Equation (3.3.2) can be seen as the residues of the differential form

$$\frac{z^2\,\mathrm{d}z}{(z-w_0)(z-w_1)(z-w_2)}.$$

However, there is yet another residue to compute: the one at ∞. This residue equals -1. The residue theorem then precisely states that

$$0 = -1 + \frac{w_0^2}{(w_0-w_1)(w_0-w_2)} + \frac{w_1^2}{(w_1-w_0)(w_1-w_2)} + \frac{w_2^2}{(w_2-w_0)(w_2-w_1)}.$$

3.3.2 The 27 lines on a smooth cubic surface

Let $\mathbb{T} = \mathbb{G}_m$ be a torus acting on $\mathbb{P}^3$ with distinct weights (w_0, w_1, w_2, w_3). This means, as ever,

$$t \cdot x_i = t^{w_i} x_i, \quad 0 \leqslant i \leqslant 3.$$

This is also equivalent to considering the vector space

$$V = \mathrm{H}^0(\mathbb{P}^3, \mathscr{O}_{\mathbb{P}^3}(1)) = \mathsf{Span}_{\mathbb{C}}\{x_0, x_1, x_2, x_3\}$$

as an equivariant vector bundle over a point, splitting into a sum of characters $t \mapsto t^{w_i}$. The torus action has four fixed points $p_0, \ldots, p_3 \in \mathbb{P}^3$ and six invariant lines $\ell_{ij} \subset \mathbb{P}^3$ which are the lines joining the fixed points (see Figure **??**). These correspond to the fixed points of the Grassmannian $\mathbb{G}(1,3)$ under the lifted $\mathbb{T}$-action. You computed the Euler characteristic of the Grassmannian in Exercise 3.2.18.

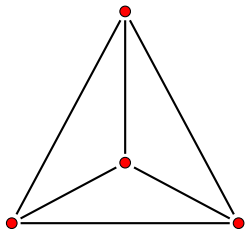


Figure 3.1: The toric polytope of $\mathbb{P}^3$. The red dots are the fixed points, and the lines joining them are the invariant lines, which correspond to the fixed points of $\mathbb{G}(1,3)$.

Let S be a general cubic hypersurface in $\mathbb{P}^3$. We know from Section 1.2 that the cycle of lines in S is Poincaré dual to the top Chern class

$$e(\mathsf{Sym}^3 \mathscr{S}^*) = c_4(\mathsf{Sym}^3 \mathscr{S}^*) \in A^4\mathbb{G}(1,3),$$

where $\mathscr{S}^*$ is the dual of the tautological subbundle. By Lemma 1.2.4, we have

$$\#\{\text{lines in } S\} = \int_{\mathbb{G}(1,3)} e(\mathsf{Sym}^3 \mathscr{S}^*).$$

According to the strategy outlined in Section 3.1.2, we will compute the latter intersection number by computing instead the equivariant integral

$$\int_{\mathbb{G}(1,3)} e^{\mathbb{T}}(\mathsf{Sym}^3 \mathscr{S}^*) \in \mathcal{H}_{\mathbb{T}}$$

via the localisation formula, and specialising the equivariant parameters appropriately (i.e. avoiding the creation of poles in the localisation formula).

Restricting the tautological exact sequence

$$0 \longrightarrow \mathscr{S} \longrightarrow \mathscr{O}_{\mathbb{G}(1,3)} \otimes_{\mathbb{C}} V^* \longrightarrow \mathscr{Q} \longrightarrow 0 \tag{3.3.3}$$

to the point $[\ell_{ij}] \in \mathbb{G}(1,3)$ corresponding to the linear subspace

$$\ell_{ij} = \mathbb{C} \cdot x_i^* \oplus \mathbb{C} \cdot x_j^* \subset V^*,$$

we obtain the sequence of $\mathbb{T}$-representations

$$0 \longrightarrow \mathbb{C}\cdot x_i^* \oplus \mathbb{C}\cdot x_j^* \longrightarrow V^* \longrightarrow \mathbb{C}\cdot x_h^* \oplus \mathbb{C}\cdot x_k^* \longrightarrow 0$$

where $\{0,1,2,3\} = \{i,j,h,k\}$. Therefore

$$\mathscr{Q}\big|_{\ell_{ij}} = \mathbb{C}\cdot x_h^* \oplus \mathbb{C}\cdot x_k^*$$

has weights $-w_h, -w_k$, and similarly

$$\mathscr{S}^*\big|_{\ell_{ij}} = \mathbb{C}\cdot x_i \oplus \mathbb{C}\cdot x_j$$

has weights w_i and w_j. Since

$$T_{\ell_{ij}}\mathbb{G}(1,3) = \mathscr{S}^*\big|_{\ell_{ij}} \otimes \mathscr{Q}\big|_{\ell_{ij}} = \mathsf{Span}_{\mathbb{C}}\left\{ x_i \otimes x_h^*, x_j \otimes x_h^*, x_i \otimes x_k^*, x_j \otimes x_k^* \right\},$$

we obtain an identity

$$e^{\mathbb{T}}(T_{\ell_{ij}}\mathbb{G}(1,3)) = (w_i - w_h)(w_j - w_h)(w_i - w_k)(w_j - w_k) \in \mathrm{H}^*_{\mathbb{T}}. \tag{3.3.4}$$

This is one of the key ingredients in the localisation formula, which reads

$$\int_{\mathbb{G}(1,3)} e^{\mathbb{T}}(\mathsf{Sym}^3 \mathscr{S}^*) = \sum_{\ell_{ij}} \frac{e^{\mathbb{T}}(\mathsf{Sym}^3 \mathscr{S}^*)|_{\ell_{ij}}}{e^{\mathbb{T}}(T_{\ell_{ij}}\mathbb{G}(1,3))}. \tag{3.3.5}$$

The numerators in (3.3.5) are computed as follows. Note that $\mathsf{Sym}^3 \mathscr{S}^*|_{\ell_{ij}} = \mathsf{Sym}^3(\mathbb{C}\cdot x_i \oplus \mathbb{C}\cdot x_j)$ is the 4-dimensional vector space generated by the classes of $x_i^3, x_i^2 x_j, x_i x_j^2, x_j^3$. Using the weights w_i and w_j of $\mathscr{S}^*|_{\ell_{ij}}$ we find

$$e^{\mathbb{T}}(\mathsf{Sym}^3 \mathscr{S}^*)\big|_{\ell_{ij}} = (3w_i)(2w_i + w_j)(w_i + 2w_j)(3w_j) \in \mathrm{H}^*_{\mathbb{T}}.$$

We are now able to write down the right hand side of (3.3.5) as follows:

$$\begin{aligned}
\sum_{0\leqslant i<j\leqslant 3} \frac{e^{\mathbb{T}}(\mathsf{Sym}^3 \mathscr{S}^*)|_{\ell_{ij}}}{e^{\mathbb{T}}(T_{\ell_{ij}}\mathbb{G}(1,3))} &= \sum_{0\leqslant i<j\leqslant 3} \frac{(3w_i)(2w_i + w_j)(w_i + 2w_j)(3w_j)}{(w_i - w_h)(w_j - w_h)(w_i - w_k)(w_j - w_k)} \\
&= 9\frac{w_0(2w_0 + w_1)(w_0 + 2w_1)w_1}{(w_0 - w_2)(w_0 - w_3)(w_1 - w_2)(w_1 - w_3)} \\
&+ 9\frac{w_0(2w_0 + w_2)(w_0 + 2w_2)w_2}{(w_0 - w_1)(w_0 - w_3)(w_2 - w_1)(w_2 - w_3)} \\
&+ 9\frac{w_0(2w_0 + w_3)(w_0 + 2w_3)w_3}{(w_0 - w_1)(w_0 - w_2)(w_3 - w_1)(w_3 - w_2)} \\
&+ 9\frac{w_1(2w_1 + w_2)(w_1 + 2w_2)w_2}{(w_1 - w_0)(w_1 - w_3)(w_2 - w_0)(w_2 - w_3)} \\
&+ 9\frac{w_1(2w_1 + w_3)(w_1 + 2w_3)w_3}{(w_1 - w_0)(w_1 - w_2)(w_3 - w_0)(w_3 - w_2)} \\
&+ 9\frac{w_2(2w_2 + w_3)(w_2 + 2w_3)w_3}{(w_2 - w_0)(w_2 - w_1)(w_3 - w_0)(w_3 - w_1)}.
\end{aligned} \tag{3.3.6}$$

This sum equals 27 for *every* choice of pairwise distinct w_i's. For instance, evaluating at $(w_0, w_1, w_2, w_3) = (0, 2, -1, 1)$ yields

$$\#\{\text{lines in } S\} = 9\left(0 + 0 + 0 + 0 + \frac{40}{12} - \frac{1}{3}\right) = 30 - 3 = 27.$$

3.3.3 The 2875 ines on the quintic 3-fold

In this section we will prove the following classical result. See also Harris' work [48].

THEOREM 3.3.3. *Let $Y \subset \mathbb{P}^4$ be a generic quintic 3-fold. Then Y contains exactly 2875 lines.*

Remark 3.3.4. Unlike the case of cubic surfaces, the statement is not true for *all* smooth quintic 3-folds. Indeed, by work of Albano–Katz [2], the Fermat quintic

$$\left\{ x_0^5 + x_1^5 + x_2^5 + x_3^5 + x_4^5 = 0 \right\} \subset \mathbb{P}^4$$

contains fifty 1-dimensional families of lines.

We will need the following auxiliary result.

THEOREM 3.3.5 ([60, Theorem 2.1]). *Let Y be a generic quintic 3-fold. Then the open subscheme of the Hilbert scheme of Y parametrising smooth irreducible rational curves of degree d is finite, nonempty and reduced. If $C \subset Y$ is any such curve, then its normal bundle is $\mathcal{N}_{C/Y} = \mathcal{O}_C(-1) \oplus \mathcal{O}_C(-1)$.*

We should mention that the result on the normal bundle also holds for other local complete intersection Calabi–Yau 3-folds and various degrees $d \leqslant 7$, see [60, Theorem 2.2] for the precise statement. See also the three appendices in Katz [62], where he works with $d \leqslant 3$. There he computes the possible bidegrees of the normal bundle to a smooth rational curve C of degree d inside a smooth quintic Y. In particular, he finds that the only admissible $\mathcal{N}_{C/Y} = \mathcal{O}_C(a) \oplus \mathcal{O}_C(-2-a)$ are

$$\begin{aligned} d = 1 &\Rightarrow a \in \{-1, 0, 1\} \\ d = 2 &\Rightarrow a \in \{-1, 0, 1, 2\} \\ d = 3 &\Rightarrow a \in \{-1, 0, 1, 2, 3, 4, 5\}. \end{aligned}$$

This variety of possible normal bundles is not available in the local (i.e. non compact) cases studied by Laufer, where the only possible normal bundles are $\mathcal{O}_C(a) \oplus \mathcal{O}_C(-2-a)$ with $a \in \{-1, 0, 1\}$ [63]. See also [19] for an explicit calculation of the normal bundle to *Laufer curves* in certain local Calabi–Yau 3-folds.

The role of Theorem 3.3.5, for us, is to make sure that the intersection number we compute via localisation really is the number we are after. The above theorem, in other words, plays the role of Lemma 1.2.4 that we needed for lines on a cubic surface. Indeed, all other possible normal bundles of smooth rational curves in Calabi–Yau 3-folds have a nonzero global section.

Back to counting lines on a generic quintic 3-fold. The ambient space we have to work with now is the 6-dimensional Grassmannian

$$\mathbb{G}(1,4) = G(2,5).$$

Let $\mathcal{S}$ be the rank 2 universal subbundle

$$\mathcal{S} \hookrightarrow \mathcal{O}_{\mathbb{G}(1,4)} \otimes_{\mathbb{C}} \mathrm{H}^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(1))^*.$$

Let $\mathbb{T} = \mathbb{G}_m^5$ be a torus acting as

$$t \cdot x_i = t_i \cdot x_i$$

on $\mathbb{P}^4$. This means, as ever, that $V = \mathrm{H}^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(1))$ is viewed as a $\mathbb{T}$-module with equivariant decomposition $\bigoplus_{0 \leqslant k \leqslant 4} \mathbb{C} \cdot x_k \otimes t_k$. This action lifts to a $\mathbb{T}$-action on $\mathbb{G}(1,4)$, with 10 fixed points corresponding to the subsets

$$I \subset \{0, 1, 2, 3, 4\}, \quad |I| = 2.$$

Let $\ell_{ij} \subset V^*$ be the linear subspace spanned by x_i^* and x_j^*. The characters of the 2-dimensional $\mathbb{T}$-representation

$$\mathscr{S}\big|_{\ell_{ij}} = \mathbb{C}\cdot x_i^* \oplus \mathbb{C}\cdot x_j^*$$

are $\{-\chi_i, -\chi_j\}$, where $\chi_i : \mathbb{T} \to \mathbb{G}_m$ sends $t \mapsto t_i$. Similarly, we have that

$$\text{characters of } \mathscr{Q}\big|_{\ell_{ij}} = \{-\chi_a, -\chi_b, -\chi_c\}$$

where $\{a,b,c,i,j\} = \{0,1,2,3,4\}$. Let $s_k \in \mathrm{H}^2_{\mathbb{T}}$ denote the weight of $-\chi_k$, for all k. The tangent space of $\mathbb{G}(1,4)$ at ℓ_{ij} is, as ever,

$$\mathscr{S}^*\big|_{\ell_{ij}} \otimes \mathscr{Q}\big|_{\ell_{ij}} = \mathsf{Span}_{\mathbb{C}}\left\{ x_i \otimes x_u^*, x_j \otimes x_u^* \,\middle|\, u \in \{a,b,c\} \right\},$$

so we obtain the Euler class

$$e^{\mathbb{T}}(T_{\ell_{ij}}\mathbb{G}(1,4)) = \prod_{k \notin I} (s_i - s_k)(s_j - s_k).$$

The rank 6 vector bundle

$$\begin{array}{ccc} \mathsf{Sym}^5(\mathscr{S}^*) & & \mathsf{Sym}^5(\mathscr{S}^*)\big|_{\ell_{ij}} \\ \downarrow & \rightsquigarrow & \downarrow \\ \mathbb{G}(1,4) & & \{\ell_{ij}\} \end{array}$$

inherits weights

$$5s_j, s_i + 4s_j, 2s_i + 3s_j, 3s_i + 2s_j, 4s_i + s_j, 5s_i.$$

Therefore, denoting i_1 and i_2 the generic elements of a subset $I \subset \{0,1,2,3,4\}$ of size 2, the localisation formula reads

$$\int_{\mathbb{G}(1,4)} e^{\mathbb{T}}(\mathsf{Sym}^5 \mathscr{S}^*) = \sum_{|I|=2} \frac{\prod_{h=0}^{5}(h s_{i_1} + (5-h) s_{i_2})}{\prod_{i \in I}\prod_{k \notin I}(s_i - s_k)} = 2875. \tag{3.3.7}$$

If you do not believe the last identity, you can copy the following code[1] in Mathematica:

```
Plus @@
(Product[h Subscript[t, #[[1]]] + (5 - h)
Subscript[t, #[[2]]], {h, 0, 5}]/
Product[Subscript[t, i] - Subscript[t, k], {i, #},
{k, DeleteCases[Range[0, 4], Alternatives @@ #]}]
& /@ Subsets[Range[0, 4], {2}])
// Together
```

Theorem 3.3.3 now follows from Theorem 3.3.5.

3.3.4 The degree of $\mathbb{G}(1, n+1)$ and its enumerative meaning

Set $n \geqslant 2$. We shall work with the $2n$-dimensional Grassmannian $\mathbb{G} = G(2, n+2) = G(2, V) = \mathbb{G}(1, n+1)$ of lines in $\mathbb{P}^{n+1}$, where $V = \mathrm{H}^0(\mathbb{P}^{n+1}, \mathscr{O}_{\mathbb{P}^{n+1}}(1))$. The goal of this section is to prove the following.

PROPOSITION 3.3.6. *Set $n \geqslant 2$. There are exactly $C_n = \frac{1}{n+1}\binom{2n}{n}$ lines $\ell \subset \mathbb{P}^{n+1}$ that are incident to $2n$ general $(n-1)$-planes $\Lambda_1, \ldots, \Lambda_{2n} \subset \mathbb{P}^{n+1}$. For instance, there are exactly 2 lines $\ell \subset \mathbb{P}^3$ intersecting 4 general lines in $\mathbb{P}^3$.*

[1] Thanks to Matteo Gallet for providing the code.

Let L be a general 2-dimensional linear subspace of V. The cycle

$$\Sigma_1(L) = \{\, \Lambda \subset V \mid \Lambda \cap L \neq 0 \,\} \subset \mathbb{G}$$

has codimension 1, and its cohomology class $\sigma_1 = [\Sigma_1(L)] \in A^1\mathbb{G}$ is independent of L. In fact, we have $\sigma_1 = c_1(\mathscr{S}^*)$, where $\mathscr{S}$ is the universal subbundle. We wish to prove (via localisation) the identity

$$\int_{\mathbb{G}(1,n+1)} \sigma_1^{2n} = C_n. \tag{3.3.8}$$

Let $\mathbb{T} = \mathbb{G}_m^{n+2}$ act with distinct weights $w_0, w_1, \dots, w_{n+1}$ on

$$V = \mathrm{H}^0(\mathbb{P}^{n+1}, \mathscr{O}_{\mathbb{P}^{n+1}}(1)) = \bigoplus_{0 \leqslant j \leqslant n+1} \mathbb{C} \cdot x_j.$$

We have, as ever, $\mathscr{S}^*|_{\ell_{ij}} = \mathbb{C} \cdot x_i \oplus \mathbb{C} \cdot x_j$ carrying the weight $w_i + w_j$ and $T_{\ell_{ij}}\mathbb{G} = \mathscr{S}^*|_{\ell_{ij}} \otimes \mathscr{Q}|_{\ell_{ij}}$, equivariantly isomorphic to

$$\mathsf{Span}_{\mathbb{C}} \left\{ x_i \otimes x_u^*, x_j \otimes x_u^* \;\middle|\; u \in I \setminus \{\, i, j \,\} \right\},$$

where $I = \{0, 1, \dots, n+1\}$. We compute, by localisation,

$$\begin{aligned}
\int_{\mathbb{G}(1,n+1)} \sigma_1^{2n} &= \int_{\mathbb{G}(1,n+1)} c_1^{\mathbb{T}}(\mathscr{S}^*)^{2n} \\
&= \sum_{\ell_{ij}} \frac{c_1^{\mathbb{T}}(\mathscr{S}^*)|_{\ell_{ij}}^{2n}}{e^{\mathbb{T}}(T_{\ell_{ij}}\mathbb{G}(1,n+1))} \\
&= \sum_{0 \leqslant i < j \leqslant n+1} \frac{(w_i + w_j)^{2n}}{\prod_{u \in I \setminus \{i,j\}} (w_i - w_u)(w_j - w_u)}.
\end{aligned}$$

Checking that the result agrees with with the n-th Catalan number

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$$

is left as a combinatorial exercise for the reader. However, as an example, set $n = 2$. We find

$$\begin{aligned}
\int_{\mathbb{G}(1,3)} \sigma_1^4 &= \sum_{0 \leqslant i < j \leqslant 3} \frac{(w_i + w_j)^4}{(w_i - w_h)(w_j - w_h)(w_i - w_k)(w_j - w_k)} \\
&= \frac{(w_2 + w_3)^4}{(w_2 - w_0)(w_2 - w_1)(w_3 - w_0)(w_3 - w_1)} \\
&\quad + \frac{(w_1 + w_3)^4}{(w_1 - w_0)(w_1 - w_2)(w_3 - w_0)(w_3 - w_2)} \\
&\quad + \frac{(w_1 + w_2)^4}{(w_1 - w_0)(w_1 - w_3)(w_2 - w_0)(w_2 - w_3)} \\
&\quad + \frac{(w_0 + w_3)^4}{(w_0 - w_1)(w_0 - w_2)(w_3 - w_1)(w_3 - w_2)} \\
&\quad + \frac{(w_0 + w_2)^4}{(w_0 - w_1)(w_0 - w_3)(w_2 - w_1)(w_2 - w_3)} \\
&\quad + \frac{(w_0 + w_1)^4}{(w_0 - w_2)(w_0 - w_3)(w_1 - w_2)(w_1 - w_3)}.
\end{aligned}$$

Specialising the equivariant parameters at, say, $(w_0, w_1, w_2, w_3) = (-2, 1, -1, 2)$, we obtain

$$\begin{aligned}
\int_{\mathbb{G}(1,3)} \sigma_1^4 &= \frac{1^4}{1 \cdot (-2) \cdot 4 \cdot 1} + \frac{3^4}{3 \cdot 2 \cdot 4 \cdot 3} + 0 + 0 \\
&\quad + \frac{(-3)^4}{(-3) \cdot (-4) \cdot (-2) \cdot (-3)} + \frac{(-1)^4}{(-1) \cdot (-4) \cdot 2 \cdot (-1)} \\
&= -\frac{1}{8} + \frac{81}{72} + \frac{81}{72} - \frac{1}{8} = \frac{-1 + 9 + 9 - 1}{8} = 2,
\end{aligned} \tag{3.3.9}$$

suggesting that there are exactly two lines in $\mathbb{P}^3$ incident to four general lines. In order to complete the proof of Proposition 3.3.6, we have to address a transversality issue. In other words, we need to exclude the following cases, which are not automatically ruled out by the computation we just completed:

1. The answer is a number smaller than C_n, i.e. some lines come with multiplicity;
2. The answer is ∞.

In other words, what we know so far is the following: *if* the number of lines through $2n$ general $(n-1)$-dimensional planes in $\mathbb{P}^{n+1}$ is finite, then it does not exceed C_n. We employ Kleiman's transversality to rule out these two degenerate cases at the same time.

THEOREM 3.3.7 (Kleiman transversality [31, Theorem 1.7]). *Let G be an algebraic group acting transitively on a variety X over an algebraically closed field of characteristic 0. Let $Y \subset X$ be a subvariety. Then*

(a) *given another subvariety $Z \subset X$, there exists a dense open subset $U \subset G$ such that $g \cdot Y$ and Z are generically transverse for all $g \in U$.*

(b) *If G is affine, then $[g \cdot Y] = [Y]$ in the Chow group $A_* X$.*

More precisely, we apply Theorem 3.3.7 to $X = \mathbb{G}(1, n+1)$, acted on by the affine algebraic group $G = \mathrm{GL}(V) = \mathrm{GL}_{n+2}$. For general $(n-1)$-planes $\Lambda_1, \Lambda_2 \subset \mathbb{P}^{n+1}$, form the codimension 1 cycles

$$\Gamma_i = \left\{ \ell \subset \mathbb{P}^{n+1} \,\middle|\, \ell \cap \Lambda_i \neq \emptyset \right\}, \quad i = 1, 2.$$

Then, in the notation of Theorem 3.3.7, set $Y = \Gamma_1$ and $Z = \Gamma_2$ and apply the result repeatedly.

The proof of Proposition 3.3.6 is now complete.

Exercise 3.3.8. Let $C_1, \ldots, C_4 \subset \mathbb{P}^3$ be general translated of curves of degree $d_1, \ldots, d_4$. Compute the number of lines $\ell \subset \mathbb{P}^3$ passing through $C_1, \ldots, C_4$. (**Hint**: Start out by proving that the cycle $\Gamma_C \subset \mathbb{G}(1,3)$ of lines meeting a curve $C \subset \mathbb{P}^3$ of degree d is a divisor in the Grassmannian, with cohomology class $d \cdot \sigma_1$).

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